Numerical checks: run report

"Conservative and absorptive Kerr scattering from a horizon-exchange law"

Generated by `run_all_checks.py` on 2026-11-02 08:59

All 57 evaluated checks passed

57 passed, 0 failed, 0 ended with an error, 1 informational, 1 skipped; wall time 7 min 51 s.

Run

Date	2026-11-02 08:59
Command	python3 run_all_checks.py -j 2
Python	CPython 3.13.18
mpmath	1.3.0 (backend: python)
SymPy	1.14.0
NumPy	2.5.3
Platform	Linux-6.18.44-fc-v185-x86_64-with-glibc2.39
CPUs	2
Scripts in parallel	2
Repository state	not a git checkout
Wall time	7 min 51 s

How to read this report

Each script is run from its own folder with the Python interpreter above, and its complete output is reproduced below its check. The output is parsed, and every number that the paper or the READMEs quote is compared with its expected value within the tolerance given in the column *Required*; a check passes when all its criteria are met. **ERROR** means that the script did not run to completion, **INFO** marks exploratory scripts without a pass/fail criterion, and **SKIP** marks checks that need data or modules that are not available. Negative controls (the opposite helicity, wrong loop orderings, deliberately wrong hypotheses) pass when the comparison fails by a wide margin. Equation and section numbers refer to the paper and its Supplemental Material.

Summary

#	Check	Paper	Key result	Status	Time
1	`proof_check.py` Symbolic proof of the exchange law	Eq. (6), S1	both loop orderings; sine identity holds	**PASS**	1 s
2	`exact_check.py` Exchange law against full MST	Eq. (6), S9(a)	law holds to 1.5e-60 (10 cases)	**PASS**	10 s
3	`test1.py` Exchange invariance of ν and of the far-zone factor	S9(b)	nu invariant to 2.5e-57; far zone to 2.8e-51	**PASS**	13 s
4	`test3.py` Super-extremal spin and the classical-like regime	Eq. (6), S9(a), S9(b)	law to 8.4e-50 at q > 1	**PASS**	15 s
5	`figure_monodromy/monodromy_loops_check.py` The loops of Fig. 1 by direct integration	Fig. 1, S1	lassos reproduce eta = +1, -1 to 1.4e-8	**PASS**	17 s
6	`classical.py` Structure of the response in the classical limit	S2, Eq. (S3)	Re X_odd/X_even = delta/u to all printed digits	**PASS**	10 s
7	`polar.py` Leading exchange-odd element for $\ell = m = 2$	Eqs. (S4), (S7); S9(i)	C5 = -64i/75 to 4.5e-6	**PASS**	15 s
8	`polar_m.py` Leading elements and slopes for all m	Eqs. (S4), (S7); S9(i)	C5 to 8.6e-5; slopes to 0.05	**PASS**	36 s
9	`slopes.py` Slopes $10m/9$ of R_{2m}	Eq. (S7), S9(i)	slopes 10m/9 to 0.007	**PASS**	58 s

#	Check	Paper	Key result	Status	Time
10	`verify_a7.py` Remainder of R_{2m} and the $\ell = 3$ coefficients	Eq. (S7), S9(i)	R_2m remainder ~ c^3; l = 3 to 0.00017	**PASS**	1 min 22 s
11	`conservative_test/analyze3.py` Taylor coefficients of the polar MST data	Eq. (18), S4	Taylor coefficients of the l = 2...19 MST data	**PASS**	8 s
12	`conservative_test/fit_contact2.py` $\eta = +1$ is the pure exponential	Eq. (18), S4	eta = +1 has no contact part (1.2e-15)	**PASS**	3 s
13	`conservative_test/closedform_check.py` Closed form (18) against MST through $O(a^{13})$	Eq. (18), Eq. (S12)	closed form to 3e-16 through O(a^10)	**PASS**	5 s
14	`conservative_test/higher_orders_eq17.py` Expansion of the closed form: Eqs. (17) and (S14)	Eq. (17), Eqs. (S13), (S14)	c^5...c^8 of Eq. (17); b_0...b_9 of (S14)	**PASS**	29 s
15	`a7_exact.py` Exact $O(c^2)$ ingredients of Eq. (17)	Eq. (17), Eq. (S7)	R_2m and the O(c), O(c^2) brackets exact	**PASS**	4 s
16	`prl_checks/a7_polar_check.py` Closed form against Eq. (17)	Eq. (17)	difference 0 at c^5, c^6, c^7	**PASS**	1 s
17	`segment_test/segment_born.py` Segment formula (S16) against MST	Eq. (S16)	(S16) vs MST to 6.8e-11	**PASS**	40 s
18	`segment_test/vs_production.py` Segment formula for $\ell = 2 \dots 6$	Eq. (S16), S4	l = 2...6 to 4.4e-16	**PASS**	2 min 31 s
19	`segment_test/nonlinear_check.py` Terms nonlinear in $\mathcal{K}$	S4	nonlinear terms -0.85 c^3 at c = 0.025	**PASS**	10 s
20	`physical_region_test/rich_phys.py` Richardson-improved coefficients	Eqs. (10), (11); S5	[q^6] = rho_2/64 to 1.7e-15; Page 4/225	**PASS**	7 s
21	`physical_region_test/pslq2.py` PSLQ identification (exploratory)	S5	informational, no pass/fail criterion	**INFO**	<1 s
22	`physical_region_test/analyze_phys.py` Coefficients of the polar projection	S5	needs phys_run.json, produced by the production drivers (–production)	**SKIP**	–
23	`prl_checks/b6_precision.py` The logarithm $B_6 = i\Pi_m/450$	S5, S9(c)	B6 = i Pi_m/450 to 1e-10	**PASS**	1 s
24	`continuation_test/d6_closed.py` Eq. (10) against the physical-region data	Eq. (10), S9(d)	Eq. (10) to 2e-16	**PASS**	<1 s
25	`continuation_test/d6_m.py` Eq. (10) for $m \neq 0$	Eq. (10), S9(d)	Eq. (10) for m != 0 to 4.8e-15	**PASS**	1 min 50 s
26	`continuation_test/d6_m0.py` Eq. (10) for $m = 0$	Eq. (10), S9(d)	D6 = -kappa^5/225 to 1.6e-17	**PASS**	53 s
27	`continuation_test/d6_general.py` $[q^6]D_6 = \rho_{\lvert m\rvert}/64$ and the large-spin limit	Eq. (10), S9(d), S9(f)	[q^6] = rho_|m|/64; large q to 3.2e-8	**PASS**	<1 s
28	`prl_checks/paper_numbers.py` Closed forms and identities of the main text	Eqs. (10), (16), (S22), (S23)	eps-constant, u0, Eq. (10), [q^5]C = 16 b_2, rho_|m|/64	**PASS**	<1 s
29	`prl_checks/se_allm_analyze.py` Classical a^5 coefficients for every m	S5(d), S9(f)	continuation per m to 1.2e-9	**PASS**	<1 s
30	`prl_checks/allm_analyze.py` Free fit of the transcendental coefficients	Eq. (12), S5	both coefficients -1/450 to 2.7e-9	**PASS**	4 s
31	`prl_checks/unified_check.py` Remainder after subtracting Eq. (S18)	Eq. (S18), S5	remainder below 1.7e-11	**PASS**	<1 s
32	`prl_checks/uniqueness_check.py` Continuation to super-extremal spin; bounds on extra terms	S5, uniqueness (c), (d)	continuation to 3.7e-12 (q $\leq$ 30)	**PASS**	1 s
33	`conservative_continuation/phys_hi_p.py` Physical-region coefficients to q^{24}	Eq. (12), S5	coefficients to q^24; [q^6] odd to 2.7e-9	**PASS**	<1 s
34	`conservative_continuation/e6_fit.py` Fit of the ψ + log term	Eq. (12)	450 lambda = -1 to 1.5e-10	**PASS**	2 s

#	Check	Paper	Key result	Status	Time
35	`conservative_continuation/e6_check_se2.py` Continued closed form against super-extremal MST	Eq. (12), S5	continuation to 1e-6	**PASS**	<1 s
36	`conservative_continuation/e6_identities.py` $[q^5]\mathcal{C} = 16b_2$ and $\mathcal{C}/q^5 \to 64/15$	Eq. (16), S9(e)	[q^5]C = 16 b_2; C/q^5 -> 64/15 (3.3e-13)	**PASS**	<1 s
37	`conservative_continuation/se_taylor_l2.py` Classical $\ell = 2$ Taylor coefficients	Eq. (17), S9(h)	odd part starts at c^5; c^5, c^6 exact	**PASS**	<1 s
38	`prl_checks/d7_phys.py` Exchange-odd ϵ^7 for $m = 2$	Eq. (S23), S9(g)	u1 = 10/9 to 5.8e-11; no q^2, q^3	**PASS**	1 s
39	`prl_checks/d7_fit2.py` Exchange-odd ϵ^7 for all m	Eq. (S23), S9(g)	u0, u1 = 5m/9 (2.9e-10) for all m	**PASS**	3 s
40	`prl_checks/e7_phys.py` Exchange-even ϵ^7 for $m = 2$	Eq. (S23), S9(g)	u0 to 2.7e-8	**PASS**	4 s
41	`prl_checks/e7_allm.py` Exchange-even ϵ^7 for all m	Eq. (S23), S9(g)	u0 to 2.5e-8, u1 = 5m/9 to 8.5e-9	**PASS**	4 s
42	`prl_checks/se7_analyze.py` Classical a^6 coefficients by extrapolation in q	S9(h)	limits commute at a^6 (1.6e-8, 3e-10)	**PASS**	<1 s
43	`prl_checks/se7_compare.py` ϵ^7 closed forms continued to super-extremal q	S9(h)	4.9e-9 (even), 2.8e-10 (odd) for q $\geq$ 50	**PASS**	<1 s
44	`prl_checks/a6_hyp.py` Negative test at a^6	S9(h)	hypothesis rejected (1.70667 vs 256/135)	**PASS**	<1 s
45	`prl_checks/analytic_odd.py` Normalization and the ϵ coefficient	S7(c)-(e), S9(j)	E coefficient pi - i nu2 to 7.3e-6	**PASS**	7 s
46	`prl_checks/series_ratio_check.py` $\mathcal{X}$ ratio and ν to $O(\epsilon^3)$	S7, S9(j)	5mq/9 to 1.3e-6; nu2 = -107/210	**PASS**	5 s
47	`prl_checks/eps4_check.py` Pole term absent from S; spheroidal coefficients	S7(f), S9(j)	E1 exact, E2 to 1.5e-9; pole term absent	**PASS**	<1 s
48	`prl_checks/h_scaling.py` Scaling of the near-zone factor	S7, S9(j)	H_even ~ eps^4, H_odd ~ eps^6; far zone 1.3e-74	**PASS**	3 s
49	`prl_checks/absorption_check.py` $\Gamma_+ - \Gamma_- = -4\epsilon^6 A_6^-$	S9(j)	Gamma_+ - Gamma_- = -4 eps^6 A6^- to 2.5e-6	**PASS**	6 s
50	`kinematics.py` BHPT against the contact-term ansatz	S6, Table S1	ratios 1 to 4.9e-31 (helicity +1)	**PASS**	<1 s
51	`fitcoeffs.py` The nine exchange-odd coefficients by a fit	S6, Table S1	nine coefficients of Table S1, all 20 digits	**PASS**	<1 s
52	`prl_checks/kinematics.py` Copy of the top-level `kinematics.py`	S6, Table S1	ratios 1 to 4.9e-31 (helicity +1)	**PASS**	<1 s
53	`prl_checks/kin_even.py` Exchange-even a^5 contact term	S6, Table S1	c3 = alpha(64/15, 16/3, 16/15) (3.4e-14)	**PASS**	<1 s
54	`prl_checks/kin_tags.py` Tagged and untagged exchange-odd terms	S6, Table S1	tagged and untagged terms of Table S1	**PASS**	<1 s
55	`prl_checks/kin_a6.py` Tagged and untagged terms at a^6	S6, Table S1	a^6 pole and polynomial parts, ratios +-1 to 3.6e-30	**PASS**	<1 s
56	`flux_test/flux_check.py` Flux balance and factorization	S10, S9(k)	flux balance to 1.7e-60	**PASS**	16 s
57	`flux_test/highfreq_check.py` Flux balance at $\epsilon = 0.6$	S10	flux balance to 1.1e-60	**PASS**	14 s
58	`flux_test/complexnu_check.py` The complex-ν branch	S10	flux balance to 2.1e-59; amplification 10.7%	**PASS**	24 s
59	`flux_test/decomp_check.py` Exchange-parity split of the absorption probability	Eq. (11), S10	Delta_odd to 2e-6; half of Page at q = 0	**PASS**	48 s

Contents

Exchange law and monodromy **5**

1. `proof_check.py` . . . 5
2. `exact_check.py` . . . 5
3. `test1.py` . . . 5
4. `test3.py` . . . 6
5. `figure_monodromy/monodromy_loops_check.py` . . . 7

Classical limit: $H_{\ell m}^{\rm odd}$ and R_{2m} **8**

6. `classical.py` . . . 8
7. `polar.py` . . . 8
8. `polar_m.py` . . . 8
9. `slopes.py` . . . 9
10. `verify_a7.py` . . . 9

Polar classical amplitude: Eqs. (17) and (18) **11**

11. `conservative_test/analyze3.py` . . . 11
12. `conservative_test/fit_contact2.py` . . . 11
13. `conservative_test/closedform_check.py` . . . 12
14. `conservative_test/higher_orders_eq17.py` . . . 12
15. `a7_exact.py` . . . 13
16. `prl_checks/a7_polar_check.py` . . . 13

Segment representation **15**

17. `segment_test/segment_born.py` . . . 15
18. `segment_test/vs_production.py` . . . 15
19. `segment_test/nonlinear_check.py` . . . 15

Physical-region double expansion in (ϵ, q) **17**

20. `physical_region_test/rich_phys.py` . . . 17
21. `physical_region_test/pslq2.py` . . . 17
22. `physical_region_test/analyze_phys.py` . . . 18
23. `prl_checks/b6_precision.py` . . . 18

Exchange-odd ϵ^6 coefficient **19**

24. `continuation_test/d6_closed.py` . . . 19
25. `continuation_test/d6_m.py` . . . 19
26. `continuation_test/d6_m0.py` . . . 19
27. `continuation_test/d6_general.py` . . . 20
28. `prl_checks/paper_numbers.py` . . . 20
29. `prl_checks/se_allm_analyze.py` . . . 21

Exchange-even ϵ^6 function and $\mathcal{C}(q)$ **22**

30. `prl_checks/allm_analyze.py` . . . 22
31. `prl_checks/unified_check.py` . . . 23
32. `prl_checks/uniqueness_check.py` . . . 23
33. `conservative_continuation/phys_hi_p.py` . . . 24

34. conservative_continuation/e6_fit.py . . . 25
35. conservative_continuation/e6_check_se2.py . . . 25
36. conservative_continuation/e6_identities.py . . . 26
37. conservative_continuation/se_taylor_l2.py . . . 26

The ϵ^7 operator and the a^6 coefficients **27**
38. prl_checks/d7_phys.py . . . 27
39. prl_checks/d7_fit2.py . . . 27
40. prl_checks/e7_phys.py . . . 28
41. prl_checks/e7_allm.py . . . 28
42. prl_checks/se7_analyze.py . . . 29
43. prl_checks/se7_compare.py . . . 29
44. prl_checks/a6_hyp.py . . . 30

Analytic control of the exchange-odd response **31**
45. prl_checks/analytic_odd.py . . . 31
46. prl_checks/series_ratio_check.py . . . 31
47. prl_checks/eps4_check.py . . . 31
48. prl_checks/h_scaling.py . . . 32
49. prl_checks/absorption_check.py . . . 32

General kinematics **34**
50. kinematics.py . . . 34
51. fitcoeffs.py . . . 34
52. prl_checks/kinematics.py . . . 35
53. prl_checks/kin_even.py . . . 35
54. prl_checks/kin_tags.py . . . 36
55. prl_checks/kin_a6.py . . . 36

Unit-flux normalization and the absorption split **37**
56. flux_test/flux_check.py . . . 37
57. flux_test/highfreq_check.py . . . 37
58. flux_test/complexnu_check.py . . . 37
59. flux_test/decomp_check.py . . . 38

Exchange law and monodromy

Paper: Eq. (6), Fig. 1; S1, S9(a), S9(b)

The exchange law, Eq. (6), relates the response $\mathcal{K}_-$ selected at the inner horizon (the image solution, continuation label $\eta = \pm 1$) to the physical response $\mathcal{K}_+$. In all numerical checks the image is computed directly from MST with $\kappa \to -\kappa$ and $\log(-\kappa) = \log\kappa + i\pi\eta$, without using the law.

1. `proof_check.py`

Symbolic proof of the exchange law. SymPy: the monodromy algebra for both orderings of the loop product gives $\mathcal{K}_-/\mathcal{K}_+ = -\lambda^{\mp 1}(\lambda a - b)/(\lambda b - a)$, and $(\lambda a - b)/(\lambda b - a) = \sin\pi(\nu - i\tau)/\sin\pi(\nu + i\tau)$.

Paper: Eq. (6), S1 Status: **PASS** Time: 1 s
Command: `python3 proof_check.py`

Criterion	Measured	Required	
Loop product M_- = M_inf M_+^(-1)	printed	"= -lam^-1 (lam a - b)/(lam b - a)"	✓
Loop product M_- = M_+^(-1) M_inf	printed	"= -lam^+1 (lam a - b)/(lam b - a)"	✓
(lam a - b)/(lam b - a) - sin pi(nu - i tau)/sin pi(nu + i tau), numerical residual	max 0 (2 values)	≤ 1e-100	✓

Output (`logs/proof-check-py.txt`):

```
M- = M_inf M+^-1:  X_-/X_+ = (a*lambda - b)/(lambda*(a - b*lambda))
   = -lam^-1 (lam a - b)/(lam b - a)
M- = M+^-1 M_inf:  X_-/X_+ = lambda*(a*lambda - b)/(a - b*lambda)
   = -lam^+1 (lam a - b)/(lam b - a)
check (lam a - b)/(lam b - a) == sin pi(nu - i tau)/sin pi(nu + i tau): 0.e-140 + 0.e-140*I
```

2. `exact_check.py`

Exchange law against full MST. Ratio of the image response to the physical one, both computed with MST at 60 digits, divided by the exchange law, for $s = 0, -1, -2$, $\ell \le 3$, $q \le 0.9$, $\epsilon \le 0.6$ and both η.

Paper: Eq. (6), S9(a) Status: **PASS** Time: 10 s
Command: `python3 exact_check.py`

Criterion	Measured	Required	
|K(image)/K(physical) / exchange law - 1|: s = 0, -1, -2; l ≤ 3; both eta	max 1.54e-60 (10 values)	≤ 1e-55	✓
|K(image)/K(physical) - 1| (the image response differs from the physical one)	min 9.2e-8 (10 values)	≥ 1e-12	✓

Output (`logs/exact-check-py.txt`):

```
l=2 m=0 s=0 q=0 eps=0.3 eta=+1:  |X_dual/X / formula - 1| = 4.27e-61   (|X_dual/X - 1| = 0.505)
l=2 m=0 s=0 q=0 eps=0.3 eta=-1:  |X_dual/X / formula - 1| = 2.65e-61   (|X_dual/X - 1| = 0.0766)
l=2 m=2 s=-2 q=0 eps=0.5 eta=+1:  |X_dual/X / formula - 1| = 1.54e-60   (|X_dual/X - 1| = 1.6)
l=2 m=2 s=-2 q=0 eps=0.5 eta=-1:  |X_dual/X / formula - 1| = 3.12e-61   (|X_dual/X - 1| = 0.0692)
l=2 m=2 s=-2 q=0.7 eps=0.3 eta=+1:  |X_dual/X / formula - 1| = 8.65e-61   (|X_dual/X - 1| = 2.57e-5)
l=2 m=2 s=-2 q=0.7 eps=0.3 eta=-1:  |X_dual/X / formula - 1| = 4.68e-61   (|X_dual/X - 1| = 0.41)
l=3 m=1 s=-1 q=0.4 eps=0.6 eta=+1:  |X_dual/X / formula - 1| = 6.24e-61   (|X_dual/X - 1| = 1.58)
l=3 m=1 s=-1 q=0.4 eps=0.6 eta=-1:  |X_dual/X / formula - 1| = 4.91e-61   (|X_dual/X - 1| = 0.401)
l=1 m=-1 s=0 q=0.9 eps=0.2 eta=+1:  |X_dual/X / formula - 1| = 2.34e-61   (|X_dual/X - 1| = 0.708)
l=1 m=-1 s=0 q=0.9 eps=0.2 eta=-1:  |X_dual/X / formula - 1| = 7.26e-61   (|X_dual/X - 1| = 9.2e-8)
```

3. `test1.py`

Exchange invariance of ν and of the far-zone factor. The renormalized angular momentum ν and the far-zone factor are the same for the physical and image solutions (Schwarzschild and Kerr, $s = 0$ and -2).

Paper: S9(b) Status: **PASS** Time: 13 s
Command: `python3 test1.py`

Criterion	Measured	Required	
|nu(image) - nu(physical)|, s = 0 and -2, 9 cases	max 2.55e-57 (9 values)	$\leq$ 1e-50	✓
|S_far(image)/S_far(physical) - 1|	max 2.84e-51 (9 values)	$\leq$ 1e-45	✓

Output (`logs/test1-py.txt`):

```
[Schwarzschild s=0] l=2 m=0 s=0 q=0 eps=0.1
  nu_phys  = 1.9962304091135874071    nu_dual - nu_phys = 1.52e-61
  |Sfar_dual/Sfar - 1| = 1.53e-51
  eta=+1: X_dual/X-1 = (-0.0051472435 - 0.1013311j) | leading formula (-0.0014649546 + 0.054108808j) | H_dual/H-1 = (-0.0051472436 - 0.10133111j)
  eta=-1: X_dual/X-1 = (-0.0014649546 - 0.054108808j) | leading formula (-0.0051472435 + 0.1013311j) | H_dual/H-1 = (-0.0014649545 - 0.054108811j)

[Schwarzschild s=0] l=2 m=0 s=0 q=0 eps=0.05
  nu_phys  = 1.9990590452343552534    nu_dual - nu_phys = 6.27e-63
  |Sfar_dual/Sfar - 1| = 6.23e-52
  eta=+1: X_dual/X-1 = (-0.00096147619 - 0.043840939j) | leading formula (-0.0005129361 + 0.032025132j) | H_dual/H-1 = (-0.0009614762 - 0.043840939j)
  eta=-1: X_dual/X-1 = (-0.0005129361 - 0.032025132j) | leading formula (-0.00096147619 + 0.043840939j) | H_dual/H-1 = (-0.0005129361 - 0.032025132j)

[Schwarzschild s=0] l=2 m=0 s=0 q=0 eps=0.025
  nu_phys  = 1.9997648510694906888    nu_dual - nu_phys = 1.33e-63
  |Sfar_dual/Sfar - 1| = 3.26e-52
  eta=+1: X_dual/X-1 = (-0.00020659669 - 0.020326109j) | leading formula (-0.00015089878 + 0.017371666j) | H_dual/H-1 = (-0.00020659669 - 0.020326109j)
  eta=-1: X_dual/X-1 = (-0.00015089878 - 0.017371666j) | leading formula (-0.00020659669 + 0.020326109j) | H_dual/H-1 = (-0.00015089878 - 0.017371666j)

[Schwarzschild s=-2] l=2 m=2 s=-2 q=0 eps=0.1
  nu_phys  = 1.9948862953636113997    nu_dual - nu_phys = 4.97e-62
  |Sfar_dual/Sfar - 1| = 1.42e-51
  eta=+1: X_dual/X-1 = (-0.0094599432 - 0.13722389j) | leading formula (-0.0026923901 + 0.073331652j) | H_dual/H-1 = (-0.0094599473 - 0.13722402j)
  eta=-1: X_dual/X-1 = (-0.0026923901 - 0.073331652j) | leading formula (-0.0094599432 + 0.13722389j) | H_dual/H-1 = (-0.0026923857 - 0.07333172j)

[Schwarzschild s=-2] l=2 m=2 s=-2 q=0 eps=0.05
  nu_phys  = 1.9987250439080176367    nu_dual - nu_phys = 2.55e-57
  |Sfar_dual/Sfar - 1| = 5.8e-52
  eta=+1: X_dual/X-1 = (-0.001764655 - 0.059381782j) | leading formula (-0.00094142244 + 0.043381547j) | H_dual/H-1 = (-0.001764655 - 0.059381785j)
  eta=-1: X_dual/X-1 = (-0.00094142244 - 0.043381547j) | leading formula (-0.001764655 + 0.059381782j) | H_dual/H-1 = (-0.0009414224 - 0.043381549j)

[Schwarzschild s=-2] l=2 m=2 s=-2 q=0 eps=0.025
  nu_phys  = 1.9996814760758549662    nu_dual - nu_phys = 1.1e-64
  |Sfar_dual/Sfar - 1| = 3.0e-52
  eta=+1: X_dual/X-1 = (-0.00037904361 - 0.027530775j) | leading formula (-0.00027685447 + 0.023529392j) | H_dual/H-1 = (-0.00037904361 - 0.027530775j)
  eta=-1: X_dual/X-1 = (-0.00027685447 - 0.023529392j) | leading formula (-0.00037904361 + 0.027530775j) | H_dual/H-1 = (-0.00027685447 - 0.023529392j)

[Kerr s=0] l=2 m=2 s=0 q=0.6 eps=0.1
  nu_phys  = 1.9962816369678494284    nu_dual - nu_phys = 1.96e-60
  |Sfar_dual/Sfar - 1| = 1.11e-51
  eta=+1: X_dual/X-1 = (-3.4205666e-11 + 8.2711143e-6j) | leading formula (-0.0010918626 - 0.046717589j) | H_dual/H-1 = (-3.362525e-11 + 8.2711068e-6j)
  eta=-1: X_dual/X-1 = (-0.0010918626 + 0.046717589j) | leading formula (-3.4205666e-11 - 8.2711143e-6j) | H_dual/H-1 = (-0.0010918564 + 0.046717546j)

[Kerr s=0] l=2 m=2 s=0 q=0.6 eps=0.05
  nu_phys  = 1.9990654490219306463    nu_dual - nu_phys = 2.55e-57
  |Sfar_dual/Sfar - 1| = 2.84e-51
  eta=+1: X_dual/X-1 = (-9.8521386e-13 + 1.4037192e-6j) | leading formula (-6.8975451e-5 - 0.011745048j) | H_dual/H-1 = (-9.7870135e-13 + 1.4037191e-6j)
  eta=-1: X_dual/X-1 = (-6.8975451e-5 + 0.011745048j) | leading formula (-9.8521386e-13 - 1.4037192e-6j) | H_dual/H-1 = (-6.8975371e-5 + 0.011745046j)

[Kerr s=0] l=2 m=2 s=0 q=0.6 eps=0.025
  nu_phys  = 1.9997656527053764305    nu_dual - nu_phys = 1.53e-62
  |Sfar_dual/Sfar - 1| = 2.92e-52
  eta=+1: X_dual/X-1 = (-4.1829583e-14 + 2.8923894e-7j) | leading formula (-4.3370518e-6 - 0.0029451799j) | H_dual/H-1 = (-4.1741979e-14 + 2.8923894e-7j)
  eta=-1: X_dual/X-1 = (-4.3370518e-6 + 0.0029451799j) | leading formula (-4.1829583e-14 - 2.8923894e-7j) | H_dual/H-1 = (-4.3370507e-6 + 0.0029451799j)
```

4. `test3.py`

Super-extremal spin and the classical-like regime. The exchange law and the invariances for $q = 1.5$, 2, 3 ($s = 0$, 50 digits); $\mathrm{Re}\,\mathcal{X}_{\mathrm{odd}}/\mathcal{X}_{\mathrm{even}} \to \delta/u$ as $\epsilon \to 0$ at fixed $a\omega$.

Paper: Eq. (6), S9(a), S9(b) Status: **PASS** Time: 15 s
Command: `python3 test3.py`

Criterion	Measured	Required	
super-extremal q = 1.5, 3, 2 (s = 0): |nu(+) - nu(-)|	max 2.5e-57 (3 values)	$\leq$ 1e-50	✓
super-extremal: |K(-)/K(+) / exchange law - 1|	max 8.42e-50 (3 values)	$\leq$ 1e-45	✓
super-extremal: |S_far(-)/S_far(+) - 1|	max 5.35e-51 (3 values)	$\leq$ 1e-45	✓
a*omega fixed, eps -> 0: Re(X_odd/X_even) vs delta/u (12 cases)	max rel. diff 0.000161 (12 values)	rel. diff $\leq$ 0.001	✓

Output (`logs/test3-py.txt`):

```
== exactness of the swap factor for super-extremal spin (s=0) ==
l=2 m=2 q=1.5 eps=0.2: nu=(1.98582673997 + 1.17387098046e-61j) (|Im nu|=1.2e-61), |nu(+)-nu(-)|=2.5e-57, |X(-)/X(+)/Phi - 1| = 6.98e-51, |Sfar(-)/Sfar(+)-1| = 0.0
l=2 m=-1 q=3 eps=0.1: nu=(1.9961050173 - 2.57293553188e-61j) (|Im nu|=2.6e-61), |nu(+)-nu(-)|=2.5e-57, |X(-)/X(+)/Phi - 1| = 8.42e-50, |Sfar(-)/Sfar(+)-1| = 7.42e-52
l=1 m=1 q=2 eps=0.3: nu=(0.951438188462 + 6.09440336765e-60j) (|Im nu|=6.1e-60), |nu(+)-nu(-)|=1.3e-57, |X(-)/X(+)/Phi - 1| = 6.06e-50, |Sfar(-)/Sfar(+)-1| = 5.35e-51

== classical-like regime: a*omega fixed, eps -> 0 (q = 2 a omega/eps) ==
l=2 m=+2 eps=0.04 q=15.0: delta=-0.000525493  u=-0.0017867  X_odd/X_even = (0.294111 - 0.00150808j)  delta/u = 0.294113  H_odd/H_even = (0.294032 - 0.00150993j)
```

```
l=2 m=+2 eps=0.02 q=30.0: delta=-0.000128972  u=-0.000445001  X_odd/X_even = (0.289824 - 0.000371143j)  delta/u = 0.289824  H_odd/H_even = (0.289743 - 0.000372123j)
l=2 m=+2 eps=0.01 q=60.0: delta=-2.98629e-5  u=-0.000111146  X_odd/X_even = (0.268682 - 8.70444e-5j)  delta/u = 0.268682  H_odd/H_even = (0.268594 - 8.75528e-5j)
l=2 m=+1 eps=0.04 q=15.0: delta=-0.000553314  u=0.000442955  X_odd/X_even = (-1.24914 + 0.000974059j)  delta/u = -1.24914  H_odd/H_even = (-1.24915 + 0.000973751j)
l=2 m=+1 eps=0.02 q=30.0: delta=-0.000133571  u=0.000111018  X_odd/X_even = (-1.20314 + 0.000187805j)  delta/u = -1.20314  H_odd/H_even = (-1.20315 + 0.000187679j)
l=2 m=+1 eps=0.01 q=60.0: delta=-2.8661e-5  u=2.7772e-5  X_odd/X_even = (-1.03201 + 5.85706e-6j)  delta/u = -1.03201  H_odd/H_even = (-1.03201 + 5.84773e-6j)
l=2 m=-1 eps=0.04 q=15.0: delta=-0.000634154  u=0.00490227  X_odd/X_even = (-0.12935 - 0.00195892j)  delta/u = -0.129359  H_odd/H_even = (-0.129332 - 0.00195805j)
l=2 m=-1 eps=0.02 q=30.0: delta=-0.000153764  u=0.00122306  X_odd/X_even = (-0.12572 - 0.000475428j)  delta/u = -0.125721  H_odd/H_even = (-0.125702 - 0.000475067j)
l=2 m=-1 eps=0.01 q=60.0: delta=-3.37094e-5  u=0.000305608  X_odd/X_even = (-0.110303 - 0.000104613j)  delta/u = -0.110303  H_odd/H_even = (-0.110282 - 0.000104449j)
l=2 m=-2 eps=0.04 q=15.0: delta=-0.000691098  u=0.00713193  X_odd/X_even = (-0.0968865 - 0.00215077j)  delta/u = -0.0969021  H_odd/H_even = (-0.0969522 - 0.00215465j)
l=2 m=-2 eps=0.02 q=30.0: delta=-0.000170329  u=0.00177908  X_odd/X_even = (-0.0957392 - 0.0005302j)  delta/u = -0.0957402  H_odd/H_even = (-0.0958054 - 0.000531727j)
l=2 m=-2 eps=0.01 q=60.0: delta=-4.01947e-5  u=0.000444525  X_odd/X_even = (-0.0904214 - 0.000125243j)  delta/u = -0.0904215  H_odd/H_even = (-0.0904914 - 0.000125911j)
```

5. `figure_monodromy/monodromy_loops_check.py`

The loops of Fig. 1 by direct integration. RK4 integration of the $s = -2$ radial Teukolsky equation ($\ell = m = 2$, $q = 0.6$, $\epsilon = 0.3$) along the loops of Fig. 1: $T_\infty = T_+ T_-^{\rm above} = T_-^{\rm below} T_+$, and the lassos passing above and below r_+ give the MST image with $\eta = +1$ and $\eta = -1$.

Paper: Fig. 1, S1 Status: **PASS** Time: 17 s

Command: `cd figure_monodromy && python3 monodromy_loops_check.py`

Criterion	Measured	Required	
|T_inf - T_+ T_-(above)| and |T_inf - T_-(below) T_+| (entries ~ 1e4, RK4)	max 0.000183 (2 values)	$\leq$ 0.01	✓
wrong orderings T_-(above) T_+ and T_+ T_-(below) (negative control)	min 6.75e4 (2 values)	$\geq$ 1e3	✓
K_-/K_+ from the lasso above r_+ vs the MST image with eta = +1	(0.99999989 + 0.00047113i) (diff 0)	diff $\leq$ 1e-6	✓
K_-/K_+ from the lasso below r_+ vs the MST image with eta = -1	(0.84690767 + 0.53173997i) (diff 1.4e-8)	diff $\leq$ 1e-6	✓
|above - below| (the two lassos select different images)	0.553	$\geq$ 0.1	✓

Output (`logs/figure-monodromy-monodromy-loops-check-py.txt`):

```
(1) |T_inf - T_+ T_-^above| = 4.2340857136531705e-05   |T_inf - T_-^below T_+| = 0.00018303155294916934   (wrong orders:  67462.95939888361 67532.08877096923 )
(2) eig T_+ : [ 0.07488408-0.e+00j 13.35397293+2.e-08j]  expected [ 0.07488408 13.35397294]
    eig T_- : [1.13700800e-02+0.0e+00j 8.79500929e+01-1.4e-07j] [1.13700800e-02-0.0e+00j 8.79500931e+01+2.7e-07j]  expected [1.13700800e-02 8.79500932e+01]
    eig T_inf: [0.9610302+0.27644339j 0.9610302-0.2764434j ]  expected [0.9610302-0.27644339j 0.9610302+0.27644339j]  (nu = (1.9554220852+0j) )
(3) K_-/K_+ from monodromy: above r+ -> (0.99999989+0.00047113j)   below r+ -> (0.84690767+0.53173997j)
    MST image, eta=+1        -> (0.99999989+0.00047113j)   eta=-1 -> (0.84690768+0.53173996j)
```

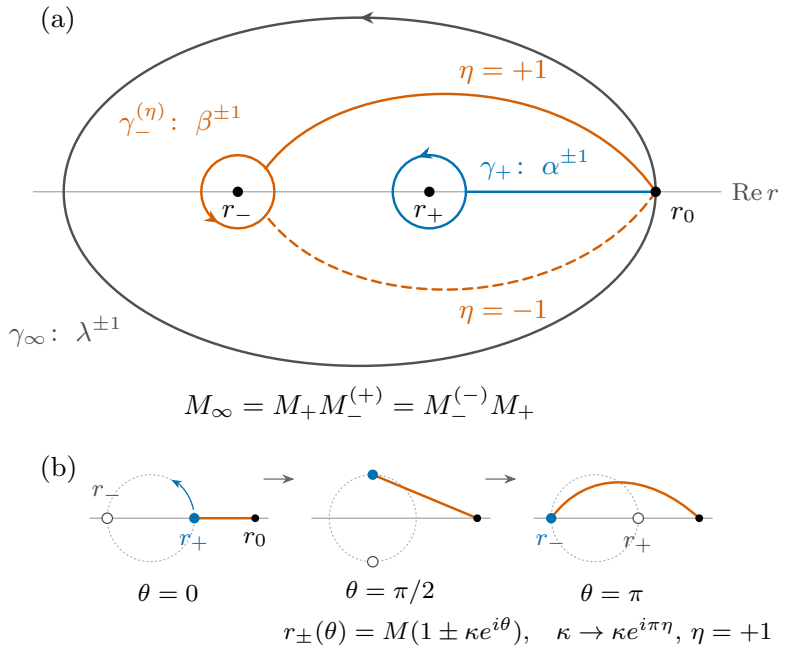


Fig. 1 of the paper (`figure_monodromy/fig_monodromy.pdf`); the check above integrates the radial equation along these loops.

Classical limit: $H^{\rm odd}_{\ell m}$ and R_{2m}

Paper: S2, S3; Eqs. (S3), (S4), (S7); S9(i)

The classical limit is $\epsilon \to 0$ at fixed $c = a\omega$, on the super-extremal branches $\kappa = \eta\, iq\sqrt{1-1/q^2}$. The exchange-odd element starts at $c^{2\ell+1}$ with the closed-form coefficient of Eq. (S4), and $R_{2m} = 1 + \frac{10m}{9}c + \frac{5(475m^2+261)}{3969}c^2 + O(c^3)$, Eq. (S7).

6. `classical.py`

Structure of the response in the classical limit. For $s = 0$, $c = 0.3$ and $m = 2, 1$: $\mathcal{X}_{\rm odd}/\mathcal{X}_{\rm even} \to \delta/u$ as $\epsilon \to 0$, with an imaginary part vanishing as ϵ^2.

Paper: S2, Eq. (S3) Status: **PASS** Time: 10 s
Command: `python3 classical.py`

Criterion	Measured	Required	
Re(X_odd/X_even) vs delta/u at eps = 0.005 (m = 2, 1)	max rel. diff 0 (2 values)	rel. diff ≤ 1e-6	✓
Im(X_odd/X_even) ~ eps^2: ratio of the values at eps = 0.01 and 0.005	4.0002 ... 4.0037 (2 values)	in [3.8, 4.2]	✓

Output (`logs/classical-py.txt`):

```
m=2: E(c=0.3) = 5.98712708737664  (O(c^2) formula: 5.98714285714286)
   eps=0.04: delta/eps^2=-0.33041572  u/eps^2=-1.1166887  X_odd/X_even=(0.29588675 - 0.0015154466j)  delta/u=0.29588884
   eps=0.02: delta/eps^2=-0.33035908  u/eps^2=-1.1125014  X_odd/X_even=(0.29695149 - 0.00037853424j)  delta/u=0.29695162
   eps=0.01: delta/eps^2=-0.33034493  u/eps^2=-1.1114584  X_odd/X_even=(0.29721752 - 9.4613094e-5j)  delta/u=0.29721753
   eps=0.005: delta/eps^2=-0.33034139  u/eps^2=-1.1111979  X_odd/X_even=(0.29728402 - 2.3651994e-5j)  delta/u=0.29728402
m=1: E(c=0.3) = 5.96139707407033  (O(c^2) formula: 5.96142857142857)
   eps=0.04: delta/eps^2=-0.3497638  u/eps^2=0.27684703  X_odd/X_even=(-1.263381 + 0.0010480681j)  delta/u=-1.2633829
   eps=0.02: delta/eps^2=-0.34969517  u/eps^2=0.277546  X_odd/X_even=(-1.2599538 + 0.00025816383j)  delta/u=-1.2599539
   eps=0.01: delta/eps^2=-0.34967802  u/eps^2=0.27771989  X_odd/X_even=(-1.2591033 + 6.4302397e-5j)  delta/u=-1.2591033
   eps=0.005: delta/eps^2=-0.34967373  u/eps^2=0.27776331  X_odd/X_even=(-1.258891 + 1.6060724e-5j)  delta/u=-1.258891
```

7. `polar.py`

Leading exchange-odd element for $\ell = m = 2$. $H^{\rm odd}/(\epsilon c^5)$ from MST at $a\omega = 0.004 \ldots 0.02$, extrapolated to $C_5 = -64i/75$; the slope $C_6/C_5 = 20/9$.

Paper: Eqs. (S4), (S7); S9(i) Status: **PASS** Time: 15 s
Command: `python3 polar.py`

Criterion	Measured	Required	
Im C5 vs -64/75, Eq. (S4) (l = m = 2)	-0.853337165371 (rel. diff 4.5e-6)	rel. diff ≤ 1e-5	✓
|Re C5| (vanishes in the limit)	2.8e-5	≤ 0.0001	✓
Re C6/C5 vs 20/9, Eq. (S7) (m = 2)	2.221018926 (diff 0.0012)	diff ≤ 0.01	✓

Output (`logs/polar-py.txt`):

```
a*w=0.004:  H_odd/(eps (aw)^5) -> (8.3853570449658e-6 - 0.86095698174802j)   (prediction at leading order: (0.0 - 0.8533333333j))
a*w=0.008:  H_odd/(eps (aw)^5) -> (2.0978117700924e-6 - 0.86865372986312j)   (prediction at leading order: (0.0 - 0.8533333333j))
a*w=0.012:  H_odd/(eps (aw)^5) -> (9.3290069367049e-7 - 0.8764270925408j)   (prediction at leading order: (0.0 - 0.8533333333j))
a*w=0.016:  H_odd/(eps (aw)^5) -> (5.2499362518601e-7 - 0.88427727986894j)   (prediction at leading order: (0.0 - 0.8533333333j))
a*w=0.02:  H_odd/(eps (aw)^5) -> (3.3610396754624e-7 - 0.89220502919837j)   (prediction at leading order: (0.0 - 0.8533333333j))

C5 = (2.79888103022e-5 - 0.853337165371j)   expected (0.0 - 0.8533333333333j)
kappa6 = C6/C5 = (2.221018926 - 0.008847049368j)   needed for BGKV polar formula: 20/9 = 2.222222222
C7/C5 = (2.8466568 + 0.91608373j)
```

8. `polar_m.py`

Leading elements and slopes for all m. C_5 against the closed form (S4) for $s = -2$ ($m = 1, 0, -1, -2$) and $s = 0$ ($m = 2, -1$); slopes $C_6/C_5 = 10m/9$ for $s = -2$ and 0 for $s = 0$, from a two-point fit in $a\omega$ (accuracy about 0.03).

Paper: Eqs. (S4), (S7); S9(i) Status: **PASS** Time: 36 s
Command: `python3 polar_m.py`

Criterion	Measured	Required	
C5 vs closed form (S4): s = -2 (m = 1, 0, -1, -2), s = 0 (m = 2, -1)	max rel. diff 8.58e-5 (6 values)	rel. diff $\leq$ 0.0002	✓
Re C6/C5 vs 10m/9 (s = -2) and 0 (s = 0), two-point fit in a*omega	max diff 0.032 (6 values)	diff $\leq$ 0.05	✓

Output (`logs/polar-m-py.txt`):

```
s=-2 l=2 m=+1: C5 = (-1.814496114e-6 + 0.2133294028j)  closed form (0.0 + 0.2133333333j) | C6/C5 = (1.1210954 - 0.0009229392j)  expected 1.1111111
s=-2 l=2 m=+0: C5 = (-1.677209675e-8 - 0.14222267j)  closed form (0.0 - 0.1422222222j) | C6/C5 = (0.002480693 + 9.7957929e-6j)  expected 0
s=-2 l=2 m=-1: C5 = (1.850864141e-6 + 0.2133294647j)  closed form (0.0 + 0.2133333333j) | C6/C5 = (-1.1012832 + 0.00091799279j)  expected -1.1111111
s=-2 l=2 m=-2: C5 = (-1.465105192e-5 - 0.8532616322j)  closed form (0.0 - 0.8533333333j) | C6/C5 = (-2.1902395 + 0.0018024114j)  expected -2.2222222
s=+0 l=2 m=+2: C5 = (4.044850418e-7 - 0.02370388851j)  closed form (0.0 - 0.0237037037j) | C6/C5 = (-0.0022914827 - 0.0018310893j)  expected 0
s=+0 l=2 m=-1: C5 = (5.160822368e-8 + 0.005925999698j)  closed form (0.0 + 0.005925925926j) | C6/C5 = (-0.0016263428 + 0.00093023709j)  expected 0
```

9. `slopes.py`

Slopes $10m/9$ of R_{2m}. Quartic fit in $a\omega$ of $H^{\rm odd}/(\epsilon c^5)$ at five values of $a\omega$ ($s = -2$, $\ell = 2$): C_5 and $C_6/C_5 = 10m/9$ for $m = 1, 0, -1, -2$.

Paper: Eq. (S7), S9(i) Status: **PASS** Time: 58 s
Command: `python3 slopes.py`

Criterion	Measured	Required	
C5 vs -(-1)^m (8/225) i (2+m)!(2-m)!, Eq. (S4)	max rel. diff 3.3e-5 (4 values)	rel. diff $\leq$ 0.0001	✓
Re C6/C5 vs 10m/9, Eq. (S7), m = 1, 0, -1, -2 (quartic fit in a*omega)	max diff 0.00698 (4 values)	diff $\leq$ 0.02	✓

Output (`logs/slopes-py.txt`):

```
m=+1: C5=(-3.468757928e-6 + 0.2133379943j)  C6/C5=(1.1052479 - 0.0043763177j)  (10m/9=1.1111111)
m=+0: C5=(-2.549532461e-8 - 0.1422259317j)  C6/C5=(-0.0069786554 + 3.6707975e-5j)  (10m/9=0.0)
m=-1: C5=(3.524017805e-6 + 0.213337946j)  C6/C5=(-1.1168795 + 0.0043929007j)  (10m/9=-1.1111111)
m=-2: C5=(-2.795524403e-5 - 0.8533369179j)  C6/C5=(-2.2233466 + 0.0086931527j)  (10m/9=-2.2222222)
```

10. `verify_a7.py`

Remainder of R_{2m} and the $\ell = 3$ coefficients. $H^{\rm odd}/H_0$ against R_{2m} through $O(c^2)$: the remainder is proportional to c^3. Leading $\ell = 3$ coefficients for $m = 2, 0, -1, 3$ against the closed form.

Paper: Eq. (S7), S9(i) Status: **PASS** Time: 1 min 22 s
Command: `python3 verify_a7.py`

Criterion	Measured	Required	
|(H_odd/H_0 - R_2m)/c^3|, Eq. (S7), m = 2, 0, -2, c = 0.01, 0.02, 0.03	max 2.43 (9 values)	$\leq$ 5	✓
relative variation of (H_odd/H_0 - R_2m)/c^3 over c = 0.01...0.03, m = +-2 (remainder ~ c^3)	max 0.0139 (2 values)	$\leq$ 0.05	✓
l = 3 leading coefficient vs closed form, m = 2, 0, -1, 3	max rel. diff 0.000168 (4 values)	rel. diff $\leq$ 0.00025	✓

Output (`logs/verify-a7-py.txt`):

```
== l=2 (s=-2): H_odd/H0 vs 1 + (10m/9)c + 5(475 m^2+261)c^2/3969, residual/c^3 should be ~const ==
  m=+2 c=0.01: ratio=(1.0224968555351 - 2.0280669264152e-13j)  series=1.0224944570421  (ratio-series)/c^3=(2.39849 - 2.02807e-7j)
  m=+2 c=0.02: ratio=(1.0455527056946 - 5.080138621169e-14j)  series=1.0455333837239  (ratio-series)/c^3=(2.41525 - 6.35017e-9j)
  m=+2 c=0.03: ratio=(1.0691824493128 - 2.2625753706157e-14j)  series=1.0691167800454  (ratio-series)/c^3=(2.4322 - 8.37991e-10j)
  m=+0 c=0.01: ratio=(1.0000328803828 - 2.1508767777162e-14j)  series=1.0000328798186  (ratio-series)/c^3=(0.000564172 - 2.15088e-8j)
  m=+0 c=0.02: ratio=(1.0001315283039 - 4.3394500054805e-15j)  series=1.0001315192744  (ratio-series)/c^3=(0.00112869 - 5.42431e-10j)
  m=+0 c=0.03: ratio=(1.0002959640821 - 1.4566942551244e-15j)  series=1.0002959183673  (ratio-series)/c^3=(0.00169314 - 5.39516e-11j)
  m=-2 c=0.01: ratio=(0.97804764703652 + 1.9363423355591e-13j)  series=0.97805001259763  (ratio-series)/c^3=(-2.36556 + 1.93634e-7j)
  m=-2 c=0.02: ratio=(0.9566256998207 4 + 4.8363143763052e-14j)  series=0.95664449483497  (ratio-series)/c^3=(-2.34938 + 6.04539e-9j)
  m=-2 c=0.03: ratio=(0.93572044554304 + 2.150418193504e-14j)  series=0.93578344671202  (ratio-series)/c^3=(-2.33338 + 7.96451e-10j)
```

```
== l=3 (s=-2): leading closed form  H0(3,m)/(eps c^7) = (8/893025) i (-1)^m (3+m)!(3-m)! ==
  m=+2: MST -> (3.902236831e-16 + 0.001074917997j)   closed form (0.0 + 0.0010749979j)
  m=+0: MST -> (1.034711694e-17 + 0.0003224993702j)   closed form (0.0 + 0.0003224993701j)
  m=-1: MST -> (6.054820647e-17 - 0.0004299913197j)   closed form (0.0 - 0.0004299991602j)
  m=+3: MST -> (-3.345786731e-15 - 0.006448903091j)   closed form (0.0 - 0.006449987402j)
```

Polar classical amplitude: Eqs. (17) and (18)

Paper: Eqs. (17), (18); S4, Eqs. (S12)-(S14)

For incidence along the spin the classical amplitude is known in closed form, Eq. (18) [Supplemental Eq. (S12)]. These checks compare it with full MST (110 digits, $\ell = 2 \ldots 19$, both η) order by order in a, and expand it symbolically to obtain Eq. (17), its a^8 term and the coefficients of Eq. (S14).

11. `conservative_test/analyze3.py`

Taylor coefficients of the polar MST data. Cauchy-integral Taylor coefficients in c of the 110-digit MST data `polar_run2.json` ($\ell = 2 \ldots 19$, both η); writes `taylor2.json`, read by the next two scripts.

Paper: Eq. (18), S4 Status: **PASS** Time: 8 s
Command: `cd conservative_test && python3 analyze3.py`

Criterion	Measured	Required	
worst relative |G| at the root nu (continued-fraction equation)	2.42e-47	$\leq$ 1e-40	✓
max |x(eta = +1) - x(eta = -1)| (nu is exchange-invariant)	2.24e-49	$\leq$ 1e-40	✓
max |Re s_1|	1.54e-15	$\leq$ 1e-13	✓
max |Re T_n|, n $\leq$ 8 (purely imaginary Taylor coefficients)	1.02e-15	$\leq$ 1e-13	✓

Output (`logs/conservative-test-analyze3-py.txt`):

```
worst relative |G| at root: 2.42e-47;  max |x(eta=+1)-x(eta=-1)| = 2.24e-49
max |Re s1| = 1.54e-15
max |l=16 term| per L: {2: '4.2e-38', 3: '6.2e-35', 4: '6.8e-33', 5: '3.3e-30', 6: '4.7e-28', 7: '1.2e-25', 8: '2.4e-23', 9: '3.4e-21'}
n=0: L2:-0.342267747316  L3:-1.30059985828  L4:-2.03893120364  L5:-2.69964829652  L6:-3.31624232674
n=1: L2:-0.252313252202  L3:-6.63157670684e-18  L4:-2.56158831599e-18  L5:-1.36376951467e-18  L6:-8.41015319382e-19
n=2: L2:0.0420522087003  L3:-0.0355406031038  L4:-1.54934777177e-18  L5:-6.88026134268e-19  L6:-3.99018059617e-19
n=3: L2:-0.00800994451435  L3:0.0118468677013  L4:-0.00447769510752  L5:-1.0438993989e-19  L6:-2.7767008373e-20
n=4: L2:0.00150186459644  L3:-0.00296171692532  L4:0.00201496279838  L5:-0.000472526555367  L6:-1.1645312519e-20
n=5: L2:-0.000266998150478  L3:0.000631832944068  L4:-0.000586170995894  L5:0.000252014162863  L6:-4.21489660205e-5
n=6: L2:4.44996917464e-5  L3:-0.000119665330316  L4:0.000135687730531  L5:-8.07737701483e-5  L6:2.50886702503e-5
n=7: L2:-6.93501689554e-6  L3:2.05140566256e-5  L4:-2.68393313138e-5  L5:1.97813314649e-5  L6:-8.6018298001e-6
n=8: L2:1.0113566306e-6  L3:-3.22175889312e-6  L4:4.69688297992e-6  L5:-4.03868850741e-6  L6:2.1952586469e-6
max |Re T_n| (n<=8): 1.02e-15
tail |T_n| for L=2 (n=10..19): ['1.78e-8', '2.16e-9', '2.47e-10', '2.68e-11', '2.77e-12', '2.72e-13', '2.55e-14', '2.28e-15', '1.96e-16', '7.03e-17', '3.3e-17', '1.12e-15', '4.44e
       ↪ -16', '1.51e-14', '3.52e-15', '1.2e-13', '1.25e-14', '4.27e-13']
```

12. `conservative_test/fit_contact2.py`

$\eta = +1$ is the pure exponential. Order by order, the $\eta = +1$ amplitude has no contact part: its Taylor coefficients are those of the exponential e^{-2cv}.

Paper: Eq. (18), S4 Status: **PASS** Time: 3 s
Command: `cd conservative_test && python3 fit_contact2.py`

Criterion	Measured	Required	
eta = +1 residual relative to the eta = -1 contact part, n = 5...11	max 1.16e-15 (7 values)	$\leq$ 1e-14	✓

Output (`logs/conservative-test-fit-contact2-py.txt`):

```
eta=+1: zeta*sigma = 1.0;  C1 = -8.8853e-18;  L>=4 residuals: ['4.96e-18', '6.95e-18', '8.2e-18', '9.14e-18', '9.93e-18', '1.06e-17']
eta=-1: zeta*sigma = 1.0;  C1 = -8.8853e-18;  L>=4 residuals: ['4.96e-18', '6.95e-18', '8.2e-18', '9.14e-18', '9.93e-18', '1.06e-17']
n=2 eta=+1: C_n=-4.5261034e-19  maxres= 1.2e-18  q= ['6.10865785e-17', '5.324167687e-16']
n=2 eta=-1: C_n=-4.5261034e-19  maxres= 1.2e-18  q= ['6.10865785e-17', '5.324167687e-16']
n=3 eta=+1: C_n=-2.4607752e-20  maxres= 8.1e-20  q= ['2.452348e-18', '-8.571586021e-17', '-3.352429712e-16']
n=3 eta=-1: C_n=-2.4607752e-20  maxres= 8.1e-20  q= ['2.452348e-18', '-8.571586021e-17', '-3.352429712e-16']
n=4 eta=+1: C_n=-3.4693581e-21  maxres= 8.1e-21  q= ['3.521345065e-19', '-1.590844562e-17', '5.458798649e-17', '1.748877739e-16']
n=4 eta=-1: C_n=-3.4693581e-21  maxres= 8.1e-21  q= ['3.521345065e-19', '-1.590844562e-17', '5.458798649e-17', '1.748877739e-16']
n=5 eta=+1: C_n=-2.6971642e-22  maxres= 4.8e-22  q= ['2.659381466e-20', '-1.095899618e-18', '1.639745146e-17', '-3.150785365e-17', '-6.843182634e-17']
n=5 eta=-1: C_n=-2.6971642e-22  maxres= 4.8e-22  q= ['8.533333333', '-1.095899618e-18', '1.639745146e-17', '-3.150785365e-17', '-6.843182634e-17']
n=6 eta=+1: C_n=-2.8786327e-23  maxres= 2.5e-23  q= ['2.885210965e-21', '-1.358846475e-19', '2.332819739e-18', '-9.748495808e-18', '1.255604766e-17', '2.301801585e-17']
n=6 eta=-1: C_n=-2.8786327e-23  maxres= 2.5e-23  q= ['22.75555556', '-11.37777778', '2.332819794e-18', '-9.748495924e-18', '1.255604778e-17', '2.30180158e-17']
n=7 eta=+1: C_n=-2.6563316e-24  maxres=     0.0  q= ['2.686738741e-22', '-1.321942237e-20', '2.118307903e-19', '-1.891024897e-18', '4.865097994e-18', '-4.756850344e-18',
       ↪ '-6.401962818e-18']
n=7 eta=-1: C_n=-2.6563315e-24  maxres=     0.0  q= ['32.50793651', '-32.50793651', '8.93968254', '-1.891024926e-18', '4.865098037e-18', '-4.756850379e-18', '-6.401962806e-18']
n=8 eta=+1: C_n=-2.8279984e-24  maxres=     0.0  q= ['2.753320064e-22', '-1.278295327e-20', '1.848420801e-19', '-1.241752464e-18', '4.218654712e-18', '-7.264321039e-18',
       ↪ '6.099766324e-18']
n=8 eta=-1: C_n=-2.8280598e-24  maxres=     0.0  q= ['32.50793651', '-48.76190476', '26.00634921', '-4.876190476', '4.218569072e-18', '-7.264253788e-18', '6.099744788e-18']
```

```
eta=+1 direct test: max_L |Im T_L^(n) - E_L^(n)/4pi - C_n Y0(L)| relative to max_L |eta=-1 contact part|
 n= 5: eta=+1 residual   1.91e-19   eta=-1 contact size       1.02   ratio 1.87e-19
 n= 6: eta=+1 residual   4.11e-20   eta=-1 contact size       2.48   ratio 1.66e-20
 n= 7: eta=+1 residual   8.34e-21   eta=-1 contact size       3.26   ratio 2.56e-21
 n= 8: eta=+1 residual    1.5e-21   eta=-1 contact size       3.01   ratio 4.97e-22
 n= 9: eta=+1 residual   2.34e-22   eta=-1 contact size       2.17   ratio 1.08e-22
 n=10: eta=+1 residual   3.59e-23   eta=-1 contact size       1.29   ratio 2.78e-23
 n=11: eta=+1 residual   7.64e-16   eta=-1 contact size      0.658   ratio 1.16e-15
```

13. conservative_test/closedform_check.py

Closed form (18) against MST through $O(a^{13})$. Partial-wave sums of the closed form against the MST Taylor coefficients, order by order from a^5 to a^{13}, both η.

Paper: Eq. (18), Eq. (S12) Status: **PASS** Time: 5 s
Command: `cd conservative_test && python3 closedform_check.py`

Criterion	Measured	Required	
eta = -1: relative deviation of MST from the closed form (S12), O(a^5)...O(a^10)	max 2.98e-16 (6 values)	≤ 5e-16	✓
eta = -1: relative deviation, O(a^11)...O(a^13)	max 1.02e-15 (3 values)	≤ 2e-15	✓
eta = +1 (pure exponential exp(-2cv)): relative deviation, O(a^5)...O(a^10)	max 3.18e-16 (6 values)	≤ 5e-16	✓
eta = +1: absolute deviation, O(a^5)...O(a^13)	max 7.64e-16 (9 values)	≤ 1e-14	✓

Output (`logs/conservative-test-closedform-check-py.txt`):

```
n=5: 2*qcons = [128/15]
n=6: 2*qcons = [1024/45, -512/45]
n=7: 2*qcons = [2048/63, -2048/63, 2816/315]
n=8: 2*qcons = [2048/63, -1024/21, 8192/315, -512/105]

n= 5 eta=+1  L<=9:  max deviation  1.91e-19  (data scale 0.000632, rel 3.02e-16)
n= 5 eta=-1  L<=9:  max deviation  3.21e-16  (data scale 1.08, rel 2.98e-16)
n= 6 eta=+1  L<=9:  max deviation  4.11e-20  (data scale 0.000136, rel 3.03e-16)
n= 6 eta=-1  L<=9:  max deviation   7.8e-16  (data scale 2.63, rel 2.96e-16)
n= 7 eta=+1  L<=9:  max deviation  8.34e-21  (data scale 2.68e-5, rel 3.11e-16)
n= 7 eta=-1  L<=9:  max deviation  1.02e-15  (data scale 3.47, rel 2.95e-16)
n= 8 eta=+1  L<=9:  max deviation   1.5e-21  (data scale 4.7e-6, rel 3.18e-16)
n= 8 eta=-1  L<=9:  max deviation  9.45e-16  (data scale 3.22, rel 2.93e-16)
n= 9 eta=+1  L<=9:  max deviation  2.34e-22  (data scale 7.42e-7, rel 3.15e-16)
n= 9 eta=-1  L<=9:  max deviation  6.81e-16  (data scale 2.33, rel 2.92e-16)
n=10 eta=+1  L<=9:  max deviation  3.59e-23  (data scale 1.14e-7, rel 3.15e-16)
n=10 eta=-1  L<=9:  max deviation  4.06e-16  (data scale 1.4, rel 2.9e-16)
n=11 eta=+1  L<=9:  max deviation  7.64e-16  (data scale 1.65e-8, rel 4.64e-8)
n=11 eta=-1  L<=9:  max deviation  7.29e-16  (data scale 0.716, rel 1.02e-15)
n=12 eta=+1  L<=8:  max deviation  1.65e-17  (data scale 2.18e-9, rel 7.55e-9)
n=12 eta=-1  L<=8:  max deviation  9.26e-17  (data scale 0.321, rel 2.88e-16)
n=13 eta=+1  L<=7:  max deviation  3.57e-19  (data scale 2.69e-10, rel 1.33e-9)
n=13 eta=-1  L<=7:  max deviation  3.45e-17  (data scale 0.128, rel 2.69e-16)
```

14. conservative_test/higher_orders_eq17.py

Expansion of the closed form: Eqs. (17) and (S14). SymPy: the elementary form (S13), the coefficients $b_0 \dots b_9$ of (S14), the factor $3 + \cos\Theta$ of the odd orders, and the $c^5 \dots c^8$ coefficients of f^{odd}/f_0, Eq. (17).

Paper: Eq. (17), Eqs. (S13), (S14) Status: **PASS** Time: 29 s
Command: `cd conservative_test && python3 higher_orders_eq17.py`

Criterion	Measured	Required	
elementary form (S13) equals the series	printed	"elementary form - series: 0"	✓
odd part of f^(eta) equals f^odd	printed	"odd part of f^(eta) - fodd: 0"	✓
odd orders b_1, b_3, ..., b_9 have the factor 3 + cos(Theta)	printed	"odd b_k at x=-3: [0, 0, 0, 0, 0]"	✓
b_3 of Eq. (S14)	printed	"b_3(x) = (x + 3)*(3*x**2 + 14*x + 23)/42"	✓

Criterion	Measured	Required	
[c^5] of Eq. (17)	printed	"[c^5] f^odd/f0 = -64*v/15"	✓
[c^6] of Eq. (17)	printed	"[c^6] f^odd/f0 = 256*v*(v - 2)/45"	✓
[c^7] of Eq. (17)	printed	"[c^7] f^odd/f0 = -128*v*(11*v**2 - 40*v + 40)/315"	✓
[c^8] of Eq. (17)	printed	"[c^8] f^odd/f0 = 256*v*(v - 2)*(3*v**2 - 10*v + 10)/315"	✓

Output (`logs/conservative-test-higher-orders-eq17-py.txt`):

```
b_0(x) = 1    = 1
b_1(x) = 2*(x + 3)/3    = 2*x/3 + 2
b_2(x) = (11*x**2 + 58*x + 91)/42    = 11*x**2/42 + 29*x/21 + 13/6
b_3(x) = (x + 3)*(3*x**2 + 14*x + 23)/42    = x**3/14 + 23*x**2/42 + 65*x/42 + 23/14
b_4(x) = (23*x**4 + 228*x**3 + 938*x**2 + 1828*x + 1463)/1512    = 23*x**4/1512 + 19*x**3/126 + 67*x**2/108 + 457*x/378 + 209/216
b_5(x) = (x + 3)*(5*x**4 + 46*x**3 + 188*x**2 + 362*x + 295)/1890    = x**5/378 + 61*x**4/1890 + 163*x**3/945 + 463*x**2/945 + 1381*x/1890 + 59/126
b_6(x) = (65*x**6 + 938*x**5 + 6175*x**4 + 22860*x**3 + 49775*x**2 + 60138*x + 32081)/166320    = 13*x**6/33264 + 67*x**5/11880 + 1235*x**4/33264 + 127*x**3/924 + 905*x**2/3024 +
        ↪ 3341*x/9240 + 4583/23760
b_7(x) = (x + 3)*(25*x**6 + 342*x**5 + 2227*x**4 + 8164*x**3 + 17743*x**2 + 21414*x + 11525)/498960    = 5*x**7/99792 + 139*x**6/166320 + 3253*x**5/498960 + 2969*x**4/99792 +
        ↪ 8447*x**3/99792 + 24881*x**2/166320 + 75767*x/498960 + 2305/33264
b_8(x) = (295*x**8 + 5576*x**7 + 50308*x**6 + 272376*x**5 + 956010*x**4 + 2212216*x**3 + 3291140*x**2 + 2878792*x + 1146727)/51891840    = 59*x**8/10378368 + 697*x**7/6486480 +
        ↪ 12577*x**6/12972960 + 97*x**5/18480 + 2897*x**4/157248 + 276527*x**3/6486480 + 164557*x**2/2594592 + 51407*x/926640 + 1146727/51891840
b_9(x) = (x + 3)*(105*x**8 + 1904*x**7 + 16996*x**6 + 91264*x**5 + 319134*x**4 + 736624*x**3 + 1095540*x**2 + 959008*x + 383905)/181621440    = x**9/1729728 + 317*x**8/25945920 +
        ↪ 811*x**7/6486480 + 3233*x**6/4127760 + 98821*x**5/30270240 + 847013*x**4/90810720 + 8347*x**3/458640 + 1061407*x**2/45405360 + 465847*x/25945920 + 76781/12108096
odd b_k at x=-3: [0, 0, 0, 0, 0]
elementary form - series: 0
odd part of f^(eta) - fodd: 0
[c^5] f^odd/f0 = -64*v/15
[c^6] f^odd/f0 = 256*v*(v - 2)/45
[c^7] f^odd/f0 = -128*v*(11*v**2 - 40*v + 40)/315
[c^8] f^odd/f0 = 256*v*(v - 2)*(3*v**2 - 10*v + 10)/315
```

15. `a7_exact.py`

Exact $O(c^2)$ ingredients of Eq. (17). SymPy: R_{2m} from the limiting MST recurrence, the spin-weighted spheroidal harmonic S_{22} to $O(c^2)$ and the $\ell = 3$ term, assembled into the $O(c)$ and $O(c^2)$ brackets of Eq. (17).

Paper: Eq. (17), Eq. (S7) Status: **PASS** Time: 4 s
Command: `python3 a7_exact.py`

Criterion	Measured	Required	
R_2m through O(c^2) from the limiting MST recurrence, Eq. (S7)	printed	"check vs 1 + (10m/9)c + 5(475m^2+261)c^2/3969 : 0"	✓
normalization of the spin-weighted basis	printed	"norm check: [1, 1, 1, 1, 1]"	✓
O(c) bracket of Eq. (17), b_1 = 2(x + 3)/3	printed	"O(c): 2*(x + 3)/3"	✓
O(c^2) bracket of Eq. (17), b_2	printed	"O(c^2): 11*x**2/42 + 29*x/21 + 13/6"	✓

Output (`logs/a7-exact-py.txt`):

```
R(c,m) = c**2*(2375*m**2/3969 + 145/441) + 10*c*m/9 + 1
   check vs 1 + (10m/9)c + 5(475m^2+261)c^2/3969 : 0
norm check: [1, 1, 1, 1, 1]
S22(0;c)S22(Theta;c)/(Y22 Y22) = c**2*(11*x**2/42 - 2*x/21 - 173/2646) + c*(2*x/3 - 2/9) + 1

polar bracket  [1 + c(...) + c^2(...)] with x = cos(Theta):
  O(c):   2*(x + 3)/3
  O(c^2): 11*x**2/42 + 29*x/21 + 13/6
```

16. `prl_checks/a7_polar_check.py`

Closed form against Eq. (17). SymPy: the exchange-odd part of the closed form (S12) has the c^5, c^6, c^7 coefficients of Eq. (17).

Paper: Eq. (17) Status: **PASS** Time: 1 s
Command: `cd prl_checks && python3 a7_polar_check.py`

Criterion	Measured	Required	
c^5, c^6, c^7: closed form expanded minus prediction of Eq. (17) = 0	3 line(s)	3 line(s)	✓

Output (`logs/prl-checks-a7-polar-check-py.txt`):

```
c^5:  MST closed form: -64*v/15    prediction: -64*v/15    difference: 0
c^6:  MST closed form: 256*v*(v - 2)/45    prediction: 256*v*(v - 2)/45    difference: 0
c^7:  MST closed form: -128*v*(11*v**2 - 40*v + 40)/315    prediction: -128*v*(11*v**2 - 40*v + 40)/315    difference: 0
```

Segment representation

Paper: S4, Eqs. (S15)-(S17)

In the classical limit the horizons sit at $r = \pm ia$, and the exchange-odd part of each partial-wave element is a finite-part integral along the segment joining them plus an endpoint term, Eq. (S16).

17. `segment_test/segment_born.py`

Segment formula (S16) against MST. Internal checks of the construction (radial equation, double pole, residue, $S = 1$ at $M = 0$, endpoint identity) and the $\ell = 2$ element at $c = 0.05$, 0.1, 0.2 against MST at $\epsilon = 10^{-12}$.

Paper: Eq. (S16) Status: **PASS** Time: 40 s
Command: `cd segment_test && python3 segment_born.py`

Criterion	Measured	Required	
residual of the M = 0 radial equation	max 3.71e-40 (3 values)	≤ 1e-30	✓
double pole vs boundary Wronskian: |C2/(A^2 i/2a) - 1|	max 2.76e-42 (3 values)	≤ 1e-30	✓
residue of the simple pole, |C1|	max 1.55e-41 (3 values)	≤ 1e-30	✓
|S(M = 0) - 1|	max 2.63e-34 (3 values)	≤ 1e-25	✓
endpoint identity	max 0 (3 values)	≤ 1e-25	✓
|segment formula (S16) / MST - 1|, l = 2, c = 0.1, 0.2, 0.05 (MST printed to 12 digits)	max 6.81e-11 (3 values)	≤ 1e-9	✓

Output (`logs/segment-test-segment-born-py.txt`):

```
l=2 c=0.1: ODE residual 3.15e-40; C2/(A^2 i/2a) = (1.0 + 2.75837838613e-42j); C1 = (1.32e-41 + 2.18e-42j); S0(M=0) = (1.0 - 5.68443782913653e-35j); endpoint identity 0.0
   S_odd/eps (segment formula) = (-6.07309597459634e-40 - 1.06837526731227e-5j)
   full MST (eps=1e-12):          (0.0 - 1.0683752673e-5j)    ratio-1 = (1.15e-11 - 5.68e-35j)
l=2 c=0.2: ODE residual 3.71e-40; C2/(A^2 i/2a) = (1.0 + 1.81575912591e-42j); C1 = (1.48e-41 + 4.68e-42j); S0(M=0) = (1.0 + 1.335363200234e-34j); endpoint identity 0.0
   S_odd/eps (segment formula) = (5.74444436039146e-38 - 0.0004301795926486 11j)
   full MST (eps=1e-12):          (0.0 - 0.000430179592639j)    ratio-1 = (2.23e-11 + 1.34e-34j)
l=2 c=0.05: ODE residual 2.02e-40; C2/(A^2 i/2a) = (1.0 - 3.38417785449e-43j); C1 = (-2.25e-42 - 4.15e-42j); S0(M=0) = (1.0 + 2.62602472450386e-34j); endpoint identity 0.0
   S_odd/eps (segment formula) = (7.83061592870768e-41 - 2.98193418117307e-7j)
   full MST (eps=1e-12):          (0.0 - 2.98193418097e-7j)    ratio-1 = (6.81e-11 + 2.63e-34j)
```

18. `segment_test/vs_production.py`

Segment formula for $\ell = 2 \ldots 6$. $\ell = 2 \ldots 6$ at $c = 0.1997$ and 0.1631 against the 110-digit MST data; the agreement is the finite-ϵ systematic of the data.

Paper: Eq. (S16), S4 Status: **PASS** Time: 2 min 31 s
Command: `cd segment_test && python3 vs_production.py`

Criterion	Measured	Required	
|segment (S16) / MST - 1|, l = 2...6 at c = 0.1997, 0.1631 (110-digit data)	max 4.44e-16 (10 values)	≤ 1e-15	✓

Output (`logs/segment-test-vs-production-py.txt`):

```
l=2 c=0.19968536: MST (-9.425634030121e-21 - 0.0004264947435263j)   segment (-8.458282690821e-38 - 0.0004264947435263j)   |rel diff| 3.15e-16
l=2 c=0.16311217: MST (-2.687409554186e-21 - 0.0001425199951056j)   segment (9.495664608177e-38 - 0.0001425199951056j)   |rel diff| 3.15e-16
l=3 c=0.19968536: MST (2.415956831526e-24 + 1.593897791675e-8j)   segment (3.010033674323e-43 + 1.593897791675e-8j)   |rel diff| 3.49e-16
l=3 c=0.16311217: MST (5.635563061904e-25 + 3.755677028455e-9j)   segment (-1.00656844611e-43 + 3.755677028455e-9j)   |rel diff| 3.48e-16
l=4 c=0.19968536: MST (-2.211215265577e-28 - 9.916537194378e-13j)   segment (1.410251942794e-47 - 9.916537194378e-13j)   |rel diff| 3.85e-16
l=4 c=0.16311217: MST (-3.519352451977e-29 - 1.584548087768e-13j)   segment (-2.958179291017e-48 - 1.584548087768e-13j)   |rel diff| 3.85e-16
l=5 c=0.19968536: MST (1.582033136584e-32 + 5.782464556057e-17j)   segment (-6.748010621646e-53 + 5.782464556057e-17j)   |rel diff| 4.17e-16
l=5 c=0.16311217: MST (1.693587980474e-33 + 6.203163103111e-18j)   segment (8.781405649309e-53 + 6.203163103111e-18j)   |rel diff| 4.16e-16
l=6 c=0.19968536: MST (-8.79149828506e-37 - 2.807055748335e-21j)   segment (2.070901151209e-56 - 2.807055748335e-21j)   |rel diff| 4.44e-16
l=6 c=0.16311217: MST (-6.302575681962e-38 - 2.014942080859e-22j)   segment (-9.310761238103e-59 - 2.014942080859e-22j)   |rel diff| 4.43e-16
```

19. `segment_test/nonlinear_check.py`

Terms nonlinear in $\mathcal{K}$. Size of the terms nonlinear in $\mathcal{K}$ in $S/F - 1 = iE\mathcal{K}/(1 + \beta\mathcal{K})$: they change the $\ell = 2$ exchange-odd element at relative order c^3 (about $-0.85c^3$ at $c = 0.025$).

Paper: S4 Status: **PASS** Time: 10 s
Command: `cd segment_test && python3 nonlinear_check.py`

Criterion	Measured	Required	
relative change of the l = 2 exchange-odd element from the terms nonlinear in K, divided by c^3	-0.987 ... -0.853 (4 values)	in [-1.1, -0.7]	✓
same at c = 0.025 vs -0.85	-0.85271044 (diff 0.0027)	diff ≤ 0.01	✓
beta K/c^3 for l = 2	0.427 ... 0.556 (4 values)	in [0.3, 0.7]	✓
|S_far(+)/S_far(-) - 1|	max 2.16e-26 (4 values)	≤ 1e-20	✓

Output (`logs/segment-test-nonlinear-check-py.txt`):

```
m=2 c=0.025: x=(-0.50129014 + 7.6691601e-33j)  |Sfar+/Sfar- - 1|=5.01e-31
   beta K(+1) = (6.6706551e-6 + 6.6878673e-18j)   beta K(-1) = (6.6530786e-6 + 6.6702455e-18j)
   beta K(+1)/c^3 = (0.42692193 + 4.2802351e-13j)   (K odd part)/c^5 = (-1.0743189e-21 - 0.89991496j)
   H_odd exact/linear - 1 = (-1.3323601e-5 - 1.3357846e-17j)   ( /c^3 = (-0.85271044 - 8.5490213e-13j) )
   S_odd/eps = (-4.874124832e-19 - 8.81079077887e-9j)
m=2 c=0.05: x=(-0.49328091 + 2.3714122e-31j)  |Sfar+/Sfar- - 1|=4.19e-29
   beta K(+1) = (5.5455379e-5 + 5.471016e-17j)   beta K(-1) = (5.4850802e-5 + 5.4113707e-17j)
   beta K(+1)/c^3 = (0.44364303 + 4.3768128e-13j)   (K odd part)/c^5 = (-2.734446e-22 - 0.96732326j)
   H_odd exact/linear - 1 = (-0.00011029706 - 1.0880586e-16j)   ( /c^3 = (-0.88237645 - 8.704469e-13j) )
   S_odd/eps = (-1.65033404686e-17 - 2.98193418097e-7j)
m=2 c=0.1: x=(-0.4781021 + 2.1860073e-30j)  |Sfar+/Sfar- - 1|=2.78e-28
   beta K(+1) = (0.00047901431 + 4.580355e-16j)   beta K(-1) = (0.00045664723 + 4.3664799e-16j)
   beta K(+1)/c^3 = (0.47901431 + 4.580355e-13j)   (K odd part)/c^5 = (-7.0312726e-23 - 1.1183544j)
   H_odd exact/linear - 1 = (-0.00093500523 - 8.9342875e-16j)   ( /c^3 = (-0.93500523 - 8.9342875e-13j) )
   S_odd/eps = (-5.91797634452e-16 - 1.0683752673e-5j)
m=2 c=0.2: x=(-0.45250713 - 1.4375773e-26j)  |Sfar+/Sfar- - 1|=2.16e-26
   beta K(+1) = (0.0044495818 + 4.026935e-15j)   beta K(-1) = (0.0034913598 + 3.1597304e-15j)
   beta K(+1)/c^3 = (0.55619773 + 5.0336687e-13j)   (K odd part)/c^5 = (-1.7199533e-23 - 1.4972219j)
   H_odd exact/linear - 1 = (-0.0078936709 - 7.1013316e-15j)   ( /c^3 = (-0.98670886 - 8.8766645e-13j) )
   S_odd/eps = (-2.38677912496e-14 - 0.000430179592639j)
```

Physical-region double expansion in (ϵ, q)

Paper: S5; data behind Eqs. (10)-(12), (16)

Two-dimensional Cauchy integrals of the $\ell = 2$, $m = 2$ S-matrix element in ϵ (two radii) and q, full MST at 90 digits, for the physical branch and its image. The data files are produced by the production drivers (option `--production`).

20. physical_region_test/rich_phys.py

Richardson-improved coefficients. Richardson extrapolation between the two radii; Page's 4/225 at $q = 0$; the exchange-odd q^6 coefficient equals $\rho_2/64$ of arXiv:2610.40101. Writes `l2_rich.json`.

Paper: Eqs. (10), (11); S5 Status: **PASS** Time: 7 s
Command: `cd physical_region_test && python3 rich_phys.py`

Criterion	Measured	Required	
q = 0, order epsˆ6: Gamma_+ - Gamma_- vs Page 4/225	0.0177777777778 (rel. diff 4.7e-15)	rel. diff ≤ 1e-10	✓
Gamma_+ vs 4/225 + 1/4096 (helicity-preserving normalization)	0.0180219184025 (rel. diff 1.6e-11)	rel. diff ≤ 1e-10	✓
exchange-odd qˆ6 coefficient vs rho_2/64 of arXiv:2610.40101	0.0062138787452 (rel. diff 1.7e-15)	rel. diff ≤ 1e-13	✓

Output (`logs/physical-region-test-rich-phys-py.txt`):

```
Richardson change |R - A(1e-5)| for k=6, p=0..8: ['3.0e-8', '0.0041', '3.2e-7', '0.012', '2.0e-8', '6.3e-13', '3.3e-13', '9.8e-14', '4.3e-15']
sign +1: Gamma(q=0) at epsˆ6 = 0.0180219184024891   (Page/Teukolsky-Press l=2: 4/225 = 0.0177777777777778)
sign -1: Gamma(q=0) at epsˆ6 = 0.000244140624711405   (Page/Teukolsky-Press l=2: 4/225 = 0.0177777777777778)

duality-odd (A+ - A-)/2 at epsˆ6 [Re, Im]:
  p= 0:     -0.0044444444444444304   -7.4e-20
  p= 1:      0.0279252680319091903   4.65e-19
  p= 2:     -0.0695976557101589719  -1.16e-18
  p= 3:      0.0837758040957276184   1.39e-18
  p= 4:     -0.0435502684133062247  -7.25e-19
  p= 5:    1.55985295216226329e-17   2.81e-22
  p= 6:      0.0062138787452044930   1.03e-19
  p= 7:   -4.17567516051808968e-18  -4.17e-23
  p= 8:  -0.00008066157949824674 94  -1.34e-21
  p= 9:    -3.4172534571792078e-19    2.2e-25
  p=10:   -0.00017271887192451265 4  -2.87e-21

duality-even (A+ + A-)/2 at epsˆ6, Im part:
  p= 0: Im       0.435717946370007866   Re      -0.756115730148074593
  p= 1: Im      -1.31839840180317889    Re        1.061170374508859 69
  p= 2: Im       0.644783865980986218   Re      -0.416229004446324187
  p= 3: Im      -0.712046524664935216   Re       0.052470368559632306 2
  p= 4: Im      0.00173310632712651618   Re    -0.004105960044491811 74
  p= 5: Im       0.0158354996606443293   Re   -6.24597884146695022e-19
  p= 6: Im    -9.05977625037558874e-18   Re    7.89617923725391997e-23
  p= 7: Im     -0.00184218444692784223   Re    3.06571870164597693e-20
  p= 8: Im     9.40382909139296344e-19   Re   -1.21113277861969168e-23
  p= 9: Im    -0.00031488602507452767 7   Re    5.24025748769961453e-21
  p=10: Im     4.04217234217730949e-19   Re   -2.26751130652395476e-24
```

21. physical_region_test/pslq2.py

PSLQ identification (exploratory). Exploratory PSLQ fits of individual coefficients in a basis of zeta values, superseded by the closed forms of the paper (digamma function). Informational: no pass/fail criterion.

Paper: S5 Status: **INFO** Time: <1 s
Command: `cd physical_region_test && python3 pslq2.py`

informational, no pass/fail criterion

Output (`logs/physical-region-test-pslq2-py.txt`):

```
duality-odd (absorption) at epsˆ6 qˆp:
  p=0 = (-1/225)1    [residual 1.4e-17]
  p=1 = (-4847308/8129131)1 + (1615205/8129131)pi    [residual -8.0e-23]
  p=2 = (2181/22007)1 + (22222/198063)pi + (-10459/198063)piˆ2    [residual -3.1e-20]
  p=3 = (2/75)pi    [residual -2.0e-16]
  p=4 = (-19093/41633)1 + (47832/41633)piˆ2 + (-4669/41633)piˆ4    [residual -1.3e-18]
  p=6 = (-1561/20)1 + (-411/20)z2 + (-117/10)z4 + (612/5)z6    [residual -9.8e-17]
duality-even, imaginary (phase) at epsˆ6 qˆp:
  p=5 = (8756/19037)1 + (-2141/19037)z3 + (-11343/38074)z5    [residual -1.0e-20]
  p=5 (test with piˆ2 added) = (-2731/2076)1 + (1337/2076)z3 + (-316/519)z5 + (125/1038)piˆ2    [residual 1.4e-19]
  p=7 = (-237/11489)1 + (10395/22978)z3 + (-34501/22978)z5 + (23515/22978)z7    [residual -9.0e-21]

l=2 partial wave alone, BBCPSS units (-160 i A_65): 2.533679945703093   minus (2048/45)(z3-z5): -4.981531057346407
```

22. physical_region_test/analyze_phys.py

Coefficients of the polar projection. Coefficients of the $L = 2$ projection compared with arXiv:2610.40101 and with the super-extremal data. Needs `phys_run.json` (about 5 MB, not shipped), produced by `run_phys.py`. Informational.

Paper: S5 Status: **SKIP** Time: –
Command: `cd physical_region_test && python3 analyze_phys.py`

needs phys_run.json, produced by the production drivers (–production)

23. prl_checks/b6_precision.py

The logarithm $B_6 = i\Pi_m/450$. Coefficient of $\epsilon^6 \ln \epsilon$ for every m: the naive two-radius fit reaches 10^{-5}–10^{-4}, the joint log bookkeeping 10^{-10}.

Paper: S5, S9(c) Status: **PASS** Time: 1 s
Command: `cd prl_checks && python3 b6_precision.py`

Criterion	Measured	Required	
B6 vs i Pi_m/450 with joint log bookkeeping, m = 2...-2, p $\leq$ 9 (max rel. deviation)	max 1.02e-10 (5 values)	$\leq$ 5e-10	✓
same with the naive two-radius fit (for comparison)	4.58e-5 ... 0.000259 (5 values)	in [1e-6, 0.001]	✓

Output (`logs/prl-checks-b6-precision-py.txt`):

```
m=+2: max relative deviation of B6 from i Pi_m/450 (p <= 9, both branches): naive two-radius 4.58e-5   joint bookkeeping 1.02e-10
m=+1: max relative deviation of B6 from i Pi_m/450 (p <= 9, both branches): naive two-radius 4.58e-5   joint bookkeeping 1.02e-10
m=+0: max relative deviation of B6 from i Pi_m/450 (p <= 9, both branches): naive two-radius 0.000259   joint bookkeeping 1.03e-12
m=-1: max relative deviation of B6 from i Pi_m/450 (p <= 9, both branches): naive two-radius 4.58e-5   joint bookkeeping 1.02e-10
m=-2: max relative deviation of B6 from i Pi_m/450 (p <= 9, both branches): naive two-radius 4.58e-5   joint bookkeeping 1.02e-10
```

Exchange-odd ϵ^6 coefficient

Paper: Eq. (10); S9(d), S9(f)

The exchange-odd ϵ^6 coefficient D_6, Eq. (10), its Boltzmann form (S22), and its continuation to super-extremal spin, where it gives the classical a^5 coefficients.

24. `continuation_test/d6_closed.py`

Eq. (10) against the physical-region data. Taylor coefficients $q^0 \dots q^{10}$ of Eq. (10) for $m = 2$ against the exchange-odd MST coefficients; $[q^6] = \rho_2/64$.

Paper: Eq. (10), S9(d) Status: **PASS** Time: <1 s
Command: `cd continuation_test && python3 d6_closed.py`

Criterion	Measured	Required	
Eq. (10) - exchange-odd MST Taylor coefficient, q^0...q^10	max 2.01e-16 (11 values)	≤ 1e-15	✓
q^6 coefficient vs rho_2/64 of arXiv:2610.40101	0.0062138787452 (rel. diff 0)	rel. diff ≤ 1e-15	✓

Output (`logs/continuation-test-d6-closed-py.txt`):

```
 p   closed form                 full-MST duality-odd part        difference
 0     -0.0044444444444444444      -0.0044444444444444304     -1.4e-17
 1       0.027925268031909273        0.027925268031909190      8.3e-17
 2      -0.069597655710159162       -0.069597655710158972     -1.9e-16
 3       0.083775804095727819        0.083775804095727618      2.01e-16
 4      -0.043550268413306303       -0.043550268413306225     -7.79e-17
 5    -3.1321461446392917e-52     1.5598529521622633e-17     -1.56e-17
 6       0.006213787452045029        0.006213787452044931      9.82e-18
 7     7.1496455994966233e-50     -4.175675160518089e-18      4.18e-18
 8   -0.00008066157949824687   -0.00008066157949824675     -1.28e-19
 9     1.2962908843947149e-47     -3.417253457179208e-19      3.42e-19
10    -0.00017271887192451293    -0.00017271887192451265     -2.73e-19

q^6 coefficient vs BBCPSS rho_2/64: 0.0062138787452045029 0.0062138787452045029
```

25. `continuation_test/d6_m.py`

Eq. (10) for $m \neq 0$. D_6 from MST (Cauchy integral in ϵ) against Eq. (10) for $m = 2, 1, -1, -2$ at $q = 0.3$ and 0.7; the lower orders vanish. Uses two worker processes.

Paper: Eq. (10), S9(d) Status: **PASS** Time: 1 min 50 s
Command: `cd continuation_test && python3 d6_m.py`

Criterion	Measured	Required	
D6 from MST vs Eq. (10), m = 2, 1, -1, -2 at q = 0.3, 0.7 (rel. diff)	max 4.8e-15 (8 values)	≤ 1e-12	✓
lower orders |D_k|, k < 6	max 3.5e-17 (8 values)	≤ 1e-15	✓

Output (`logs/continuation-test-d6-m-py.txt`):

```
m=+2 q=0.3: D6 MST = (-0.00041693994406214 - 6.62206623466475e-21j)   conjecture =  -0.00041693994062141   rel.diff 3.4e-15   (max |D_k<6| 1.5e-19)
m=+2 q=0.7: D6 MST = (-4.31915691793035e-7 - 6.85849535185242e-24j)   conjecture =   -4.31915691793037e-7   rel.diff 4.8e-15   (max |D_k<6| 1.6e-22)
m=+1 q=0.3: D6 MST = (-0.00125725400567449 - 1.99696307052969e-20j)   conjecture =   -0.00125725400567449   rel.diff 3.3e-15   (max |D_k<6| 4.5e-19)
m=+1 q=0.7: D6 MST = (-2.62037065554497e-5 - 4.16157677176242e-22j)   conjecture =   -2.62037065554499e-5   rel.diff 4.2e-15   (max |D_k<6| 9.4e-21)
m=-1 q=0.3: D6 MST = (-0.00906934773760109 - 1.44071204156554e-19j)   conjecture =   -0.00906934773760111   rel.diff 2.9e-15   (max |D_k<6| 3.3e-18)
m=-1 q=0.7: D6 MST = (-0.0123901161276832 - 1.9683652296678e-19j)   conjecture =    -0.0123901161276833   rel.diff 3.0e-15   (max |D_k<6| 4.5e-18)
m=-2 q=0.3: D6 MST = (-0.0216959941843769 - 3.44673324340398e-19j)   conjecture =    -0.0216959941843777   rel.diff 2.7e-15   (max |D_k<6| 7.8e-18)
m=-2 q=0.7: D6 MST = (-0.0965660087700338 - 1.53433391347868e-18j)   conjecture =    -0.0965660087700341   rel.diff 2.5e-15   (max |D_k<6| 3.5e-17)
```

26. `continuation_test/d6_m0.py`

Eq. (10) for $m = 0$. For $m = 0$: $D_6 = -\kappa^5/225$, and D_4, D_5 vanish.

Paper: Eq. (10), S9(d) Status: **PASS** Time: 53 s
Command: `cd continuation_test && python3 d6_m0.py`

Criterion	Measured	Required	
m = 0: D6 vs -kappa^5/225, q = 0.3, 0.7	max rel. diff 1.59e-17 (2 values)	rel. diff ≤ 1e-10	✓
|D4|, |D5|	max 7.31e-23 (4 values)	≤ 1e-20	✓

Output (`logs/continuation-test-d6-m0-py.txt`):

```
m=0 q=0.3: D4=(5.79835216322e-28 - 2.98104149246e-30j)  D5=(-7.30520805593e-23 + 3.75575089597e-25j)  D6=(-0.00351092023419 - 5.57692793054e-20j)  D7=(-0.0225923384461 +
       ↪ 0.0001161546983l8j)   | -kappa^5/225 = -0.00351092023419
m=0 q=0.7: D4=(1.59770083066e-28 - 7.00933817709e-31j)  D5=(-2.01290658606e-23 + 8.83093089375e-26j)  D6=(-0.00082554912634 - 1.31131090136e-20j)  D7=(-0.00622522017869 +
       ↪ 2.73123293383e-5j)   | -kappa^5/225 = -0.00082554912634
```

27. `continuation_test/d6_general.py`

$[q^6]D_6 = \rho_{|m|}/64$ **and the large-spin limit.** The q^6 coefficients of the coth part equal $\rho_{|m|}/64$ of arXiv:2610.40101 for all m, and $32D_6/q^5$ continued along the upper path tends to the classical a^5 coefficient $(-1)^m \frac{8}{225} i(2+m)!(2-m)!$.

Paper: Eq. (10), S9(d), S9(f) Status: **PASS** Time: <1 s
Command: `cd continuation_test && python3 d6_general.py`

Criterion	Measured	Required	
q^6 Taylor coefficient of the coth part vs rho_|m|/64, m = 2...-2	max rel. diff 0 (5 values)	rel. diff ≤ 1e-15	✓
32 D6/q^5 at q = 1e4 vs (-1)^m (8/225) i (2+m)!(2-m)!	max rel. diff 3.17e-8 (5 values)	rel. diff ≤ 1e-7	✓

Output (`logs/continuation-test-d6-general-py.txt`):

```
q^6 Taylor coefficient of the coth-part vs BBCPSS rho_m/64:
  m=+2:   0.006213878745204502878 9   rho_|m|/64 = 0.006213878745204502878 9
  m=+1:  -0.001966450104009737742 9   rho_|m|/64 = -0.001966450104009737742 9
  m=+0:   0.001388888888888888888 9   rho_|m|/64 = 0.001388888888888888888 9
  m=-1:  -0.001966450104009737742 9   rho_|m|/64 = -0.001966450104009737742 9
  m=-2:   0.006213878745204502878 9   rho_|m|/64 = 0.006213878745204502878 9

large-spin limit 32*D6/q^5 (upper path) vs closed-form eta-odd a^5 coefficient (-1)^m (8/225) i (2+m)!(2-m)!:
  m=+2: q=1e2,1e3,1e4 -> ['(0.0002680915092 + 0.853297747j)', '(2.680826625e-6 + 0.8533329778j)', '(2.68082574e-8 + 0.8533333298j)']   target (0.0 + 0.8533333333j)
  m=+1: q=1e2,1e3,1e4 -> ['(-3.35058536e-5 - 0.2132888891j)', '(-3.351027696e-7 - 0.2133328889j)', '(-3.351032119e-9 - 0.2133333289j)']   target (0.0 - 0.2133333333j)
  m=+0: q=1e2,1e3,1e4 -> ['(0.0 + 0.1421866693j)', '(0.0 + 0.1422218667j)', '(0.0 + 0.1422222187j)']   target (0.0 + 0.1422222222j)
  m=-1: q=1e2,1e3,1e4 -> ['(3.35058536e-5 - 0.2132888891j)', '(3.351027696e-7 - 0.2133328889j)', '(3.351032119e-9 - 0.2133333289j)']   target (0.0 - 0.2133333333j)
  m=-2: q=1e2,1e3,1e4 -> ['(-0.0002680915092 + 0.853297747j)', '(-2.680826625e-6 + 0.8533329778j)', '(-2.68082574e-8 + 0.8533333298j)']   target (0.0 + 0.8533333333j)
```

28. `prl_checks/paper_numbers.py`

Closed forms and identities of the main text. The ϵ-constant $i(2\gamma_E + 2\ln 2 - \frac{37}{12})$, u_0, Eq. (10) in its Bose and Boltzmann forms, $\pi(\tau+\epsilon) = (\omega - m\Omega_H)/2T_H$, $160[q^5]F = 16b_2$, $[q^6]D_6 = \rho_{|m|}/64$ and the large-q limits, and the pole terms pole_m.

Paper: Eqs. (10), (16), (S22), (S23) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 paper_numbers.py`

Criterion	Measured	Required	
(1) eps-constant s_1(c = 0) vs i(2 gamma_E + 2 ln 2 - 37/12)	(-1.784615268e-17 - 0.5426076424i) (diff 1.1e-16)	diff ≤ 1e-15	✓
(1) u0 vs pi + i(2 gamma_E + 2 ln 2 - 37/12 + 107/210)	(3.141592654 - 0.03308383289i) (diff 3.1e-15)	diff ≤ 1e-12	✓
(2) Eq. (10): Bose-factor form = Boltzmann form (S22)	1 line(s)	1 line(s)	✓
(3) pi(tau + eps) = (omega - m Omega_H)/(2 T_H)	-4.62049739527 (rel. diff 0)	rel. diff ≤ 1e-12	✓
(4) 160 [q^5] F = 16 b_2 of arXiv:2610.40101	2.53743322527 (rel. diff 0)	rel. diff ≤ 1e-15	✓
(4) 160 F/q^5 at q = 1e8 vs 64/15	4.26666666667 (rel. diff 3.1e-15)	rel. diff ≤ 1e-12	✓
(5) [q^6] D6 = rho_|m|/64, m = 2...-2	max rel. diff 0 (5 values)	rel. diff ≤ 1e-15	✓
(5) 32 D6/q^5 at large q vs the classical a^5 coefficient	max rel. diff 3.14e-12 (5 values)	rel. diff ≤ 1e-9	✓

Criterion	Measured	Required	
(6) 75 pole_m/i = 64, -16, 16, -64 for m = 2, 1, -1, -2	max diff 0 (4 values)	diff $\leq$ 1e-9	✓

Output (`logs/prl-checks-paper-numbers-py.txt`):

```
(1) s1(c=0) = (-1.7846152680107368789e-17 - 0.542607642410376833 21j)   i(2gE+2ln2-37/12) = (0.0 - 0.5426076424103769 9329j)
    u0 = pi + i(2gE + 2ln2 - 37/12 - nu2) = (3.14159265358979 - 0.03308383288656 75j)
(2) Pi_m = m q [1+(m^2-1)q^2][4+(m^2-4)q^2];  D6 = -(pi/900)Pi(coth(pi w)-eta) = -(k^5/900)|G(3+iw)|^2 e^{-eta pi w}: OK
(3) pi(tau+eps) = -4.62049739526719   (w - m OmegaH)/(2TH) = -4.62049739526719
(4) 160[q^5]F = 2.5374332252717219416   16 b2 = 2.5374332252717219416
    160 F/q^5 at q = 1e8 (continued): 4.26666666666668   64/15 = 4.26666666666667
(5) m=+2: [q^6]D6 =   0.006213878745204503  rho/64 =   0.006213878745204503 | 32 D6/q^5 -> (2.680825731e-12 + 0.8533333333j)  target (0.0 + 0.8533333333j)
(5) m=+1: [q^6]D6 =  -0.001966450104009738  rho/64 =  -0.001966450104009738 | 32 D6/q^5 -> (-3.351032164e-13 - 0.2133333333j)  target (0.0 - 0.2133333333j)
(5) m=+0: [q^6]D6 =   0.001388888888888889  rho/64 =   0.001388888888888889 | 32 D6/q^5 -> (0.0 + 0.1422222222j)  target (0.0 + 0.1422222222j)
(5) m=-1: [q^6]D6 =  -0.001966450104009738  rho/64 =  -0.001966450104009738 | 32 D6/q^5 -> (3.351032164e-13 - 0.2133333333j)  target (0.0 - 0.2133333333j)
(5) m=-2: [q^6]D6 =   0.006213878745204503  rho/64 =   0.006213878745204503 | 32 D6/q^5 -> (-2.680825731e-12 + 0.8533333333j)  target (0.0 + 0.8533333333j)
(6) m=+2: pole_m = (0.0 + 0.853333333333j)   x75/i = (64.0 + 0.0j)
(6) m=+1: pole_m = (0.0 - 0.213333333333j)   x75/i = (-16.0 + 0.0j)
(6) m=-1: pole_m = (0.0 + 0.213333333333j)   x75/i = (16.0 + 0.0j)
(6) m=-2: pole_m = (0.0 - 0.853333333333j)   x75/i = (-64.0 + 0.0j)
```

29. `prl_checks/se_allm_analyze.py`

Classical a^5 coefficients for every m. Super-extremal exchange-even c^5 coefficient for every m against $32ia_5^{(m)} + \text{pole}_m$ from the physical region; per-branch coefficients used in S5(d). Reads `allm_phys_result.json`.

Paper: S5(d), S9(f) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 se_allm_analyze.py`

Criterion	Measured	Required	
super-extremal exchange-even c^5 (classical MST) - [32 i a_5 + pole_m], m = 2...-2	max 1.24e-9 (5 values)	$\leq$ 2e-9	✓
m = 2, eta = +1 classical c^5 coefficient / 32 vs -2.3457997302e-5 (S5(d))	-2.34579973039e-5 (diff 1.9e-15)	diff $\leq$ 1e-14	✓

Output (`logs/prl-checks-se-allm-analyze-py.txt`):

```
 m   SE eta-even c^5 (MST, classical)        32 i a5 (phys fit) + pole_m                diff      | pole_m      | eta-odd c^5   (-1)^m (8/225) i (2+m)!(2-m)!
+2      0.852582677419608344 54i     0.852582676179962359 48886i     1.24e-9   0.8533333333331i   -0.8533333333333i  0.85333333333333
+1   -0.181282659600735858 7969i    -0.181282659208423211 4362i    -3.92e-10   -0.2133333333331i    0.2133333333333i  -0.21333333333333
+0  -9.388245966426240277 02e-77i   4.817771075981396702 98e-57i   -4.82e-57              0.0i   -0.142222222222221i  0.14222222222222
-1    0.181282659600735858 7969i     0.181282659208423211 4362i    3.92e-10   0.2133333333331i    0.2133333333333i  -0.21333333333333
-2     -0.852582677419608344 54i    -0.852582676402137114 9966i    -1.02e-9   -0.8533333333331i   -0.8533333333333i  0.85333333333333

per-branch c^5 (Im):
  m=+2: eta=+1 -0.000705065591372525 6876   eta=-1      1.70591601075294195   32 a5 = -0.000705065715337097 3845
  m=+1: eta=+1    0.032050673732597541 6   eta=-1    -0.394615992934069259   32 a5 = 0.032050674124910121 9
  m=+0: eta=+1    -0.142222222222222267   eta=-1     0.142222222222222267   32 a5 = 4.817771075981396 7e-57
  m=-1: eta=+1     0.394615992934069259   eta=-1   -0.032050673732597541 6   32 a5 = -0.032050674124910121 9
  m=-2: eta=+1     -1.70591601075294195   eta=-1  0.000705065591372525 6876   32 a5 = 0.000705065693119621 8337
```

Exchange-even ϵ^6 function and $\mathcal{C}(q)$

Paper: Eqs. (12), (16); S5; S9(c), S9(e)

The exchange-even ϵ^6 function, Eq. (12): a polynomial of degree five plus $-(i/450)P^{(m)}[\ln(\epsilon\kappa)+\mathrm{Re}\,\psi(1+i\tau_0)]$; the uniqueness of this form, and the contact coefficient $\mathcal{C}(q)$ of Eq. (16).

30. `prl_checks/allm_analyze.py`

Free fit of the transcendental coefficients. Exchange-even ϵ^6 function for every m: a free fit gives $-1/450$ for both the $\Pi_m \ln\kappa$ and the $\Pi_m \mathrm{Re}\,\psi$ coefficients; residuals of the closed form; for $m=0$ a polynomial of degree four. Writes `allm_phys_result.json`.

Paper: Eq. (12), S5 Status: **PASS** Time: 4 s
Command: `cd prl_checks && python3 allm_analyze.py`

Criterion	Measured	Required	
free fit: 450 x coefficients of Pi_m ln(kappa) and Pi_m Re psi(1 + i varpi), m = 1, -1, -2	max diff 2.71e-9 (6 values)	diff ≤ 1e-8	✓
residual of the closed form, Taylor coefficients p = 6...15, m = 1, -1, -2	max 5e-12 (30 values)	≤ 1e-11	✓
m = 0: Taylor coefficients p ≥ 5 vanish (E6 is a polynomial)	max 9.73e-17 (11 values)	≤ 1e-15	✓

Output (`logs/prl-checks-allm-analyze-py.txt`):

```
===== m = +1 =====
 log coefficient B6 (self-dual) vs -(i/450) P^(m):
   p=0: (-6.6640031406385e-14 + 1.2970993053824e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (1.03e-12 + 5.6e-19j))
   p=1: (-4.29130157323e-9 + 0.0088884814187161j)   pred (5.8593780510281e-96 + 0.0088888888888889j)   (anti-self-dual log: (-3.22e-12 - 1.76e-18j))
   p=2: (4.2397165644972e-13 + 1.5493130592802e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (2.09e-12 + 1.14e-18j))
   p=3: (3.2184761928869e-9 - 0.0066663610623523j)   pred (0.0 - 0.0066666666666667j)   (anti-self-dual log: (2.42e-12 + 1.32e-18j))
   p=4: (-3.0548373645955e-13 - 1.4051909136316e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (-2.81e-12 - 1.53e-18j))
   p=5: (-7.8671927455746e-18 + 1.0439192592153e-12j)   pred (-5.4545482947753e-94 - 2.3181830252795e-92j)   (anti-self-dual log: (1.68e-17 + 2.12e-22j))
   p=6: (1.4352449179272e-22 - 1.5882883316615e-17j)   pred (0.0 + 0.0j)   (anti-self-dual log: (4.54e-13 + 2.48e-19j))
   p=7: (3.7576483922687e-20 - 6.8839850613297e-14j)   pred (8.7272772716404e-93 - 2.9018196928204e-91j)   (anti-self-dual log: (-5.05e-18 - 3.57e-23j))
 free fit (odd p<=15): a1=-0.549682295656536 a3=0.109477843145636 a5=0.00100158355166457  450*lam_log=-0.999999997288  450*lam_psi=-0.999999997288
 fixed lam=-1/450: a1=-0.549682295642621 a3=0.109477843118276 a5=0.00100158356640344
   p   Im(self-dual)              closed-form (p>=7)        residual     Re(self-dual)
   0       0.4357194696801278                     0.0       0.436   -0.756116
   1       -0.5548131015528572       -0.5548131015528572        0.0   0.530585
   2      -0.08092639139119176                     0.0     -0.0809   0.0332273
   3       0.1195664533568157        0.1195664533568157        0.0   -0.0267898
   4      -0.01592625975617983                     0.0     -0.0159   0.0034925
   5     -0.004433203361182971     -0.004433203361182971        0.0   -9.13534e-17
   6    -1.844312483758952e-16                     0.0   -1.84e-16   1.65902e-21
   7     0.000298107933199224     0.000298107934003762   -8.08e-13   4.36336e-19
   8    -1.057047882528379e-17                     0.0   -1.06e-17   -5.13356e-23
   9     0.000105778438451692     0.000105778438738558   -2.87e-13   1.54826e-19
  10    -7.179608555318982e-19                     0.0   -7.18e-19   -1.99631e-23
  11      4.370983446425289e-5      4.370983458279172e-5   -1.19e-13   6.39774e-20
  12     1.291290913177378e-18                     0.0    1.29e-18   -9.16416e-24
  13       1.92466329056234e-5       1.924663295781926e-5   -5.22e-14   2.8171e-20
  14     1.489720121033478e-18                     0.0    1.49e-18   -4.566e-24
  15      8.523024732461648e-6       8.52302475557563e-6   -2.31e-14   1.2475e-20

===== m = -1 =====
 log coefficient B6 (self-dual) vs -(i/450) P^(m):
   p=0: (-6.6640031406385e-14 + 1.2970993053824e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (1.03e-12 + 5.6e-19j))
   p=1: (4.29130157323e-9 - 0.0088884814187161j)   pred (-5.8593780510281e-96 - 0.0088888888888889j)   (anti-self-dual log: (3.22e-12 + 1.76e-18j))
   p=2: (4.2397165644972e-13 + 1.5493130592802e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (2.09e-12 + 1.14e-18j))
   p=3: (-3.2184761928869e-9 + 0.0066663610623523j)   pred (0.0 + 0.0066666666666667j)   (anti-self-dual log: (-2.42e-12 - 1.32e-18j))
   p=4: (-3.0548373645955e-13 - 1.4051909136316e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (-2.81e-12 - 1.53e-18j))
   p=5: (7.8671927455746e-18 - 1.0439192592153e-12j)   pred (5.4545482947753e-94 + 2.3181830252795e-92j)   (anti-self-dual log: (-1.68e-17 - 2.12e-22j))
   p=6: (1.4352449179272e-22 - 1.5882883316615e-17j)   pred (0.0 + 0.0j)   (anti-self-dual log: (4.54e-13 + 2.48e-19j))
   p=7: (-3.7576483922687e-20 + 6.8839850613297e-14j)   pred (-8.7272772716404e-93 + 2.9018196928204e-91j)   (anti-self-dual log: (5.05e-18 + 3.57e-23j))
 free fit (odd p<=15): a1=0.549682295656536 a3=-0.109477843145636 a5=-0.00100158355166457  450*lam_log=-0.999999997288  450*lam_psi=-0.999999997288
 fixed lam=-1/450: a1=0.549682295642621 a3=-0.109477843118276 a5=-0.00100158356640344
   p   Im(self-dual)              closed-form (p>=7)        residual     Re(self-dual)
   0       0.4357194696801278                     0.0       0.436   -0.756116
   1       0.5548131015528572        0.5548131015528572        0.0   -0.530585
   2      -0.08092639139119176                     0.0     -0.0809   0.0332273
   3       -0.1195664533568157       -0.1195664533568157        0.0   0.0267898
   4      -0.01592625975617983                     0.0     -0.0159   0.0034925
   5      0.004433203361182971      0.004433203361182971        0.0   9.13534e-17
   6    -1.844312483758952e-16                     0.0   -1.84e-16   1.65902e-21
   7    -0.000298107933199224    -0.000298107934003762    8.08e-13   -4.36336e-19
   8    -1.057047882528379e-17                     0.0   -1.06e-17   -5.13356e-23
   9    -0.000105778438451692    -0.000105778438738585    2.87e-13   -1.54826e-19
  10    -7.179608555318982e-19                     0.0   -7.18e-19   -1.99631e-23
  11     -4.370983446425289e-5     -4.370983458279172e-5    1.19e-13   -6.39774e-20
  12     1.291290913177378e-18                     0.0    1.29e-18   -9.16416e-24
  13      -1.92466329056234e-5      -1.924663295781926e-5    5.22e-14   -2.8171e-20
  14     1.489720121033478e-18                     0.0    1.49e-18   -4.566e-24
  15     -8.523024732461648e-6      -8.52302475557563e-6    2.31e-14   -1.2475e-20

===== m = -2 =====
 log coefficient B6 (self-dual) vs -(i/450) P^(m):
   p=0: (-6.6640031406385e-14 + 1.2970993053824e-7j)   pred (0.0 + 0.0j)   (anti-self-dual log: (1.03e-12 + 5.6e-19j))
   p=1:  (8.58260314646e-9 - 0.017776962837432j)   pred (-2.9829560987052e-95 - 0.017777777777778j)   (anti-self-dual log: (6.45e-12 + 3.52e-18j))
   p=2: (1.2960464376676e-12 + 1.3979848067579e-6j)   pred (0.0 + 0.0j)   (anti-self-dual log: (1.61e-11 + 8.77e-18j))
   p=3: (2.5747809284075e-8 - 0.053330888473302j)   pred (-2.0454556105407e-94 - 0.053333333333333j)   (anti-self-dual log: (1.93e-11 + 1.06e-17j))
   p=4: (6.0872603434714e-12 - 8.6473286366697e-8j)   pred (0.0 + 0.0j)   (anti-self-dual log: (1.01e-11 + 5.49e-18j))
   p=5: (-4.7309013708019e-18 + 2.707426611439e-12j)   pred (1.0909096589551e-93 + 3.7636383233949e-92j)   (anti-self-dual log: (1.18e-16 + 1.52e-21j))
```

```
  p=6: (-4.2719495994257e-22 + 6.8621868786177e-17j)   pred (0.0 + 0.0j)   (anti-self-dual log: (-1.43e-12 - 7.83e-19j))
  p=7: (2.3220721582015e-19 - 4.2540196897837e-13j)   pred (-8.7272772716404e-93 + 1.723637261149e-91j)   (anti-self-dual log: (-3.16e-17 - 2.25e-22j))
free fit (odd p<=15): a1=1.09936459131307 a3=0.131535806558426 a5=2.34579860911748e-5  450*lam_log=-0.999999997288  450*lam_psi=-0.999999997288
fixed lam=-1/450: a1=1.09936459128524 a3=0.131535806682649 a5=2.34580290988818e-5
  p   Im(self-dual)            closed-form (p>=7)        residual     Re(self-dual)
  0       0.4357194696801278                     0.0       0.436   -0.756116
  1        1.109626203105714        1.109626203105714         0.0   -1.06117
  2       0.6448002838789433                     0.0       0.645   -0.416229
  3       0.0857299290304934       0.0857299290304934    1.53e-92   -0.0524701
  4     0.001732090787054319                     0.0     0.00173   -0.00410596
  5     -0.01583549962884838     -0.01583549962884838         0.0   -5.4935e-17
  6     7.968337026431478e-16                    0.0    7.97e-16   -4.938e-21
  7      0.001842184441931932     0.001842184446927845    -5.0e-12   2.69638e-18
  8     -8.27091323640316e-17                    0.0   -8.27e-17   7.57402e-22
  9     0.0003148860242205729    0.0003148860250745282   -8.54e-13   4.60894e-19
 10    -3.555203039807281e-17                    0.0   -3.56e-17   1.41803e-22
 11      1.328260160875823e-5     1.328260164477996e-5    -3.6e-14   1.94416e-20
 12    -1.0523835457227 52e-17                   0.0   -1.05e-17   1.11814e-23
 13     -2.886334924423931e-5    -2.886334932251531e-5    7.83e-14   -4.22469e-20
 14    -8.137650438045826e-19                    0.0   -8.14e-19   -1.06514e-23
 15     -2.086353965523677e-5    -2.086353971181768e-5    5.66e-14   -3.05377e-20

===== m = +0 =====
 log coefficient B6 (self-dual) vs -(i/450) P^(m):
  p=0: (-6.6640031406385e-14 + 1.2970993053824e-7j)   pred 0   (anti-self-dual log: (1.03e-12 + 5.6e-19j))
  p=1: (-1.3052732460635e-61 - 3.2518482836582e-62j)   pred 0   (anti-self-dual log: (1.43e-61 + 1.49e-62j))
  p=2: (1.3328006271042e-13 - 2.5941986101526e-7j)   pred 0   (anti-self-dual log: (-2.57e-12 - 1.4e-18j))
  p=3: (-6.9835955903522e-61 - 7.1677027989549e-61j)   pred 0   (anti-self-dual log: (-1.51e-60 + 6.74e-62j))
  p=4: (-6.6640031230643e-14 + 1.2970993046583e-7j)   pred 0   (anti-self-dual log: (1.92e-12 + 1.05e-18j))
  p=5: (-1.5619444039186e-59 + 1.3917343980034e-59j)   pred 0   (anti-self-dual log: (1.6e-59 - 2.15e-61j))
  p=6: (-7.1763462993743e-23 + 8.3761221909853e-18j)   pred 0   (anti-self-dual log: (-3.21e-13 - 1.75e-19j))
  p=7: (7.0121312626552e-59 - 2.5105668205843e-58j)   pred 0   (anti-self-dual log: (-1.1e-58 + 2.64e-59j))
  p   self-dual coefficient (should terminate: polynomial)
  0  (-0.7561157301488572 + 0.4357194696801278j)
  1  (-1.412895988057674e-60 - 3.514235707432713e-61j)
  2  (0.1830460967000012 - 0.3228352831479035j)
  3  (-7.55945629153408e-60 - 7.764576970546053e-60j)
  4  (-0.002170879889990421 + 0.02727608989285813j)
  5  (-1.688912934933161e-58 + 1.505553461244186e-58j)
  6  (-8.295236784820559e-22 + 9.7263114821771e-17j)
  7  (7.587133492585004e-58 - 2.715325964393345e-57j)
  8  (-9.441574164338469e-24 + 1.638560597122644e-17j)
  9  (-2.312661494578 38e-56 + 2.432024002646284e-56j)
 10  (-3.776642765948879e-24 + 6.554263801173189e-18j)
 11  (1.883391800229967e-55 - 3.118135434701264e-55j)
 12  (-1.888324656424191e-24 + 3.277137253515164e-18j)
 13  (-2.035692970694269e-54 + 3.582774534606989e-54j)
 14  (-1.079043829904595e-24 + 1.872651770783047e-18j)
 15  (3.332595096156453e-53 - 3.480410172677199e-53j)
```

31. prl_checks/unified_check.py

Remainder after subtracting Eq. (S18). Physical branch: after subtracting the closed form (S18), the Taylor coefficients $6 \le p \le 20$ vanish for every m (uniqueness (b)).

Paper: Eq. (S18), S5 Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 unified_check.py`

Criterion	Measured	Required	
Taylor coefficients 6 ≤ p ≤ 20 after subtracting Eq. (S18), m = 2, 1, -1, -2, 0	max 1.69e-11 (5 values)	≤ 2e-11	✓

Output (`logs/prl-checks-unified-check-py.txt`):

```
m=+2: |remainder Taylor coeff| p=0..5: ['0.873', '1.55', '0.767', '0.189', '0.00446', '2.35e-5']  max p=6..20: 1.69e-11  | max|B6 - (i/450)Pi| = 2.44e-6
m=+1: |remainder Taylor coeff| p=0..5: ['0.873', '0.774', '0.0875', '0.116', '0.0163', '0.001']  max p=6..14: 5.33e-12  | max|B6 - (i/450)Pi| = 4.07e-7
m=-1: |remainder Taylor coeff| p=0..5: ['0.873', '0.774', '0.0875', '0.116', '0.0163', '0.001']  max p=6..14: 5.33e-12  | max|B6 - (i/450)Pi| = 4.07e-7
m=-2: |remainder Taylor coeff| p=0..5: ['0.873', '1.55', '0.767', '0.189', '0.00446', '2.35e-5']  max p=6..14: 1.69e-11  | max|B6 - (i/450)Pi| = 2.44e-6
m=+0: |remainder Taylor coeff| p=0..5: ['0.873', '5.64e-52', '0.371', '3.22e-50', '0.0274', '5.45e-49']  max p=6..14: 3.77e-12  | max|B6 - (i/450)Pi| = 2.59e-7
```

32. prl_checks/uniqueness_check.py

Continuation to super-extremal spin; bounds on extra terms. The $m = 2$ polynomial fixed at $|q| \le 0.3$ and continued to $1.5 \le q \le 5000$ reproduces direct MST; bounds on additional terms $c_p q^p$; the q^5 coefficient $-2.3457997302 \times 10^{-5} i$ of S5(d).

Paper: S5, uniqueness (c), (d) Status: **PASS** Time: 1 s
Command: `cd prl_checks && python3 uniqueness_check.py`

Criterion	Measured	Required	
polynomial fixed at |q| ≤ 0.3, continued to 50 ≤ q ≤ 5000, vs direct MST	max 3.3e-10 (9 values)	≤ 4e-10	✓
same for 1.5 ≤ q ≤ 30, joint log bookkeeping on both sides	max 3.7e-12 (8 values)	≤ 5e-12	✓

Criterion	Measured	Required	
bound on an additional term c_6 q^6	3.16e-18	$\leq$ 1e-17	✓
bound on an additional term c_7 q^7	6.33e-22	$\leq$ 1e-20	✓
q^5 coefficient (joint bookkeeping), Im, vs -2.3457997302e-5 (S5(d))	-2.34579973024e-5 (diff 3.9e-16)	diff $\leq$ 1e-14	✓

Output (`logs/prl-checks-uniqueness-check-py.txt`):

```
remainder Taylor coefficients |p| for p = 6..20 (should vanish): 3.5e-9
   q     rel. difference (continued closed form vs direct super-extremal MST)
   1.5   0.00019
     2   0.00017
     3   9.2e-5
     5   4.3e-5
     8   2.0e-5
    12   1.0e-5
    20   4.6e-6
    30   2.5e-6
    50   3.2e-10
    80   3.3e-10
   120   3.3e-10
   200   3.3e-10
   300   1.5e-10
   600   3.9e-11
  1000   1.5e-11
  2000   3.8e-12
  5000   5.9e-13

bounds on an additional term c_p q^p from the agreement at q = 5000 (|E6(q)|/q^5 = 0.026644):
   |c_6| <~ 3.16e-18
   |c_7| <~ 6.33e-22
   |c_8| <~ 1.27e-25

 p   joint-bookkeeping poly coeff                naive poly coeff (ev - T6)                 difference
 0    (-0.75611573014807 + 0.43571788642838j)    (-0.75611573014886 + 0.43571946968013j)   1.58e-6
 1       (1.061170384751 - 1.09935464404471j)       (1.061170273715 - 1.09936459128523j)   9.95e-6
 2    (-0.41622900444632 + 0.64478321994339j)     (-0.4162290044311 + 0.64480028387894j)   1.71e-5
 3    (0.052470380458234 - 0.13150596451028j)    (0.052470066178016 - 0.13153580668265j)   2.98e-5
 4   (-0.0041059600444953 + 0.0017331462882208j)   (-0.0041059599730032 + 0.0017320907870543j)   1.06e-6
 5   (1.2470632619581e-16 - 2.3457997302387e-5j)   (5.4934987295581e-17 - 2.3458029099882e-5j)   3.18e-11

   q     rel. difference with joint bookkeeping on both sides
   1.5   3.7e-12
     2   2.9e-12
     3   1.4e-12
     5   7.2e-13
     8   5.0e-13
    12   5.0e-13
    20   6.8e-13
    30   9.6e-13
```

33. `conservative_continuation/phys_hi_p.py`

Physical-region coefficients to q^{24}. Coefficients up to q^{24} with the $\epsilon^6 \ln \epsilon$ term separated, from `../prl_checks/phys_l2_*.json`; writes `phys_e6_hi.json`.

Paper: Eq. (12), S5 Status: **PASS** Time: <1 s
Command: `cd conservative_continuation && python3 phys_hi_p.py`

Criterion	Measured	Required	
exchange-odd q^6 coefficient (one radius, no extrapolation) vs rho_2/64	0.00621387872835 (rel. diff 2.7e-9)	rel. diff $\leq$ 1e-8	✓

Output (`logs/conservative-continuation-phys-hi-p-py.txt`):

```
p= 0: even A6 = (-0.7561157301488572 + 0.4357194696801278j)  odd A6 = (-0.00444444443239 + 6.50491271402e-18j)  even log-coeff B6 = (-6.664003141e-14 + 1.297099305e-7j)
p= 1: even A6 = (1.061170273714987 - 1.109626203105714j)  odd A6 = (0.0279252679562 - 4.08735520113e-17j)  even log-coeff B6 = (-8.582603146e-9 + 0.01777696284j)
p= 2: even A6 = (-0.4162290044311035 + 0.6448002838789433j)  odd A6 = (-0.0695976555214 + 1.01875764947e-16j)  even log-coeff B6 = (1.296046438e-12 + 1.397984807e-6j)
p= 3: even A6 = (0.052470066178016 - 0.0857299290304934j)  odd A6 = (0.0837758038685 - 1.22644747366e-16j)  even log-coeff B6 = (-2.574780928e-8 + 0.05333088847j)
p= 4: even A6 = (-0.004105959973003181 + 0.0017320907870543 19j)  odd A6 = (-0.0435502682952 + 6.3775708224e-17j)  even log-coeff B6 = (6.087260343e-12 - 8.647328637e-8j)
p= 5: even A6 = (5.493498729558084e-17 + 0.01583549962884838j)  odd A6 = (-1.3719356319e-15 - 1.75690462335e-20j)  even log-coeff B6 = (4.730901371e-18 - 2.707426611e-12j)
p= 6: even A6 = (-4.938004934468363e-21 + 7.96833706431477e-16j)  odd A6 = (0.00621387872835 - 9.09516369113e-18j)  even log-coeff B6 = (-4.271949599e-22 + 6.862186879e-17j)
p= 7: even A6 = (-2.696378440132939e-18 - 0.001842184441931932j)  odd A6 = (3.67262277486e-16 + 2.60565954519e-21j)  even log-coeff B6 = (-2.322072158e-19 + 4.25401969e-13j)
p= 8: even A6 = (7.574016205023368e-22 - 8.270913236403163e-17j)  odd A6 = (-8.06615792795e-5 + 1.18063124646e-19j)  even log-coeff B6 = (6.552406484e-23 - 7.12276028e-18j)
p= 9: even A6 = (-4.608941033795153e-19 - 0.0003148860242205729j)  odd A6 = (3.00557795532e-17 - 1.3758230039e-23j)  even log-coeff B6 = (-3.969136342e-20 + 7.271429069e-14j)
p=10: even A6 = (1.418025304404415e-22 - 3.555203039807282e-17j)  odd A6 = (-0.0001727188871456 + 2.52806093465e-19j)  even log-coeff B6 = (1.226757104e-23 - 3.061676229e-18j)
p=11: even A6 = (-1.944155639519076e-20 - 1.328260160875821e-5j)  odd A6 = (-6.86327778675e-18 - 7.04985936024e-23j)  even log-coeff B6 = (-1.674271543e-21 + 3.06725257e-15j)
p=12: even A6 = (1.118138235753203e-23 - 1.052383545722752e-17j)  odd A6 = (-6.86798680161e-5 + 1.00525724288e-19j)  even log-coeff B6 = (9.673198418e-25 - 9.0629357e-19j)
p=13: even A6 = (4.224686242874499e-20 + 2.886334924423934e-5j)  odd A6 = (-8.23621353981e-18 - 3.12062912422e-23j)  even log-coeff B6 = (3.638223091e-21 - 6.665198823e-15j)
p=14: even A6 = (-1.065136410369412e-23 - 8.137650438045848e-19j)  odd A6 = (-1.54371917744e-5 + 2.25951942736e-20j)  even log-coeff B6 = (-9.214670988e-25 - 7.007996068e-20j)
```

```
p=15: even A6 = (3.053765916584682e-20 + 2.086353965523679e-5j)  odd A6 = (-4.50348561781e-18 - 8.54249202246e-24j)  even log-coeff B6 = (2.629847765e-21 - 4.817862231e-15j)
p=16: even A6 = (-9.066956589764496e-24 + 1.785456754180775e-18j)  odd A6 = (2.52788482427e-6 - 3.7000274588e-21j)  even log-coeff B6 = (-7.843973861e-25 + 1.537602929e-19j)
p=17: even A6 = (1.383061448860638e-20 + 9.449171096061616e-6j)  odd A6 = (-1.58943200453e-18 + 5.80760749765e-26j)  even log-coeff B6 = (1.19106741e-21 - 2.182026892e-15j)
p=18: even A6 = (-4.658152937207071e-24 + 1.766093552818085e-18j)  odd A6 = (5.75684521908e-6 - 8.42621027282e-21j)  even log-coeff B6 = (-4.029845016e-25 + 1.520927676e-19j)
p=19: even A6 = (3.726066015597188e-21 + 2.545673819755955e-6j)  odd A6 = (-9.53981846729e-20 + 2.16298664737e-24j)  even log-coeff B6 = (3.208820406e-22 - 5.878535483e-16j)
p=20: even A6 = (-1.613944429185959e-24 + 1.098841572267932e-18j)  odd A6 = (4.4089274271e-6 - 6.45328287498e-21j)  even log-coeff B6 = (-1.39624997e-25 + 9.463023909e-20j)
p=21: even A6 = (-7.000325089826857e-22 - 4.782671738881488e-7j)  odd A6 = (4.40353691847e-19 + 1.95237702812e-24j)  even log-coeff B6 = (-6.028552863e-23 + 1.104426832e-16j)
p=22: even A6 = (-9.774658132912203e-26 + 4.894206757867454e-19j)  odd A6 = (2.41547440811e-6 - 3.53549480281e-21j)  even log-coeff B6 = (-8.456217829e-27 + 4.214801887e-20j)
p=23: even A6 = (-1.942704519964402e-21 - 1.327269200586325e-6j)  odd A6 = (4.94524887536e-19 + 1.21556777413e-24j)  even log-coeff B6 = (-1.673021862e-22 + 3.064964155e-16j)
p=24: even A6 = (4.517449149541934e-25 + 1.034617568238909e-19j)  odd A6 = (9.37287565593e-7 - 1.37189425084e-21j)  even log-coeff B6 = (3.908119866e-26 + 8.909938206e-21j)
```

34. conservative_continuation/e6_fit.py

Fit of the $\psi+\log$ term. The odd coefficients $q^7 \dots q^{23}$ are reproduced by the $\psi+\log$ ansatz with $\lambda = -1/450$. Writes `e6_fit.json`.

Paper: Eq. (12) Status: **PASS** Time: 2 s
Command: `cd conservative_continuation && python3 e6_fit.py`

Criterion	Measured	Required	
fitted coefficient of the psi + log term, 450 lambda, vs -1	-1.00000000015 (diff 1.5e-10)	diff ≤ 1e-8	✓
fit residuals of the odd coefficients p = 7...23	max 2.86e-12 (9 values)	≤ 1e-11	✓

Output (`logs/conservative-continuation-e6-fit-py.txt`):

```
fit: a1 = -1.09936459128373  a3 = -0.131535806690734  a5 = -2.34580360428429e-5  lambda = -0.0022222222225507
 p   data Im E6               model                      residual
 1      -1.10962620310571       -1.10962620310571         0.0
 3    -0.0857299290304934     -0.0857299290304934         0.0
 5     0.0158354996288484      0.0158354996288484    4.18e-53
 7   -0.00184218444193193    -0.00184218444479129    2.86e-12
 9  -0.000314886024220573   -0.000314886023478666   -7.42e-13
11   -1.32826016087582e-5    -1.32826003985134e-5   -1.21e-12
13    2.88633492442393e-5     2.88633503341253e-5   -1.09e-12
15    2.08635396552368e-5     2.08635405595686e-5   -9.04e-13
17    9.44917109606162e-6      9.4491718504361e-6   -7.54e-13
19    2.54567381975595e-6     2.54567446575016e-6   -6.46e-13
21   -4.78267173888149e-7    -4.78266605887818e-7   -5.68e-13
23   -1.32726920058633e-6    -1.32726869075627e-6    -5.1e-13
lambda * 450 = -1.00000000015    lambda*900 = -2.0000000003
a1 = -1.09936459128373  identify: None
a3 = -0.131535806690734  identify: None
a5 = -2.34580360428429e-5  identify: None
```

35. conservative_continuation/e6_check_se2.py

Continued closed form against super-extremal MST. The closed form continued to $1.5 \le q \le 5000$ against direct MST at super-extremal spin, with the naive log bookkeeping (10^{-6} to 10^{-8}; the refined test is `uniqueness_check.py`).

Paper: Eq. (12), S5 Status: **PASS** Time: <1 s
Command: `cd conservative_continuation && python3 e6_check_se2.py`

Criterion	Measured	Required	
continued closed form vs direct super-extremal MST, 1.5 ≤ q ≤ 5000 (naive log bookkeeping)	max 1e-6 (17 values)	≤ 2e-6	✓

Output (`logs/conservative-continuation-e6-check-se2-py.txt`):

```
  q      E6 from MST at super-extremal spin (log-subtracted)        continued physical closed form                  rel.diff
 1.5      (0.380055717533373014 - 0.183926245679859229j)      (0.380055712312061166 - 0.183926651554966221j)  9.6e-7
   2      (0.781430949913782639 + 0.919975230769561963j)       (0.781430938242421 1 + 0.919974202472915205j)  8.5e-7
   3        (2.111165632456317 86 + 8.02612775188146387j)        (2.1111655947897258 + 8.02612392810989707j)  4.6e-7
   5        (8.748145838929201 42 + 92.49295463072037 78j)       (8.748145668899026 32 + 92.49293465202855 23j)  2.2e-7
   8        (34.25786653960882 76 + 914.7854599712077 56j)       (34.257865850971494 5 + 914.7853678048927 87j)  1.0e-7
  12        (102.667734273080065 + 6789.738944014478 06j)        (102.667731965570345 + 6789.738594061874 21j)  5.2e-8
  20         (287.54757041702785 + 86195.2305727604742j)          (287.547559841789195 + 86195.228637506907j)  2.2e-8
  30         (10.121601080661089 + 651395.13248091909 1j)         (10.1215657927710515 + 651395.124773805498j)  1.2e-8
  50       (-9618.35843265238045 + 8351127.93909330667j)        (-9618.38989677246912 + 8351119.57369572087j)  1.0e-6
  80       (-100999.601246394781 + 87444493.6537785087j)        (-100999.729014802933 + 87444446.0919984707j)  5.4e-7
 120       (-621842.332219026245 + 663593554.805393291j)         (-621842.75848343878 + 663593334.396363003j)  3.3e-7
 200        (-5495999.16413626299 + 8530031753.9395205j)        (-5496001.09179976452 + 8530030176.58012483j)  1.8e-7
 300       (-29616765.1088963333 + 64762423647.7320579j)        (-29616771.9217866912 + 64762415405.3763559j)  1.3e-7
 600       (-502852440.230498972 + 2072051381506.84201j)        (-502852494.262469516 + 2072051244941.56328j)  6.6e-8
1000      (-3970129013.78261378 + 26645204092230.6506j)        (-3970129243.39444261 + 26645203013680.2806j)  4.0e-8
2000      (-64607054029.1076253 + 852612799131772.835j)        (-64607055344.4437075 + 852612781140685.545j)  2.1e-8
5000      (-2549204655382.40978 + 83261153757540985.2j)        (-2549204649623.15954 + 83261152979208328.9j)  9.3e-9
```

36. conservative_continuation/e6_identities.py

$[q^5]\mathcal{C} = 16b_2$ **and** $\mathcal{C}/q^5 \to 64/15$**.** $160[q^5]F = 16b_2$ of arXiv:2610.40101 to all printed digits, and $160F/q^5 \to 64/15$ at large q.

Paper: Eq. (16), S9(e) Status: **PASS** Time: <1 s
Command: `cd conservative_continuation && python3 e6_identities.py`

Criterion	Measured	Required	
160 [q^5] F = 16 b_2 of arXiv:2610.40101 (all printed digits)	2.537433225271721941600605	2.537433225271721941600605	✓
160 F/q^5 at q = 1e7 vs 64/15	4.26666666667 (rel. diff 3.3e-13)	rel. diff ≤ 1e-11	✓

Output (`logs/conservative-continuation-e6-identities-py.txt`):

```
160 * [q^5] F (Taylor, physical)   = 2.537433225271721941600605
16 b2 (2610.40101, eq. 4.2)        = 2.537433225271721941600605
160 * F(q)/q^5 continued, q=1.0e+3: 4.26672957616662
160 * F(q)/q^5 continued, q=1.0e+5: 4.26666667688736
160 * F(q)/q^5 continued, q=1.0e+7: 4.26666666666808
64/15 = 4.26666666666667
```

37. conservative_continuation/se_taylor_l2.py

Classical $\ell = 2$ Taylor coefficients. Classical ($\epsilon \to 0$ at fixed $a\omega$) $\ell = 2$ Taylor coefficients for both branches, from `polar_run2.json`: the exchange-odd part starts at c^5 with $-64i/75$, followed by $-256i/135$. Writes `se_taylor_l2.json`.

Paper: Eq. (17), S9(h) Status: **PASS** Time: <1 s
Command: `cd conservative_continuation && python3 se_taylor_l2.py`

Criterion	Measured	Required	
exchange-odd coefficients of c^0...c^4	max 1.37e-31 (5 values)	≤ 1e-25	✓
exchange-odd c^5: Im vs -64/75	-0.853333333333 (diff 2.2e-16)	diff ≤ 1e-14	✓
exchange-odd c^6: Im vs -256/135	-1.8962962963 (diff 6.7e-16)	diff ≤ 1e-14	✓

Output (`logs/conservative-continuation-se-taylor-l2-py.txt`):

```
c^0: eta-even (-1.7846152680107368789e-17 - 0.542607642410376833321j)   eta-odd (6.1132608721008790e-39 + 2.208682364137816e-39j)
c^1: eta-even (-4.4507176160691794563e-17 - 0.666666666666666777139j)   eta-odd (1.3691300711978150e-39 + 6.058075721102301e-40j)
c^2: eta-even (-2.6530265236682651165e-17 + 0.137566137566137613233j)   eta-odd (-1.078677781239896e-34 - 2.164460762157156e-35j)
c^3: eta-even (1.7529269334294766296e-17 - 0.025279247501469738202j)   eta-odd (-2.409112758725990e-35 + 8.821351442429499e-34j)
c^4: eta-even (-6.5695360711869660755e-18 + 0.003980584983356465091 5j)   eta-odd (-1.290140578957835e-31 + 4.603148345122169e-32j)
c^5: eta-even (4.5200954639604111426e-18 + 0.852582677419608344 54j)   eta-odd (-2.823153739653773e-18 - 0.85333333333333336j)
c^6: eta-even (9.1290614313280501457e-17 + 1.8964816078651064584j)   eta-odd (-9.160700831031984e-17 - 1.896296296296297j)
c^7: eta-even (1.7283439480670931045e-16 + 2.3230364974864529475j)   eta-odd (-1.727837850399611e-16 - 2.323070462753003j)
c^8: eta-even (1.9060448491452592921e-16 + 2.032584779336302961j)   eta-odd (-1.906158516297064e-16 - 2.032582141224117j)
```

The ϵ^7 operator and the a^6 coefficients

Paper: Eq. (S23); S9(g), S9(h)

At order ϵ^7 both sectors follow from ϵ^6 by the operator $(1/\kappa)\partial_{\tau_0} + u_0 + \frac{5m}{9}q$, with $u_0 = \pi + i(2\gamma_E + 2\ln 2 - \frac{37}{12} - \nu_2)$ and $\nu_2 = -107/210$, Eq. (S23).

38. `prl_checks/d7_phys.py`

Exchange-odd ϵ^7 for $m = 2$. Fit of $D_7 - (1/\kappa)\partial_{\tau_0} D_6$ to $(u_0 + u_1 q + u_2 q^2 + u_3 q^3) D_6 + \mathrm{poly}_7(q)$: $u_1 = 10/9$, no q^2 and q^3 terms.

Paper: Eq. (S23), S9(g) Status: **PASS** Time: 1 s
Command: `cd prl_checks && python3 d7_phys.py`

Criterion	Measured	Required	
fit residual of D7 - (1/kappa) dD6/dtau0 = (u0 + u1 q + u2 q^2 + u3 q^3) D6 + poly_7(q)	7.2e-15	≤ 1e-12	✓
Re u0 vs pi (fit with u3 free)	3.14160296124 (diff 1e-5)	diff ≤ 0.0001	✓
Im u0 vs 2 gamma_E + 2 ln 2 - 37/12 - nu2	-0.0330838328848 (diff 1.8e-12)	diff ≤ 1e-9	✓
u1 vs (5/9) m = 10/9	(1.111111111 + 5.60562815e-15i) (diff 5.8e-11)	diff ≤ 1e-8	✓
|u2|, |u3| (no q^2, q^3 terms)	max 6.61e-12 (2 values)	≤ 1e-9	✓
D6 closed form vs data, p = 0...8	max 4.47e-12 (9 values)	≤ 1e-11	✓

Output (`logs/prl-checks-d7-phys-py.txt`):

```
radius stability of the anti-self-dual eps^7 Taylor coefficients (r=2e-5 vs 1e-5):
  p= 0  (-0.027925313842958 + 0.00014703925726579j)   diff (-4.58e-8 - 2.35e-14j)
  p= 1   (0.16350059995477 - 0.0009238749008311j)   diff (2.88e-7 + 1.48e-13j)
  p= 2  (-0.36913436788251 + 0.0023025572106889j)   diff (-7.17e-7 - 3.69e-13j)
  p= 3   (0.37735660604538 - 0.002771624702493j)   diff (8.63e-7 + 4.44e-13j)
  p= 4  (-0.12750922953192 + 0.0014408098022741j)   diff (-4.49e-7 - 2.31e-13j)
  p= 5  (-0.045255689153202 + 8.6024038143658e-18j)   diff (4.96e-12 + 6.02e-17j)
  p= 6  (0.019521539866741 - 0.0002055789259727 7j)   diff (6.4e-8 + 3.29e-14j)
  p= 7  (0.010333895407046 - 1.2758113690315e-18j)   diff (-1.33e-12 - 8.93e-18j)
  p= 8  (-0.00025340665701025 + 2.6685942163442e-6j)   diff (-8.31e-10 - 4.27e-16j)
  p= 9  (0.00065297801311867 + 6.7370462844382e-21j)   diff (-1.09e-13 + 4.72e-20j)
  p=10  (-0.00054261411950066 + 5.7142022948021e-6j)   diff (-1.78e-9 - 9.15e-16j)
  p=11  (-0.0002116278282829 + 3.4508112842477e-20j)   diff (2.48e-14 + 2.42e-19j)

check D6 closed form vs data (p=0..8): ['(2.37e-13 + 5.55e-20j)', '(-1.49e-12 - 3.49e-19j)', '(3.71e-12 + 8.69e-19j)', '(-4.47e-12 - 1.05e-18j)', '(2.32e-12 + 5.44e-19j)', '(-1.17
       ↪ e-17 - 7.02e-23j)', '(-3.32e-13 - 7.76e-20j)', '(3.13e-18 + 1.04e-23j)', '(4.3e-15 + 1.01e-21j)']

fit D7 - (1/kappa) dD6/dtau0 = (u0+u1 q+u2 q^2+u3 q^3) D6 + poly_7(q):  residual 7.196e-15
   u0 = (3.141602961243327 - 0.033083832888478939j)
   u1 = (1.111111111052867 + 5.605628150354486e-15j)
   u2 = (3.1225763062028 98e-16 + 1.879684196283626e-16j)
   u3 = (-6.614208576399409e-12 + 7.957902292362233e-17j)
   v0 = (7.449777898124785e-13 + 5.73382084447713e-17j)
   v1 = (-4.37917646439228e-12 - 3.343831750577241e-16j)
   v2 = (1.01430757028987e-11 + 7.361783521236152e-16j)
   v3 = (-1.138910097065556e-11 - 6.808519184341756e-16j)
   v4 = (6.029635842202 65e-12 + 8.580490447708534e-17j)
   v5 = (-1.342338737539814e-12 + 2.425203223277093e-16j)
   v6 = (5.541199896987521e-13 - 7.840295146967225e-17j)
   v7 = (-3.690512905373 63e-13 - 3.264281717764691e-17j)
   (5/9) m = 1.111111111111111
```

39. `prl_checks/d7_fit2.py`

Exchange-odd ϵ^7 for all m. u_0, $u_1 = 5m/9$ and $u_2 = 0$ for $m = 2, 1, -1, -2, 0$.

Paper: Eq. (S23), S9(g) Status: **PASS** Time: 3 s
Command: `cd prl_checks && python3 d7_fit2.py`

Criterion	Measured	Required	
Re u0 vs pi, m = 2, 1, -1, -2, 0	max diff 6.47e-10 (5 values)	diff ≤ 1e-8	✓
Im u0 vs 2 gamma_E + 2 ln 2 - 37/12 - nu2	max diff 3.53e-12 (5 values)	diff ≤ 1e-11	✓
u1 vs (5/9) m	max diff 2.93e-10 (5 values)	diff ≤ 1e-9	✓
|u2| (no q^2 term)	max 7.75e-11 (5 values)	≤ 1e-9	✓
fit residual	max 7.64e-13 (5 values)	≤ 1e-11	✓

Output (`logs/prl-checks-d7-fit2-py.txt`):

```
m=+2: D7 - D6*dlog/dtau0/kappa = (u0 + u1 q + u2 q^2) D6 :  u0 = (3.141592654237 - 0.0330838328901j)  u1 = (1.111111111293 + 1.259738833407e-15j)  u2 = (5.2e-11 + 3.65e-17j)
        ↪ resid 7.64e-13   [(5/9)m = 1.111111111]
m=+1: D7 - D6*dlog/dtau0/kappa = (u0 + u1 q + u2 q^2) D6 :  u0 = (3.14159265423 - 0.0330838328901j)  u1 = (0.5555555556441 + 6.279242075841e-16j)  u2 = (7.75e-11 + 5.42e-17j)
        ↪ resid 6.71e-13   [(5/9)m = 0.5555555556]
m=-1: D7 - D6*dlog/dtau0/kappa = (u0 + u1 q + u2 q^2) D6 :  u0 = (3.14159265423 - 0.0330838328901j)  u1 = (-0.5555555556441 - 6.279242075841e-16j)  u2 = (7.75e-11 + 5.42e-17j)
        ↪ resid 6.71e-13   [(5/9)m = -0.5555555556]
m=-2: D7 - D6*dlog/dtau0/kappa = (u0 + u1 q + u2 q^2) D6 :  u0 = (3.141592654237 - 0.0330838328901j)  u1 = (-1.111111111293 - 1.259730207046e-15j)  u2 = (5.2e-11 + 3.65e-17j)
        ↪ resid 7.64e-13   [(5/9)m = -1.111111111]
m=+0: D7 - D6*dlog/dtau0/kappa = (u0 + u1 q + u2 q^2) D6 :  u0 = (3.141592654219 - 0.0330838328901j)  u1 = (1.13697196049e-57 - 3.657179768086e-57j)  u2 = (7.09e-11 + 4.97e-17j)
        ↪  resid 3.47e-13   [(5/9)m = 0.0]

u0 - pi = (6.47242573231288e-10 - 0.0330838328900975j)
identify Im u0: None -0.0330838328900975
```

40. prl_checks/e7_phys.py

Exchange-even ϵ^7 for $m = 2$. The same operator acting on T_6, plus a polynomial of degree six; the $\epsilon^7 \ln\epsilon$ coefficient B_7. Writes `e7_fit.json` and `e7_phys.json`.

Paper: Eq. (S23), S9(g) Status: **PASS** Time: 4 s
Command: `cd prl_checks && python3 e7_phys.py`

Criterion	Measured	Required	
B6 at p = 1, 3 vs 4i/225, 12i/225	max diff 5.46e-12 (2 values)	diff ≤ 1e-10	✓
u0 (fit) - [pi + i(2 gamma_E + 2 ln 2 - 37/12 - nu2)]	2.68e-8	≤ 1e-7	✓
u1 vs (5/9) m = 10/9	(1.111111132 - 7.005052117e-13i) (diff 2.1e-8)	diff ≤ 1e-7	✓
|u2| (no q^2 term)	2.71e-10	≤ 1e-9	✓
fit residual	4.45e-12	≤ 1e-11	✓
B7 data vs -(i/450)[P'/kappa + (u0 + 5mq/9) P], p = 0...9	max 1.59e-6 (10 values)	≤ 5e-6	✓

Output (`logs/prl-checks-e7-phys-py.txt`):

```
B6 check (p=1,3):  (1.19015170061892e-16 + 0.0177777777759569j) (4.864404835776736e-16 + 0.05333333333278723j)   expected 4i/225 = 0.0177777777777778  12i/225 = 0.0533333333333333

A7 fit:  u0 = (3.14159268039 - 0.03308383256j)   u1 = (1.11111113166 - 7.00505211698e-13j)   u2 = (-2.498e-10 - 1.047e-10j)   residual 4.45e-12
        poly coeffs: [’(-1.15467 + 0.121635j)’, ’(1.75002 - 1.0114j)’, ’(-0.855729 + 1.12824j)’, ’(0.213209 - 0.332937j)’, ’(-0.0112077 - 0.0109312j)’, ’(0.000530294 - 5.84927e-5j
        ↪ )’, ’(-1.33487e-14 + 2.89531e-6j)’, ’(-2.8164e-12 + 6.71058e-12j)’, ’(1.63388e-15 + 2.2794e-11j)’]
u0 closed form pi + i(2 gamma_E + 2 ln2 - 37/12 + 107/210) = (3.1415926535898 - 0.033083832886567j)   fit - closed: (2.68e-8 + 3.61e-10j)
A7 with u0,u1 fixed: residual 5.03e-11  poly: [’(-1.15467102831 + 0.121635114927j)’, ’(1.75002041177 - 1.01139989545j)’, ’(-0.855729117632 + 1.12824252865j)’, ’(0.213208723181 -
        ↪ 0.332937339444j)’, ’(-0.0112076797637 - 0.01093122437j)’, ’(0.000530294298848 - 5.84922931057e-5j)’, ’(-2.23940875373e-15 + 2.89563833555e-6j)’]
64*[q^6]poly = (-1.433221602390e-13 + 0.0001853208534753j)   classical eta=+1 c^6 (m=2) = 0.0001853115688095661
B7 data vs -(i/450)[P'/kappa + (u0 + 5m q/9) P]:
   p=0: (6.26722423682e-13 - 0.008888888877279j)   (-1.04832608518e-63 - 0.008888888888889j)   diff (6.267e-13 + 1.161e-10j)
   p=1: (0.000588157054919 + 0.0558500071886j)   (0.000588157029095 + 0.0558505360638j)   diff (2.582e-11 - 5.289e-7j)
   p=2: (-1.21666863447e-11 - 0.095802467917 5j)   (8.88160136167e-63 - 0.0958024691358j)   diff (-1.217e-11 + 1.218e-9j)
   p=3: (0.0017644719291 + 0.167550021932j)   (0.0017644708728 + 0.167551608191j)   diff (1.056e-10 - 1.586e-6j)
   p=4: (-5.71846089486e-11 + 0.005925925952 61j)   (7.03707622979e-62 + 0.005925925925 93j)   diff (-5.718e-11 + 2.669e-11j)
   p=5: (8.57544975149e-13 + 2.5436073088e-11j)   (-2.63665519488e-61 - 2.98685818223e-61j)   diff (8.575e-13 + 2.544e-11j)
   p=6: (-1.94825386152e-16 + 1.24385330014e-11j)   (-1.04153170082e-61 + 6.62368869592e-61j)   diff (-1.948e-16 + 1.244e-11j)
   p=7: (-4.20909354018e-14 - 3.99686125927e-12j)   (3.94253902245e-60 - 5.3095975586e-60j)   diff (-4.209e-14 - 3.997e-12j)
   p=8: (2.98772684894e-17 - 1.29110188223e-12j)   (-1.67072104946e-59 + 9.66718345064e-60j)   diff (2.988e-17 - 1.291e-12j)
   p=9: (-7.19464650678e-15 - 6.83110332778e-13j)   (6.77162492953e-59 + 6.42414590485e-59j)   diff (-7.195e-15 - 6.831e-13j)
(5/9) m = 1.11111111111   anti-self-dual u0 = pi - 0.0330838328901 i
```

41. prl_checks/e7_allm.py

Exchange-even ϵ^7 for all m. u_0 and $u_1 = 5m/9$ for $m = 1, -1, -2$; for $m = 0$, $B_7 = -(2i/225)\kappa^4$.

Paper: Eq. (S23), S9(g) Status: **PASS** Time: 4 s
Command: `cd prl_checks && python3 e7_allm.py`

Criterion	Measured	Required	
u0 vs closed form, m = 1, -1, -2	max diff 2.53e-8 (3 values)	diff ≤ 5e-8	✓
u1 vs 5m/9	max diff 8.5e-9 (3 values)	diff ≤ 2e-8	✓
fit residual	max 4.2e-12 (3 values)	≤ 1e-10	✓
m = 0: residual of A7 - (1/kappa) dT6/dtau0 = polynomial	3.3e-12	≤ 1e-10	✓
m = 0: |B7 - prediction|, B7 = -(2i/225) kappa^4	2.2e-10	≤ 1e-9	✓

Output (`logs/prl-checks-e7-allm-py.txt`):

```
m=+1: u0 = (3.1415926789 - 0.033083833153j) (closed (3.1415926536 - 0.033083832887j))  u1 = (0.55555554989 + 7.9493124052e-14j) (5m/9 = 0.555555555556)  resid 7.1e-13 | max|B7 -
       ↪ pred| = 2.6e-7  (|B7| ~ 0.0279)
m=-1: u0 = (3.1415926789 - 0.033083833153j) (closed (3.1415926536 - 0.033083832887j))  u1 = (-0.55555554989 - 7.9493124052e-14j) (5m/9 = -0.555555555556)  resid 7.1e-13 | max|B7 -
       ↪ pred| = 2.6e-7  (|B7| ~ 0.0279)
m=-2: u0 = (3.1415926789 - 0.033083833153j) (closed (3.1415926536 - 0.033083832887j))  u1 = (-1.1111111196 - 1.8699687035e-13j) (5m/9 = -1.1111111111)  resid 4.2e-12 | max|B7 -
       ↪ pred| = 1.6e-6  (|B7| ~ 0.168)
m=+0: A7 - (1/kappa) dT6/dtau0 = poly6: resid 3.3e-12;  max|B7 - pred| = 2.2e-10;  B7(p=0,2,4) = ['(6.267224237e-13 - 0.008888888773j)', '(-1.253491522e-12 + 0.01777777756j)',
       ↪ '(6.26802572e-13 - 0.008888888786j)']  vs -(2i/225) kappa^4 -> ['(-6.194374272e-66 - 0.008888888889j)', '(1.922852469e-64 + 0.01777777778j)', '(1.488905023e-62 -
       ↪ 0.008888888889j)']
```

42. prl_checks/se7_analyze.py

Classical a^6 coefficients by extrapolation in q. ϵ^7 coefficient at super-extremal $q = 1.5 \ldots 5000$ and the extrapolation of $64A_7/q^6$ to $q \to \infty$, against the classical values: the limits commute at a^6. Writes `se7_A7.json`.

Paper: S9(h) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 se7_analyze.py`

Criterion	Measured	Required	
eta-even: 64 A7/q^6 extrapolated from 300 ≤ q ≤ 5000 vs classical c^6	1.896481592 (diff 1.6e-8)	diff ≤ 1e-7	✓
eta-odd: same vs 256/135	1.896296296 (diff 3e-10)	diff ≤ 1e-8	✓
classical eta-odd c^6 = -256/135	-1.8962962963 (diff 3.7e-12)	diff ≤ 1e-9	✓

Output (`logs/prl-checks-se7-analyze-py.txt`):

```
targets (classical c^6, Im): eta-even 1.89648160787   eta-odd -1.8962962963
   q     nr   64*A7_even/q^6 (Im)        64*A7_odd/q^6 (Im)       B7_even/q^6            B7_odd/q^6
   1.5   2          6.7941209753548          4.6187986312784   (0.0006005 + 0.03992j)   (4.217e-7 - 2.738e-7j)
     2   2          4.3785686410202          3.1439481847354    (0.000239 + 0.01804j)   (1.679e-7 + 1.145e-7j)
     3   2           3.540591743796          2.1262318745975   (6.777e-5 + 0.005899j)   (4.761e-8 + 1.188e-7j)
     5   2          2.8845188441687          2.7805597790794    (1.43e-5 + 0.001441j)   (1.005e-8 + 7.712e-8j)
     8   2          2.5275056422462          2.4949814328257   (3.46e-6 + 0.0003981j)   (2.434e-9 + 4.895e-8j)
    12   2          2.3246305313659          2.3117939401419   (1.021e-6 + 0.0001337j)   (7.19e-10 + 3.277e-8j)
    20   2          2.1575245722084          2.1532636214975   (2.199e-7 + 3.519e-5j)   (1.551e-10 + 1.97e-8j)
    30   2          2.0719984581797          2.0701177197449   (6.498e-8 + 1.268e-5j)   (4.592e-11 + 1.314e-8j)
    50   5           2.002528358312          2.0017727472321   (1.412e-8 + 3.696e-6j)   (3.489e-16 + 6.379e-13j)
    80   5          1.9630273225691          1.9626333716055   (3.446e-9 + 1.251e-6j)   (2.419e-16 + 6.204e-13j)
   120   3          1.9409446500446          1.9406735776586    (1.021e-9 + 5.08e-7j)   (1.721e-16 + 6.106e-13j)
   200   3          1.9232066949828          1.9229956147609   (2.206e-10 + 1.69e-7j)   (1.083e-16 + 6.025e-13j)
   300   3          1.9143137720879          1.9141200942437   (6.535e-11 + 7.204e-8j)   (3.318e-17 + 2.66e-13j)
   600   3          1.9054051291833          1.9052202996853   (8.169e-12 + 1.724e-8j)   (4.292e-18 + 6.607e-14j)
  1000   3           1.901837403137          1.9016535996979   (1.764e-12 + 6.093e-9j)   (9.401e-19 + 2.372e-14j)
  2000   3          1.8991600931149          1.8989760353127   (2.206e-13 + 1.502e-9j)   (1.187e-19 + 5.919e-15j)
  5000   3          1.8975531276563          1.8973684527433   (1.412e-14 + 2.384e-10j)   (7.648e-21 + 9.459e-16j)
extrapolation q -> infinity (3 points, q >= 1000.0):  eta-even 1.896481576   eta-odd 1.896296296   (targets 1.896481608, 1.896296296 = 256/135)
extrapolation q -> infinity (4 points, q >= 600.0):  eta-even 1.896481588   eta-odd 1.896296296   (targets 1.896481608, 1.896296296 = 256/135)
extrapolation q -> infinity (5 points, q >= 300.0):  eta-even 1.896481592   eta-odd 1.896296296   (targets 1.896481608, 1.896296296 = 256/135)
```

43. prl_checks/se7_compare.py

ϵ^7 closed forms continued to super-extremal q. The physical-region ϵ^7 closed forms continued to $1.5 \le q \le 5000$ against the direct MST coefficients.

Paper: S9(h) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 se7_compare.py`

Criterion	Measured	Required	
eta-even, 50 ≤ q ≤ 5000: continued closed form vs direct MST (rel.)	max 4.9e-9 (9 values)	≤ 1e-8	✓
eta-odd, 50 ≤ q ≤ 5000	max 2.8e-10 (9 values)	≤ 1e-9	✓
both sectors, q ≤ 30 (less precise direct data)	max 0.00038 (16 values)	≤ 0.001	✓

Output (`logs/prl-checks-se7-compare-py.txt`):

```
   q      E7 closed-form (continued)          MST eps^7 (even)            rel.diff  |  D7 closed-form            MST (odd)            rel.diff
   1.5   (0.926724345835 + 1.20978237738j)  (0.926752836404 + 1.20920756617j)   0.00038 | (0.7074543178 + 0.8220849889j) (0.7075129476 + 0.8220469244j)    6.4e-5
   2.0    (2.31712752191 + 4.37973624421j)   (2.31716995889 + 4.37856864102j)   0.00024 |   (1.830393341 + 3.143858714j)   (1.830524469 + 3.143948185j)    4.4e-5
   3.0    (8.40773600406 + 40.3328505112j)   (8.40773745814 + 40.3295528317j)    8.0e-5 |   (6.948607019 + 35.60867782j)   (6.949030665 + 35.60973495j)    3.1e-5
   5.0    (46.0893646985 + 704.235780293j)    (46.0881561749 + 704.22823344j)    1.1e-5 |   (38.07344711 + 678.8328943j)    (38.0753636 + 678.8476023j)    2.2e-5
   8.0    (233.843451791 + 10352.6166877j)   (233.831169885 + 10352.6631106j)    4.6e-6 |   (192.6321105 + 10219.28732j)   (192.6398975 + 10219.44395j)    1.5e-5
  12.0    (1020.28195632 + 108457.262125j)   (1020.20760371 + 108457.962071j)    6.5e-6 |   (846.0344297 + 107857.8635j)   (846.0606353 + 107859.0581j)    1.1e-5
  20.0    (7709.32329154 + 2157513.19186j)   (7708.67449802 + 2157524.57221j)    5.3e-6 |    (6356.558633 + 2153248.23j)   (6356.679774 + 2153263.621j)    7.1e-6
  30.0     (44981.3375019 + 23601265.006j)   (44977.8627046 + 23601357.4377j)    3.9e-6 |   (35941.42103 + 23579817.72j)   (35941.82967 + 23579934.65j)    5.0e-6
  50.0    (487635.419204 + 488898525.285j)   (487635.420096 + 488898524.979j)   6.3e-10 |   (366356.9788 + 488714049.5j)   (366356.9789 + 488714049.6j)   2.8e-10
  80.0    (4792784.49445 + 8040559942.24j)   (4792784.48945 + 8040559913.24j)    3.6e-9 |  (3405790.849 + 8038946288.0j)   (3405790.85 + 8038946290.0j)   2.8e-10
 120.0    (35579391.7396 + 90556713982.5j)   (35579391.6411 + 90556713592.5j)    4.3e-9 | (24335037.3 + 9.054406641e+10j) (24335037.31 + 9.054406644e+10j)   2.8e-10
 200.0  (452823387.255 + 1.92320670373e+12j) (452823385.39 + 1.92320669498e+12j)    4.5e-9 | (299305422.5 + 1.922995614e+12j) (299305422.6 + 1.922995615e+12j)   2.8e-10
 300.0  (3430130133.95 + 2.18052304136e+13j) (3430130117.19 + 2.18052303102e+13j)    4.7e-9 | (2226313266.0 + 2.18030242e+13j) (2226313267.0 + 2.18030242e+13j)   1.3e-10
 600.0  (109697332648.0 + 1.38904034592e+15j) (109697332066.0 + 1.38904033917e+15j)    4.9e-9 | (6.986572285e+10 + 1.388905598e+15j) (6.986572285e+10 + 1.388905598e+15j)   3.3e-11
```

```
1000.0  (1.411118985e+12 + 2.9716209569e+16j) (1.41111897793e+12 + 2.9716209424e+16j)    4.9e-9 | (8.918066378e+11 + 2.971333749e+16j) (8.918066378e+11 + 2.97133375e+16j)   1.2e
        ↪ -11
2000.0  (4.51744877158e+13 + 1.8991601024e+18j) (4.517448755e+13 + 1.89916009311e+18j)    4.9e-9 | (2.838257413e+13 + 1.898976035e+18j) (2.838257413e+13 + 1.898976035e+18j)   3.2e
        ↪ -12
5000.0  (4.41303019272e+15 + 4.63269808824e+20j) (4.41303019687e+15 + 4.63269806557e+20j)    4.9e-9 | (2.762836687e+15 + 4.632247199e+20j) (2.762836687e+15 + 4.632247199e+20j)
        ↪ 5.4e-13
```

44. prl_checks/a6_hyp.py

Negative test at a^6. Shifting only the singular function (trigamma or csch^2) gives the wrong large-q limit; the check passes when this hypothesis is rejected.

Paper: S9(h) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 a6_hyp.py`

Criterion	Measured	Required	
negative control: shifted singular function, |64 X7/qˆ6 (m = 2, large q) - 256/135|	0.19	≥ 0.1	✓

Output (`logs/prl-checks-a6-hyp-py.txt`):

```
m=+2:  64 X7_even/qˆ6 -> [’(0.0 + 1.706665529j)’, ’(0.0 + 1.706666655j)’, ’(0.0 + 1.706666667j)’]
        64 X7_odd/qˆ6  -> [’(0.0 + 1.706665529j)’, ’(0.0 + 1.706666655j)’, ’(0.0 + 1.706666667j)’]
m=+1:  64 X7_even/qˆ6 -> [’(0.0 - 0.8533313422j)’, ’(0.0 - 0.8533333134j)’, ’(0.0 - 0.8533333331j)’]
        64 X7_odd/qˆ6  -> [’(0.0 - 0.8533313422j)’, ’(0.0 - 0.8533333134j)’, ’(0.0 - 0.8533333331j)’]
m=-1:  64 X7_even/qˆ6 -> [’(0.0 - 0.8533313422j)’, ’(0.0 - 0.8533333134j)’, ’(0.0 - 0.8533333331j)’]
        64 X7_odd/qˆ6  -> [’(0.0 + 0.8533313422j)’, ’(0.0 + 0.8533333134j)’, ’(0.0 + 0.8533333331j)’]
m=-2:  64 X7_even/qˆ6 -> [’(0.0 + 1.706665529j)’, ’(0.0 + 1.706666655j)’, ’(0.0 + 1.706666667j)’]
        64 X7_odd/qˆ6  -> [’(0.0 - 1.706665529j)’, ’(0.0 - 1.706666655j)’, ’(0.0 - 1.706666667j)’]
targets: pole part at aˆ6 = (10m/9) x aˆ5 pole part;  m=2: 256/135 = 1.896296296 ; m=1: (10/9)(16/75) = 0.237037037
```

Analytic control of the exchange-odd response

Paper: S7; S9(j)

Full-MST checks of the analytic structure behind Eqs. (10) and (11): the normalization of $\delta^2\mathcal{K}_+$, the ϵ coefficient $\pi - i\nu_2$ with $\nu_2 = -107/210$, the absence of the near-zone pole term from S, and the absorption difference $\Gamma_+ - \Gamma_- = -4\epsilon^6 A_6^-$.

45. prl_checks/analytic_odd.py

Normalization and the ϵ coefficient. $\delta^2\mathcal{K}_+/[(i/1800)\epsilon^7\kappa^5\tau(1+\tau^2)(4+\tau^2)] \to 1 + \frac{5mq}{9}\epsilon$, and $(e^{i\pi\nu} + e^{-i\pi\nu}\rho)\epsilon/(2i\delta) \to 1 + (\pi - i\nu_2)\epsilon$.

Paper: S7(c)-(e), S9(j) Status: **PASS** Time: 7 s
Command: `cd prl_checks && python3 analytic_odd.py`

Criterion	Measured	Required	
normalization ratio: O(eps) coefficient c1 vs 5mq/9 (finite eps)	max diff 0.00142 (4 values)	diff $\leq$ 0.005	✓
E coefficient vs pi - i nu2	max diff 7.3e-6 (4 values)	diff $\leq$ 1e-5	✓
exchange-law residual	max 1.17e-60 (4 values)	$\leq$ 1e-55	✓

Output (`logs/prl-checks-analytic-odd-py.txt`):

```
m=+2 q=0.3: c1 = (0.3347396886 + 1.749298677e-6j)  (5mq/9 = 0.3333333333) | E-coefficient = (3.141594214 + 0.5095172798j)  (pi - i nu2 = (3.141592654 + 0.5095238095j)) | law
        ↪ residual 2.07e-61 | x = (-0.509474 + 1.8569335e-65j)
m=+1 q=0.3: c1 = (0.1680759988 + 8.630466183e-6j)  (5mq/9 = 0.1666666667) | E-coefficient = (3.141594265 + 0.5095171327j)  (pi - i nu2 = (3.141592654 + 0.5095238095j)) | law
        ↪ residual 1.11e-60 | x = (-0.50949899 - 7.7403631e-65j)
m=-1 q=0.5: c1 = (-0.2763604356 + 1.831873013e-6j)  (5mq/9 = -0.2777777778) | E-coefficient = (3.141594402 + 0.5095167185j)  (pi - i nu2 = (3.141592654 + 0.5095238095j)) | law
        ↪ residual 1.17e-60 | x = (-0.50956565 + 2.2454321e-64j)
m=+2 q=0.5: c1 = (0.5569578188 + 1.401707289e-6j)  (5mq/9 = 0.5555555556) | E-coefficient = (3.141594145 + 0.5095174786j)  (pi - i nu2 = (3.141592654 + 0.5095238095j)) | law
        ↪ residual 1.09e-60 | x = (-0.50944067 + 7.4186597e-65j)
```

46. prl_checks/series_ratio_check.py

$\mathcal{X}$ ratio and ν to $O(\epsilon^3)$. Full MST at 60 digits: $\mathcal{X}$ ratio $1 + \frac{5mq}{9}\epsilon + \ldots$ and $\nu = 2 - \frac{107}{210}\epsilon^2 + \frac{mq}{12}\epsilon^3$ for four (m, q); ν_2 depends on neither m nor q.

Paper: S7, S9(j) Status: **PASS** Time: 5 s
Command: `cd prl_checks && python3 series_ratio_check.py`

Criterion	Measured	Required	
X-ratio coefficient c1 vs 5mq/9	max diff 1.32e-6 (4 values)	diff $\leq$ 2e-6	✓
(nu - 2)/epsˆ2 at eps = 1e-3 vs nu2 = -107/210, four (m, q)	max diff 6.69e-5 (4 values)	diff $\leq$ 0.0002	✓
nu3 vs mq/12	max diff 0.000189 (4 values)	diff $\leq$ 0.001	✓

Output (`logs/prl-checks-series-ratio-check-py.txt`):

```
m=+2 q=0.3: c1 = (0.333332458298 + 7.19155658477e-58j)  (5mq/9 = 0.333333333333)   c2 = (-0.575842 - 2.09438e-55j)  (-5/9 + ... )   (nu-2)/epsˆ2 at eps=1e-3: (-0.5094739959 +
        ↪ 1.856933518e-65j)  -> nu3 ˜ (0.0498136 + 1.85693e-62j) (mq/12 = 0.05)
m=+1 q=0.5: c1 = (0.277777064559 + 1.62831209239e-59j)  (5mq/9 = 0.277777777778)   c2 = (-0.579267 - 2.349e-57j)  (-5/9 + ... )   (nu-2)/epsˆ2 at eps=1e-3: (-0.50948232 -
        ↪ 5.647196457e-65j)  -> nu3 ˜ (0.0414895 - 5.6472e-62j) (mq/12 = 0.0416667)
m=-2 q=0.4: c1 = (-0.444443128977 + 4.25144586621e-58j)  (5mq/9 = -0.444444444444)   c2 = (-0.529844 - 1.09433e-55j)  (-5/9 + ... )   (nu-2)/epsˆ2 at eps=1e-3: (-0.5095906653 -
        ↪ 4.022178875e-64j)  -> nu3 ˜ (-0.0668558 - 4.02218e-61j) (mq/12 = -0.0666667)
m=+0 q=0.6: c1 = (2.28742888163e-8 - 6.22287728452e-56j)  (5mq/9 = 0.0)   c2 = (-0.608177 + 1.55571e-53j)  (-5/9 + ... )   (nu-2)/epsˆ2 at eps=1e-3: (-0.5095239769 - 2.879983176e
        ↪ -68j)  -> nu3 ˜ (-0.000167337 - 2.87998e-65j) (mq/12 = 0.0)
```

47. prl_checks/eps4_check.py

Pole term absent from S; spheroidal coefficients. The pole term $(i/900)\epsilon^4\Pi_m/\nu_2$ of the near-zone factor is absent from S (ϵ^4 coefficient is cubic in q); $E_1 = -4m/3$ and $E_2 = \frac{10}{189}(9 - m^2) - 1$.

Paper: S7(f), S9(j) Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 eps4_check.py`

Criterion	Measured	Required	
spheroidal coefficient E1 vs -4m/3	max diff 8.42e-14 (4 values)	diff ≤ 1e-10	✓
spheroidal coefficient E2 vs (10/189)(9 - m^2) - 1	max diff 1.5e-9 (4 values)	diff ≤ 5e-9	✓
eps^4 coefficient of S: q^4 and q^5 terms (cubic in q: the pole term is absent)	max 2.19e-18 (4 values)	≤ 1e-15	✓
exchange-odd part of S through eps^5: |S(+) - S(-)|/(|S| + 1e-8), q^0...q^5	1.32e-13	≤ 1e-6	✓

Output (`logs/prl-checks-eps4-check-py.txt`):

```
eps^3 sign +1: q^0..q^5 = ['(0.86856087 + 0.82376289j)', '(-0.22253588 - 0.52359878j)', '(-2.0771773e-27 + 0.034391534j)', '(5.3361577e-24 - 1.974564e-17j)', '(-1.9308558e-27 +
    ↪ 1.7913563e-23j)', '(7.9415383e-33 + 3.2381462e-28j)']
eps^3 sign -1: q^0..q^5 = ['(0.86856087 + 0.82376289j)', '(-0.22253588 - 0.52359878j)', '(1.7671569e-27 + 0.034391534j)', '(5.3315302e-24 - 1.974564e-17j)', '(4.7475432e-28 +
    ↪ 1.7913563e-23j)', '(-8.6040538e-33 + 3.2381462e-28j)']
eps^4 sign +1: q^0..q^5 = ['(-0.84813736 + 0.28940463j)', '(0.24946339 - 0.52987294j)', '(-0.066325663 + 0.11774047j)', '(-6.5351951e-19 - 0.0031599059j)', '(2.1535949e-22 -
    ↪ 2.19396e-18j)', '(-8.2217064e-28 - 3.6116907e-23j)']
eps^4 sign -1: q^0..q^5 = ['(-0.84813736 + 0.28940463j)', '(0.24946339 - 0.52987294j)', '(-0.066325663 + 0.11774047j)', '(-6.5300338e-19 - 0.0031599059j)', '(-5.2952118e-23 -
    ↪ 2.19396e-18j)', '(8.9075944e-28 - 3.6116907e-23j)']
eps^5 sign +1: q^0..q^5 = ['(-1.0065728 + 1.0680959j)', '(0.79959073 - 1.0184172j)', '(-0.16569714 + 0.35314742j)', '(0.021911587 - 0.018138728j)', '(-2.6376114e-17 +
    ↪ 0.00024878656j)', '(9.1701452e-23 + 4.4234108e-18j)']
eps^5 sign -1: q^0..q^5 = ['(-1.0065728 + 1.0680959j)', '(0.79959073 - 1.0184172j)', '(-0.16569714 + 0.35314742j)', '(0.021911587 - 0.018138728j)', '(6.4853053e-18 + 0.00024878656
    ↪ j)', '(-9.9351556e-23 + 4.4234108e-18j)']
pole-term prediction (i/900) Pi_2/nu2: q^1 coeff = (0.0 - 0.017445483j)   q^3 coeff = (0.0 - 0.052336449j)
superradiant eps^5 prediction Pi_2/900: q^1 = 0.0088888889  q^3 = 0.026666667
m=+2: E1 = -2.66666666667 (-4m/3 = -2.66666666667)   E2 = -0.7354497358  ((10/189)(9-m^2)-1 = -0.7354497354)
m=+1: E1 = -1.33333333333 (-4m/3 = -1.33333333333)   E2 = -0.5767195758  ((10/189)(9-m^2)-1 = -0.5767195767)
m=+0: E1 = -8.42241654288e-14 (-4m/3 = 0.0)   E2 = -0.5238095223  ((10/189)(9-m^2)-1 = -0.5238095238)
m=-1: E1 = 1.33333333333 (-4m/3 = 1.33333333333)   E2 = -0.5767195758  ((10/189)(9-m^2)-1 = -0.5767195767)
```

48. `prl_checks/h_scaling.py`

Scaling of the near-zone factor. The exchange-even part of the near-zone factor scales as ϵ^4 (the pole term, cancelled by the far zone), the exchange-odd part as ϵ^6; the far-zone factor is exchange-invariant.

Paper: S7, S9(j) Status: **PASS** Time: 3 s
Command: `cd prl_checks && python3 h_scaling.py`

Criterion	Measured	Required	
variation of |H_even|/eps^4 over eps = 1e-3, 5e-4, 2.5e-4	0.000504	≤ 0.01	✓
variation of |H_odd|/eps^6	0.00545	≤ 0.01	✓
|S_far(+) - S_far(-)| (far zone exchange-invariant)	max 1.33e-74 (3 values)	≤ 1e-60	✓

Output (`logs/prl-checks-h-scaling-py.txt`):

```
eps=1e-3: |H_even|/eps^4 = 0.0066511877  |H_even|/eps^5 = 6.6511877  |H_odd|/eps^6 = 0.00041999007  |Sfar_+ - Sfar_-| = 1.33e-74  Sfar-1 ~ (-1.0 + 5.98291e-16j)
eps=5e-4: |H_even|/eps^4 = 0.006648944  |H_even|/eps^5 = 13.297888  |H_odd|/eps^6 = 0.00041846127  |Sfar_+ - Sfar_-| = 5.53e-76  Sfar-1 ~ (-1.0 + 2.05014e-17j)
eps=2.5e-4: |H_even|/eps^4 = 0.0066478326  |H_even|/eps^5 = 26.591331  |H_odd|/eps^6 = 0.00041769966  |Sfar_+ - Sfar_-| = 6.75e-80  Sfar-1 ~ (-1.0 + 6.97068e-19j)
```

49. `prl_checks/absorption_check.py`

$\Gamma_+ - \Gamma_- = -4\epsilon^6 A_6^-$. Physical solution against its image, including $q = 0$ (Page, 4/225), after linear extrapolation in ϵ.

Paper: S9(j) Status: **PASS** Time: 6 s
Command: `cd prl_checks && python3 absorption_check.py`

Criterion	Measured	Required	
(Gamma_+ - Gamma_-)/eps^6 extrapolated vs -4 A6^-, (m, q) = (2, 0), (0, 0.3), (2, 0.3), (-1, 0.5)	max 2.46e-6 (4 values)	≤ 5e-6	✓
q = 0: -4 A6^- vs Page 4/225	0.01777777778 (rel. diff 1.3e-10)	rel. diff ≤ 1e-9	✓

Output (`logs/prl-checks-absorption-check-py.txt`):

```
m=+2 q=0 eps=2e-3: Gamma+/eps^6 =   3906249023.0  Gamma-/eps^6 =   3906249023.0  (G+ - G-)/eps^6 =  0.01800323556   -4 A6^- = 0.01777777778
m=+2 q=0 eps=1e-3: Gamma+/eps^6 = 6.249999609e+10  Gamma-/eps^6 = 6.249999609e+10  (G+ - G-)/eps^6 =  0.01789000363   -4 A6^- = 0.01777777778
   linear extrapolation eps -> 0: 0.0177767717   |difference| = 1.01e-6
m=+0 q=0.3 eps=2e-3: Gamma+/eps^6 =   3906249157.0  Gamma-/eps^6 =   3906249157.0  (G+ - G-)/eps^6 =  0.01422609913   -4 A6^- = 0.01404368094
m=+0 q=0.3 eps=1e-3: Gamma+/eps^6 = 6.249999663e+10  Gamma-/eps^6 = 6.249999663e+10  (G+ - G-)/eps^6 =  0.01413447869   -4 A6^- = 0.01404368094
   linear extrapolation eps -> 0: 0.01404285824   |difference| = 8.23e-7
m=+2 q=0.3 eps=2e-3: Gamma+/eps^6 =   3909375343.0  Gamma-/eps^6 =   3909375343.0  (G+ - G-)/eps^6 =   0.0016922774   -4 A6^- = 0.001667759776
m=+2 q=0.3 eps=1e-3: Gamma+/eps^6 = 6.252500138e+10  Gamma-/eps^6 = 6.252500138e+10  (G+ - G-)/eps^6 = 0.001679959963   -4 A6^- = 0.001667759776
   linear extrapolation eps -> 0: 0.001667642527   |difference| = 1.17e-7
m=-1 q=0.5 eps=2e-3: Gamma+/eps^6 =   3903646058.0  Gamma-/eps^6 =   3903646058.0  (G+ - G-)/eps^6 =  0.04717962616   -4 A6^- = 0.04661765471
m=-1 q=0.5 eps=1e-3: Gamma+/eps^6 = 6.247916756e+10  Gamma-/eps^6 = 6.247916756e+10  (G+ - G-)/eps^6 =  0.04689740936   -4 A6^- = 0.04661765471
   linear extrapolation eps -> 0: 0.04661519256   |difference| = 2.46e-6
Page: 4/225 = 0.01777777778
```

General kinematics

Paper: S6, Table S1

The exchange-odd BHPT amplitude in general kinematics against the contact-term ansatz of Bautista, Guevara, Kavanagh and Vines, the nine exchange-odd coefficients of Table S1, and the tagged and untagged exchange-even terms. The helicity -1 comparisons are negative controls.

50. `kinematics.py`

BHPT against the contact-term ansatz. Exchange-odd BHPT amplitude over the ansatz at random kinematic points: ratios 1 (a^5) and -1 (a^6, sign convention $a \to -a$) for helicity $+1$; failure for helicity -1.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `python3 kinematics.py`

Criterion	Measured	Required	
addition theorem for the spin-weighted harmonics	1 (diff 0)	diff $\leq$ 1e-12	✓
helicity +1, a^5: BHPT / contact-term ansatz at 6 random kinematic points	max diff 4.91e-31 (6 values)	diff $\leq$ 1e-25	✓
helicity +1, a^6: BHPT / ansatz = -1 (sign convention a -> -a)	max diff 3.79e-31 (6 values)	diff $\leq$ 1e-25	✓
helicity -1 (negative control): |ratio - 1| at a^5 and |ratio + 1| at a^6	min 0.332 (12 values)	$\geq$ 0.001	✓

Output (`logs/kinematics-py.txt`):

```
addition theorem check: 1.0

=== O(a^5) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(1.0 + 5.092615752e-33j)', '(1.0 - 4.911255482e-31j)', '(1.0 - 7.465622617e-32j)', '(1.0 - 4.792543642e-32j)', '(1.0 +
    ↪ 4.248888145e-33j)', '(1.0 - 1.88407939e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.06256355679 - 0.9980409818j)', '(-0.3269262284 + 0.9450498618j)', '(0.9385035227 + 0.3452696598j)', '(0.7291539887 +
    ↪ 0.6843496627j)', '(-0.736915553 - 0.6759848133j)', '(0.1965747281 - 0.9804888456j)']

=== O(a^6) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(-1.0 - 3.089289559e-31j)', '(-1.0 + 3.794871056e-31j)', '(-1.0 + 8.859119745e-32j)', '(-1.0 + 1.896233436e-31j)', '(-1.0
    ↪ + 7.930416306e-32j)', '(-1.0 + 1.97694001e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.05163961636 + 0.9986657849j)', '(0.1726873036 - 0.9849766978j)', '(-0.944851179 - 0.3274999992j)', '(-0.7563099249 -
    ↪ 0.6542134954j)', '(0.6148258526 + 0.7886629007j)', '(-0.03013458634 + 0.9995458502j)']
```

51. `fitcoeffs.py`

The nine exchange-odd coefficients by a fit. A least-squares fit of the ansatz structures to the BHPT amplitude at random points gives the nine coefficients of Table S1; perturbing one of them spoils the agreement.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `python3 fitcoeffs.py`

Criterion	Measured	Required	
nine exchange-odd coefficients: least-squares fit vs Table S1	max diff 0 (9 values)	diff $\leq$ 1e-15	✓
fit residuals at a^5 and a^6	max 1.02e-23 (2 values)	$\leq$ 1e-20	✓
negative control: c4^(1) perturbed by 0.01, |BHPT/ansatz - 1|	0.0145	$\geq$ 0.001	✓

Output (`logs/fitcoeffs-py.txt`):

```
addition theorem check: 1.0

=== O(a^5) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(1.0 + 5.092615752e-33j)', '(1.0 - 4.911255482e-31j)', '(1.0 - 7.465622617e-32j)', '(1.0 - 4.792543642e-32j)', '(1.0 +
    ↪ 4.248888145e-33j)', '(1.0 - 1.88407939e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.06256355679 - 0.9980409818j)', '(-0.3269262284 + 0.9450498618j)', '(0.9385035227 + 0.3452696598j)', '(0.7291539887 +
    ↪ 0.6843496627j)', '(-0.736915553 - 0.6759848133j)', '(0.1965747281 - 0.9804888456j)']

=== O(a^6) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(-1.0 - 3.089289559e-31j)', '(-1.0 + 3.794871056e-31j)', '(-1.0 + 8.859119745e-32j)', '(-1.0 + 1.896233436e-31j)', '(-1.0
    ↪ + 7.930416306e-32j)', '(-1.0 + 1.97694001e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.05163961636 + 0.9986657849j)', '(0.1726873036 - 0.9849766978j)', '(-0.944851179 - 0.3274999992j)', '(-0.7563099249 -
    ↪ 0.6542134954j)', '(0.6148258526 + 0.7886629007j)', '(-0.03013458634 + 0.9995458502j)']
O(a^5): residual 6.07e-26
```

```
   c4^(0)   fitted =    4.2666666666666666667   table = 4.2666666666666666667
   c4^(1)   fitted =                      3.2   table = 3.2
   c4^(2)   fitted =   0.26666666666666666667   table = 0.26666666666666666667
O(a^6): residual 1.02e-23
   c9^(0)   fitted =  -2.8444444444444444444   table = -2.8444444444444444444
   c9^(1)   fitted =  -2.1333333333333333333   table = -2.1333333333333333333
   c9^(2)   fitted =  -0.17777777777777777778   table = -0.17777777777777777778
   c10^(0)  fitted =  -5.6888888888888888889   table = -5.6888888888888888889
   c10^(1)  fitted =  -7.8222222222222222222   table = -7.8222222222222222222
   c10^(2)  fitted =  -2.1333333333333333333   table = -2.1333333333333333333

perturbed c4^(1) by 0.01 -> BHPT/ansatz = (1.01433134848 + 0.0021388176235j)
```

52. prl_checks/kinematics.py

Copy of the top-level `kinematics.py`. Same as the top-level script.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 kinematics.py`

Criterion	Measured	Required	
addition theorem for the spin-weighted harmonics	1 (diff 0)	diff $\leq$ 1e-12	✓
helicity +1, a^5: BHPT / contact-term ansatz at 6 random kinematic points	max diff 4.91e-31 (6 values)	diff $\leq$ 1e-25	✓
helicity +1, a^6: BHPT / ansatz = -1 (sign convention a -> -a)	max diff 3.79e-31 (6 values)	diff $\leq$ 1e-25	✓
helicity -1 (negative control): |ratio - 1| at a^5 and |ratio + 1| at a^6	min 0.332 (12 values)	$\geq$ 0.001	✓

Output (`logs/prl-checks-kinematics-py.txt`):

```
addition theorem check: 1.0

=== O(a^5) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(1.0 + 5.092615752e-33j)', '(1.0 - 4.911255482e-31j)', '(1.0 - 7.465622617e-32j)', '(1.0 - 4.792543642e-32j)', '(1.0 +
        ↪ 4.248888145e-33j)', '(1.0 - 1.88407939e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.06256355679 - 0.9980409818j)', '(-0.3269262284 + 0.9450498618j)', '(0.9385035227 + 0.3452696598j)', '(0.7291539887 +
        ↪ 0.6843496627j)', '(-0.736915553 - 0.6759848133j)', '(0.1965747281 - 0.9804888456j)']

=== O(a^6) ===
helicity +1: BHPT/ansatz at 6 random kinematic points: ['(-1.0 - 3.089289559e-31j)', '(-1.0 + 3.794871056e-31j)', '(-1.0 + 8.859119745e-32j)', '(-1.0 + 1.896233436e-31j)', '(-1.0
        ↪ + 7.930416306e-32j)', '(-1.0 + 1.976940001e-31j)']
helicity -1: BHPT/ansatz at 6 random kinematic points: ['(-0.05163961636 + 0.9986657849j)', '(0.1726873036 - 0.9849766978j)', '(-0.944851179 - 0.3274999992j)', '(-0.7563099249 -
        ↪ 0.6542134954j)', '(0.6148258526 + 0.7886629007j)', '(-0.03013458634 + 0.9995458502j)']
```

53. prl_checks/kin_even.py

Exchange-even a^5 contact term. The ψ-pole contributions are the α-tagged terms $c_3 = \alpha(64/15, 16/3, 16/15)$; helicity -1 is a negative control.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 kin_even.py`

Criterion	Measured	Required	
helicity +1: exchange-odd BHPT / ansatz (c4, alpha = 1) at 8 points	max diff 3.99e-30 (8 values)	diff $\leq$ 1e-25	✓
fit residual of the exchange-even a^5 contact term	4.9e-30	$\leq$ 1e-25	✓
fitted c3 vs -(64/15, 16/3, 16/15): the alpha-tagged psi-pole terms	max diff 3.38e-14 (3 values)	diff $\leq$ 1e-12	✓
helicity -1 (negative control): fit residual	10.1	$\geq$ 0.001	✓

Output (`logs/prl-checks-kin-even-py.txt`):

```
helicity +1
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 + 1.15550030966e-30j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 - 3.99095745504e-30j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 - 3.51372406425e-32j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 + 3.51531771552e-31j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 - 6.78634721435e-31j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 - 1.41104934934e-30j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 - 5.75602927942e-33j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (1.0 + 8.17436029701e-33j)
   fitted c3 (in units of the eta-odd normalization): ['(-4.2666666666667 - 6.5489590225383e-30j)', '(-5.3333333333333 - 1.0466248282701e-29j)', '(-1.0666666666667 -
        ↪ 5.1051063037823e-30j)']   residual 4.9e-30
```

```
   BGKV alpha=1: [64/15, 16/3, 4/3] = ['4.2666666666667', '5.3333333333333', '1.3333333333333'];  alpha=0: [0, 0, 4/15]
helicity -1
   eta-odd BHPT/ansatz(c4, alpha=1) = (0.976014894749 - 0.217703755661j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (-0.415657583811 + 0.909521177884j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (-0.978428189516 + 0.206587216351j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (0.760893838448 - 0.648876387775j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (0.96285775596 + 0.27000915130 5j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (0.954760172082 + 0.297376888487j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (-0.999558543294 + 0.0297105793807j)
   eta-odd BHPT/ansatz(c4, alpha=1) = (0.916430889158 + 0.400192985192j)
   fitted c3 (in units of the eta-odd normalization): ['(5.678624678112 + 0.029087110708328j)', '(10.922778500611 - 0.42211545757021j)', '(5.2095492705866 - 0.61719445246363j)']
        ↪  residual 10.1
   BGKV alpha=1: [64/15, 16/3, 4/3] = ['4.2666666666667', '5.3333333333333', '1.3333333333333'];  alpha=0: [0, 0, 4/15]
```

54. prl_checks/kin_tags.py

Tagged and untagged exchange-odd terms. The $m \neq 0$ (coth-pole) part gives the α-tagged $c_4 = \eta\alpha(64/15, 16/5, 0)$, the $m = 0$ part the untagged $c_4 = \eta(0, 0, 4/15)$; the exchange-even ψ-pole part gives c_3.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 kin_tags.py`

Criterion	Measured	Required	
eta-odd full part: coefficients vs (64/15, 16/5, 4/15)	max diff 3.33e-12 (3 values)	diff $\leq$ 1e-10	✓
eta-odd full part: fit residual	6.59e-31	$\leq$ 1e-25	✓
eta-odd m != 0 part: coefficients vs (64/15, 16/5, 0)	max diff 3.33e-12 (3 values)	diff $\leq$ 1e-10	✓
eta-odd m != 0 part: fit residual	6.37e-31	$\leq$ 1e-25	✓
eta-odd m = 0 part: coefficients vs (0, 0, 4/15)	max diff 3.33e-13 (3 values)	diff $\leq$ 1e-10	✓
eta-odd m = 0 part: fit residual	1.39e-31	$\leq$ 1e-25	✓
eta-even psi-pole part: coefficients vs (64/15, 16/3, 16/15)	max diff 3.33e-12 (3 values)	diff $\leq$ 1e-10	✓
eta-even psi-pole part: fit residual	8.23e-31	$\leq$ 1e-25	✓
negative control: untagged c3^(2) = 4/15 as a pure l = 2 pattern, fit residual	1.18	$\geq$ 0.1	✓

Output (`logs/prl-checks-kin-tags-py.txt`):

```
eta-odd  full (-1)^m(2+m)!(2-m)!      -> (['4.26666666667', '3.2', '0.266666666667'], '6.59e-31')
eta-odd  m != 0 part (coth pole)     -> (['4.26666666667', '3.2', '-3.55055050272e-32'], '6.37e-31')
eta-odd  m = 0 part (polynomial eta) -> (['-1.07262097027e-31', '-2.35309856925e-32', '0.266666666667'], '1.39e-31')
eta-even psi-pole part               -> (['4.26666666667', '5.33333333333', '1.06666666667'], '8.23e-31')
BGKV: c4 = eta*(alpha 64/15, alpha 16/5, 4/15);  c3 = (alpha 64/15, alpha 16/3, 4/15 + alpha 16/15)
untagged c3^(2)=4/15 as l=2 pattern (m=2 antisym, m=1 antisym, m=0): ['(-3.567902848 + 8.470069125j)', '(2.156401397 - 0.3672993525j)', '(0.5935290311 - 2.98743298j)']  residual
        ↪ 1.18
```

55. prl_checks/kin_a6.py

Tagged and untagged terms at a^6. At a^6 the $m \neq 0$ (pole) part equals the α-tagged terms and the $m = 0$ part the untagged $c_9^{(2)} = -8/45$; all ratios are ± 1.

Paper: S6, Table S1 Status: **PASS** Time: <1 s
Command: `cd prl_checks && python3 kin_a6.py`

Criterion	Measured	Required	
eta-odd full / (tagged+untagged): ratio = -1	max diff 1.25e-30 (6 values)	diff $\leq$ 1e-25	✓
eta-odd m!=0 part / alpha-tagged: ratio = -1	max diff 1.45e-30 (6 values)	diff $\leq$ 1e-25	✓
eta-odd m=0 part / untagged: ratio = -1	max diff 1.35e-30 (6 values)	diff $\leq$ 1e-25	✓
eta-even pole part/ alpha-tagged: ratio = +1	max diff 3.56e-30 (6 values)	diff $\leq$ 1e-25	✓

Output (`logs/prl-checks-kin-a6-py.txt`):

```
ratios BHPT / BGKV-piece at 6 random kinematic points (helicity +1):
  eta-odd full      / (tagged+untagged): ['(-1.0 + 1.2535716e-30j)', '(-1.0 - 2.9666957e-31j)', '(-1.0 + 1.9609062e-31j)', '(-1.0 - 8.6009829e-31j)', '(-1.0 + 1.0224889e-30j)',
        ↪ '(-1.0 + 4.399991e-31j)']
  eta-odd m!=0 part / alpha-tagged     : ['(-1.0 + 1.4532493e-30j)', '(-1.0 - 3.494217e-31j)', '(-1.0 + 1.9727416e-31j)', '(-1.0 - 9.1524127e-31j)', '(-1.0 + 1.022579e-30j)',
        ↪ '(-1.0 + 3.3187587e-31j)']
  eta-odd m=0 part  / untagged         : ['(-1.0 + 5.5574103e-31j)', '(-1.0 - 3.7165272e-31j)', '(-1.0 - 1.3510145e-30j)', '(-1.0 - 5.4366543e-31j)', '(-1.0 + 3.1464012e-31j)',
        ↪ '(-1.0 - 6.456657e-32j)']
  eta-even pole part/ alpha-tagged     : ['(1.0 - 3.557073e-30j)', '(1.0 + 3.8335771e-31j)', '(1.0 - 5.4069023e-32j)', '(1.0 + 1.7462968e-30j)', '(1.0 - 2.5843393e-30j)', '(1.0 -
        ↪ 4.5657541e-31j)']
```

Unit-flux normalization and the absorption split

Paper: Eq. (11); S9(k), S10

Unit-flux normalization of the MST far-zone factor, flux balance $|S_P|^2 + \mathcal{P}_H = 1$ including superradiant cases and the complex-ν branch, and the exchange-parity split of the absorption probability, Eq. (11).

56. `flux_test/flux_check.py`

Flux balance and factorization. Floquet amplitudes evaluated separately from the continued-fraction coefficients: MST relations, factorized against direct S_P, $|\mathcal{C}|^2 = D^2 + 144M^2\omega^2$, flux balance; all m, $q = 0 \ldots 0.95$, $\epsilon = 0.01 \ldots 0.4$.

Paper: S10, S9(k) Status: **PASS** Time: 16 s
Command: `cd flux_test && python3 flux_check.py`

Criterion	Measured	Required	
MST continued-fraction residual g(nu)	max 1.3e-60 (17 values)	≤ 1e-55	✓
B[R_Cˆ(-nu-1)]/B[R_Cˆnu] relations (incoming and outgoing)	max 3.8e-59 (17 values)	≤ 1e-55	✓
factorized vs direct S_P	max 3.2e-61 (17 values)	≤ 1e-55	✓
|C|ˆ2 of S10 vs Dˆ2 + 144 Mˆ2 omegaˆ2	max 3.1e-61 (17 values)	≤ 1e-55	✓
flux balance |S_P|ˆ2 + P_H - 1	max 1.7e-60 (17 values)	≤ 1e-55	✓
|S/S_P| - D/|C| (D-normalized element)	max 2.3e-61 (17 values)	≤ 1e-55	✓
superradiant cases included (P_H < 0)	6 of 17	≥ 1	✓

Output (`logs/flux-test-flux-check-py.txt`):

```
Flux balance and factorization, s=-2, l=2 (60 digits).  Columns:
 m    q     eps   |  g(nu)   A-identities  S_fact/S_dir-1 | |C|ˆ2(S10)/(Dˆ2+144wˆ2)-1 | |S_P|ˆ2+P_H-1   P_H           | old norm: |S_old|ˆ2+P_H-1, |S_old/S_P| - D/|C|
+2   0.0  0.01 |  7.9e-62  3.8e-59      0.0 |      0.0 |  6.2e-61    1.8367776e-14                |    -6.25e-6  -7.8e-62
+2   0.0   0.3 |  1.0e-62  8.2e-61  1.6e-61 |      0.0 |  6.2e-61     7.0011989e-5                |   -0.0055931  -7.8e-62
+2   0.3  0.05 |  5.7e-61  2.7e-60  3.7e-62 |  1.6e-61 |  1.6e-60    -1.8585372e-9 (superradiant) |  -0.00015938       0.0
+1   0.3  0.05 |  2.9e-61  6.2e-60  1.6e-61 |  1.6e-61 |  1.1e-60   -5.8138304e-10 (superradiant) |   -0.0001578       0.0
+0   0.3  0.05 |  1.9e-61  5.8e-60  9.3e-62 |  1.6e-61 |  5.4e-61     2.702887e-10                |  -0.00015623   7.8e-62
-1   0.3  0.05 |  1.5e-61  2.1e-60  1.6e-61 |  1.6e-61 |  1.4e-60     1.3354313e-9                |  -0.00015467       0.0
-2   0.3  0.05 |  5.9e-61  2.1e-60  3.7e-62 |      0.0 |  1.4e-60     3.3402725e-9                |  -0.00015313   1.6e-61
+2   0.7   0.2 |  1.9e-61  1.4e-60  3.2e-61 |      0.0 |  1.4e-60    -7.6838938e-6 (superradiant) |   -0.0030071  -1.6e-61
+1   0.7   0.2 |  4.9e-61  2.4e-60  1.7e-61 |  1.6e-61 |  7.8e-61     -7.422609e-7 (superradiant) |   -0.0027416   7.8e-62
+0   0.7   0.2 |  9.3e-61  2.1e-60  1.1e-61 |  3.1e-61 |  1.4e-60       7.55259e-7                |   -0.0024984  -7.8e-62
-1   0.7   0.2 |  4.8e-61  1.2e-60      0.0 |  1.6e-61 |  6.2e-61     8.9878902e-6                |   -0.0022758  -7.8e-62
-2   0.7   0.2 |  6.5e-62  1.9e-60  5.1e-62 |  1.6e-61 |  1.4e-60     5.5184216e-5                |    -0.002072  -7.8e-62
+2  0.95   0.4 |  3.4e-61  2.3e-61  1.6e-61 |  1.6e-61 |  7.8e-62   -0.00049210855 (superradiant) |   -0.016456  -7.8e-62
+1  0.95   0.4 |  4.9e-61  1.1e-60  2.6e-61 |      0.0 |  1.7e-60    -7.3202258e-6 (superradiant) |   -0.012873   2.3e-61
+0  0.95   0.4 |  2.7e-61  4.5e-61  5.2e-62 |      0.0 |      0.0     2.8466504e-5                |   -0.010036   7.8e-62
-1  0.95   0.4 |  1.3e-60  9.1e-61  9.1e-62 |      0.0 |  9.3e-61     0.0021655341                |   -0.0077799       0.0
-2  0.95   0.4 |  4.6e-61  1.0e-60  1.6e-61 |  3.1e-61 |  1.2e-60       0.01993915                |   -0.0059147   1.6e-61
```

57. `flux_test/highfreq_check.py`

Flux balance at $\epsilon = 0.6$. The same flux balance at $M\omega = 0.3$, with ν followed by continuation in ϵ.

Paper: S10 Status: **PASS** Time: 14 s
Command: `cd flux_test && python3 highfreq_check.py`

Criterion	Measured	Required	
flux balance R + P_H - 1 at eps = 0.6	max 1.1e-60 (3 values)	≤ 1e-55	✓
continued-fraction residual |g|	max 2.7e-61 (3 values)	≤ 1e-55	✓

Output (`logs/flux-test-highfreq-check-py.txt`):

```
m=+2 q=0.95 eps=0.6: nu=(1.81864268655 - 7.65308406782e-69j) |g|=2.4e-61  R=  1.00359215401  P_H=-0.00359215401245 (superradiant)  R+P_H-1=-5.4e-61  |a_N|=5.5e-186
m=+2 q= 0.0 eps=0.6: nu=(1.77928054242 + 4.92735485789e-70j) |g|=2.7e-61  R= 0.945209438963  P_H=  0.054790561037               R+P_H-1=-1.1e-60  |a_N|=1.7e-149
m=-2 q=0.95 eps=0.6: nu=(1.69032267024 - 1.43037616791e-69j) |g|=1.2e-61  R= 0.496806846685  P_H=  0.503193153315               R+P_H-1= 1.1e-60  |a_N|=1.2e-183
```

58. `flux_test/complexnu_check.py`

The complex-ν branch. Flux balance for $\nu = \frac{1}{2} + iy$, $\frac{3}{2} + iy$, $-\frac{1}{2} + iy$ at $\epsilon = 0.8 \ldots 1.2$, with superradiant amplification up to about 11%.

Paper: S10 Status: **PASS** Time: 24 s
Command: `cd flux_test && python3 complexnu_check.py`

Criterion	Measured	Required	
flux balance R + P_H - 1 on the complex-nu branch	max 2.1e-59 (6 values)	$\leq$ 1e-55	✓
continued-fraction residual |g|	max 3.1e-60 (6 values)	$\leq$ 1e-55	✓
Re nu half-integer on the complex branch	max diff 0 (6 values)	diff $\leq$ 1e-12	✓
largest superradiant amplification -P_H	0.107	> 0	✓

Output (`logs/flux-test-complexnu-check-py.txt`):

```
m=+2 q=0.95 eps=0.8: nu=          (0.5 - 0.143853892834j) |g|=3.1e-60  R=  1.01479615442  P_H=-0.0147961544203 (superradiant)  R+P_H-1= 2.1e-59
m=+2 q=0.95 eps=1.0: nu=         (-0.5 + 0.516726953208j) |g|=3.1e-61  R=  1.04421285245  P_H=-0.0442128524475 (superradiant)  R+P_H-1= 4.7e-61
m=+2 q=0.95 eps=1.2: nu=          (-0.5 + 0.84476870908j) |g|=3.2e-61  R=   1.1069295934  P_H= -0.106929593396 (superradiant)  R+P_H-1=-6.3e-60
m=+2 q= 0.0 eps=0.8: nu=          (1.5 + 0.186246853145j) |g|=7.8e-62  R= 0.310178668422  P_H=  0.689821331578               R+P_H-1=-7.8e-61
m=+2 q= 0.0 eps=1.0: nu=          (0.5 + 0.361880615394j) |g|=9.1e-61  R=0.0154845487133  P_H=  0.984515451287               R+P_H-1= 9.3e-61
2 0.0 1.2 no complex root found
m=-2 q=0.95 eps=0.8: nu=          (1.5 + 0.383244031659j) |g|=1.3e-60  R=0.0324565514006  P_H=  0.967543448599               R+P_H-1= 4.7e-61
-2 0.95 1.0 no complex root found
-2 0.95 1.2 no complex root found
```

59. `flux_test/decomp_check.py`

Exchange-parity split of the absorption probability. $1 - |S_+|^2 = [1 - |S_{\rm even}|^2] + \Delta_{\rm odd} - |S_{\rm odd}|^2$ with the image computed directly: $\Delta_{\rm odd} \to \frac{\pi}{225}\kappa^5\varpi(1+\varpi^2)(4+\varpi^2)\bar{n}_\eta$, one half of Page's value at $q = 0$, and $\Delta_{\rm odd}/\mathcal{P}_H = \kappa/(1+\kappa)$ for $m = 0$.

Paper: Eq. (11), S10 Status: **PASS** Time: 48 s
Command: `cd flux_test && python3 decomp_check.py`

Criterion	Measured	Required	
Delta_odd/epsˆ6 vs (pi/225) kappaˆ5 varpi (1 + varpiˆ2)(4 + varpiˆ2) n_eta (rel.), both eta	max 2e-6 (12 values)	$\leq$ 5e-6	✓
Delta_odd/P_H vs kappa/(1 + kappa) (m = 0; 1/2 at q = 0)	max diff 2.3e-7 (6 values)	diff $\leq$ 1e-6	✓
flux balance and the decomposition identity	max 1.7e-60 (24 values)	$\leq$ 1e-55	✓

Output (`logs/flux-test-decomp-check-py.txt`):

```
 m    q   eta |  Delta_odd/eps^6      prediction           |  P_H/eps^6 (eps->0)   [1-|S_even|^2]/eps^6   Delta_odd/P_H   | flux res.
+2  0.0 +1 | 0.00888887857326 0.00888888888889 ( -1.2e-6) |  0.01777775068   0.008888872103   0.50000018 (k/(1+k) = 0.5) | 9.3e-61 id 1.0e-61
+2  0.0 -1 | 0.00888887210301 0.00888888888889 ( -1.9e-6) |  0.01777775068   0.008888878573   0.49999982 (k/(1+k) = 0.5) | 9.3e-61 id 1.8e-61
+0  0.6 +1 | 0.00291270829372 0.00291271111111 ( -9.7e-7) | 0.006553590313    0.00364088202   0.44444467 (k/(1+k) = 0.44444444) | 1.7e-60 id 1.5e-61
+0  0.6 -1 | 0.00291270547356 0.00291271111111 ( -1.9e-6) | 0.006553590313    0.00364088484   0.44444424 (k/(1+k) = 0.44444444) | 1.7e-60 id 7.0e-62
+0  0.3 +1 | 0.00702183256651 0.00702184046839 ( -1.1e-6) |  0.01438270756   0.007360874991   0.48821354 (k/(1+k) = 0.48821335) | 7.0e-61 id 9.3e-62
+0  0.3 -1 | 0.00702182714527 0.00702184046839 ( -1.9e-6) |  0.01438270756   0.007360880412   0.48821316 (k/(1+k) = 0.48821335) | 7.0e-61 id 7.5e-62
+2  0.3 +1 | 0.000833879124627 0.000833879888124 ( -9.2e-7) |   (eps^5 term)     -11.83587664    | 1.0e-60 id 1.6e-61
+2  0.3 -1 |  0.0433919080621  0.0433919883688 ( -1.9e-6) |   (eps^5 term)     -11.87843467    | 1.0e-60 id 2.4e-61
+1  0.5 +1 | 0.000619546565484 0.000619547080548 ( -8.3e-7) |   (eps^5 term)      -6.30850651    | 1.6e-60 id 3.0e-61
+1  0.5 -1 |  0.023308783696  0.0233088273565 ( -1.9e-6) |   (eps^5 term)     -6.331195747    | 1.6e-60 id 1.3e-61
-2  0.4 +1 |  0.0664025816226  0.0664026700448 ( -1.3e-6) |   (eps^5 term)      18.44185322    | 1.1e-60 id 2.0e-61
-2  0.4 -1 | 0.000275634801402 0.000275635345281 ( -2.0e-6) |   (eps^5 term)      18.50798017    | 1.1e-60 id 2.2e-61
```